

Higher Teichmüller theory: barycenter parametrisation of Hitchin components

Álvaro Sánchez Hernández

Born 24th February 1998 in Salamanca, Spain

August 19, 2022

Master's Thesis Mathematics

Advisor: Prof. Dr. Ursula Hamenstädt

Second Advisor: Dr. Alexander Thomas

MATHEMATISCHES INSTITUT

MATHEMATISCH-NATURWISSENSCHAFTLICHE FAKULTÄT DER
RHEINISCHEN FRIEDRICH-WILHELMS-UNIVERSITÄT BONN

Abstract:

In this Master thesis, we describe a parametrisation of Hitchin components with a geometric flavor. For this, we employ the barycenter method introduced by work of Douady-Earle and Besson-Courtois-Gallot. We explain the procedure followed by Slegers to apply such construction to higher Teichmüller theory and, basing ourselves on his stratagem, suggest some lines of improvement and further research.

The main goal of this discussion is to provide a comprehensive pathway into the topic for a graduate student. Therefore, the focus is on setting up the theory from the foundations and showing all the relevant ideas, intuitions and results that take place in the development. We attempt to segue smoothly through sections without dwelling on details that would not affect our subsequent formalism. We also contribute with a slight generalisation of the final statement.

The main background ideas are representations of semisimple Lie algebras and symmetric spaces of non-compact type. We start by presenting them and highlighting their interactions in the first two sections. Afterwards, we tackle the barycenter construction, that allows us to extend maps between boundaries of spaces to maps between the spaces themselves, and illustrate its geometric relevance. Finally, we briefly review the fundamental pillars supporting higher Teichmüller theory and apply our construction to this framework. The key is that representations in Hitchin components have a natural associated boundary map between symmetric spaces.

Given a Fuchsian representation and a Hitchin representation, the resulting parametrisation yields a map between symmetric spaces that intertwines them. The methodology grants geometric information about the maps obtained and, consequently, we hope this could shed some light on current strives within the field that might originate from lack of explicit knowledge of the parametrising maps, fulfilling a request by Hitchin.

Contents

0	Introduction	3
0.1	Structure	3
0.2	Narrative	4
1	Lie theory	11
1.1	The study of Lie algebras	11
1.2	Semisimplicity	14
1.2.1	Root systems, the Weyl group and Dynkin diagrams	14
1.2.2	Cartan subalgebras, root decomposition and the Killing form	17
1.2.3	$\mathfrak{sl}(2, \mathbb{C})$ and the principal TDS	23
1.3	Reality	26
2	Symmetric spaces	30
2.1	Nonpositive curvature and convexity	30
2.1.1	Boundary at infinity	32
2.1.2	Tits metric	34
2.2	Symmetric spaces	37
2.3	Metric, building and flags at infinity	40
2.4	Hierarchy in the non-compact regime and examples	44
3	The barycenter method	52
3.1	How to: barycenter maps	52
3.2	Why: Jacobian bounds	56
4	Hitchin components	60
5	Geometric parametrisation	66
5.1	Slegers parametrisation and its extensions	66
5.2	Further developments	72

0 Introduction

0.1 Structure

This master thesis tries to serve two purposes. On the one hand, it is intended as an introduction for graduate students to the topic of **higher Teichmüller theory**, perhaps from the eyes of a geometer. Thus, it is supposed to be as self-contained as possible within the time restrictions in hand (concerning proofs, we will avoid those for which there is extensive literature). On the other hand, it is a slight generalisation of a result from I. Slegers in this area: the parametrisation of Hitchin components using maps with geometric significance [Sle21].

For this goal, I chose to structure it in a matter similar to a book, treating concepts in thematical blocks. In the start, these blocks could appear slightly unrelated or unmotivated, so I provide a brief description of its role in the big picture of the theory in the beginning of the chapters or intertwining comments. As we get further into the exposition, the ideas will start to fit and its motivation will hopefully become clear. In this way, I intend to both provide enough direction and incentive to the development and also highlight the importance of the tools used as interesting in and of themselves.

Since the topic is one with connections to many areas, and this is part of its appeal and powerfulness, I also try to address it from as many perspectives as possible, even if this is done on a superficial manner.

The first part is going to be concerned with **Lie theory**: the representation theory of Lie groups, which is fundamented in the representation theory of Lie algebras. Our main object of study is a representation of a surface group into a semisimple Lie group. We then need to understand the structure of the latter and how this affects the properties of its representations. This is the most algebraic section, but permeates the whole theory.

The second part treats the geometry of **symmetric spaces**. Here we will find an interplay between Riemannian geometry and, again, Lie groups. We will carry out a series of geometrical constructions on certain manifolds associated to each of our Lie groups which will lay the proper ground for our parametrisation of Hitchin components.

In the third chapter we reveal the key to Slegers idea: the **barycenter method**. We want to present it and find a place for it in our context and purpose, as well as show its extraordinariness in modern differential geometry.

The fourth chapter will formalise the problem at hand and provide us with context and motivation: we present the study of **Hitchin components** and the reason behind our choice of parametrisation. Here we will also give an outline of the previous work in the subject.

During the fifth chapter we collect all the pieces of the puzzle and start joining them to fulfill our program. We describe Slegers result and at the same time point out our way of extending the scope of his **geometric parametrisation** of Hitchin components.

I would like to particularly thank Prof Ursula Hamenstädt for agreeing to supervise this thesis, granting me the opportunity to carry out this discussion, offering the topic and providing key insight into the subject, as well as for her courses on previous semesters on advanced geometry that motivated the topics and methods used and got me engaged in the area. Also thanks to my second supervisor, Dr Alexander Thomas, for bringing a new perspective into the topic, his help and energy, and the ideas from his course on non-abelian Hodge correspondence. I am also grateful for all the small conversations and moments with many other mathematicians and friends that helped me in different manners along the way, contributing with, as we say in Spanish, their *own grain of sand*; special mention to my Bachelor thesis supervisor Dr Luis Giraldo Suárez that introduced me one fortuitous afternoon years ago to the topic of Teichmüller theory.

0.2 Narrative

History is instructive. And what it suggests to people is that even if they do little things, if they walk on the picket line, if they join a vigil, if they write a letter to their local newspaper. Anything they do, however small, becomes part of a much, much larger sort of flow of energy. And when enough people do enough things, however small they are, then change takes place.

Howard Zinn

Consider a smooth orientable surface Σ , that is, a manifold of dimension 2. This means we have a topological surface equipped with a smooth structure. We can improve the chart transition functions from smooth functions to holomorphic functions to obtain a Riemann surface.

Definition 0.1. A *Riemann surface* is a topological manifold of dimension 2 equipped with a *complex structure*.

Notice in particular that we do not add the condition of *oriented* in the definition: this is because a complex structure already provides an orientation, since biholomorphic maps are orientation preserving.

Besides, such a structure allows us to define complex functions and meromorphic functions on our surface. Historically, Riemann surfaces arose as spaces in which multivalued functions (e.g. the logarithm) on the complex plane become univalued. What's more, its motivation can be dated back to classical mechanics: when trying to solve the dynamics of certain classical systems, we need to consider the inverse functions of certain integrals, called elliptic integrals (they were related to the problem of computing the arc length of an ellipse). These inverse functions are, in general, multivalued, and behave properly in a Riemann surface.

Example 0.2. Consider the Weierstrass $\wp$ function of periods $\alpha, \beta \in \mathbb{C} \setminus 0$, where $\text{Im}\left(\frac{\alpha}{\beta}\right) > 0$:

$$\wp(z) = \frac{1}{z^2} + \sum_{(m,n) \in \mathbb{Z}^2 \setminus (0,0)} \left(\frac{1}{(z - m\alpha - n\beta)^2} - \frac{1}{(m\alpha + n\beta)^2} \right)$$

It is periodic: $\wp(z + \alpha) = \wp(z + \beta) = \wp(z)$. If we consider the lattice Λ with basis $\{\alpha, \beta\}$, one can check that $\wp$ is meromorphic on the torus $\mathbb{C}/\Lambda$, which has an associated Riemann surface structure.

This function satisfies the following differential equation:

$$\wp'^2 = 4\wp^3 - 60G_4\wp - 140G_6 \tag{1}$$

where G_k are the Eisenstein series of weight k associated to the lattice Λ in a certain normalised form (any Eisenstein series can be expressed as a polynomial on G_4 and G_6):

$$G_k = \sum_{(m,n) \in \mathbb{Z}^2 \setminus (0,0)} \frac{1}{(m\alpha + n\beta)^k}$$

A further connection with algebraic geometry can be drawn by interpreting the previous differential equation as the one describing an elliptic curve (substituting $\wp'$ by y and $\wp$ by x).

From (1) one can obtain the elliptic integral:

$$z = \int \frac{d\wp}{\wp'} = \int \frac{d\wp}{\sqrt{4\wp^3 - 60G_4\wp - 140G_6}}$$

Studying the propagation of waves in shallow waters (following experiments from Scott Russell after observing the navigation of a towboat in the Union Canal connecting Glasgow and Edinburgh [Jag06]) one can arrive at the Korteweg-de Vries equation:

$$u_t = \frac{1}{4}u_{xxx} + 3uu_x$$

We can try to find a wavelike solution [Mul22] propagating with constant velocity c of the form $u(x, t) = -f(x + ct) + \frac{c}{3}$, which after direct substitution yields:

$$\frac{1}{4}f''' = 3ff' \implies 2f'' = 12f^2 - g_2 \implies 2f''f' = 12f^2f' - g_2f' \implies f'^2 = 4f^3 - g_2f - g_3$$

where g_2 and g_3 are constants of integration (we integrated, then multiplied by f' , then integrated again). Thus, by properly adjusting the constants, the Weierstrass function gives a solution of the KdV equation that can be inverted.

Now the question is:

In how many ways can we equip a certain smooth orientable surface with a Riemann surface structure (modulo biholomorphic equivalence)?

The short answer is: *infinitely many*.

The long answer is: we can attempt to parametrise them, giving birth to moduli spaces of Riemann surfaces and Teichmüller theory.

But let us first get back to the notion of complex structure. In particular, the differential of a holomorphic map is a complex linear map. Thus, given a smooth structure, we would need to define a way to *multiply by i* in our tangent spaces, so as to be able to consider complex scalar multiplication in the differential of our biholomorphic chart maps. This information is contained in what is called an almost complex structure.

Definition 0.3. *An **almost complex structure** on a smooth manifold M is a vector bundle isomorphism $J : TM \longrightarrow TM$ with $J^2 = -id$.*

We thus equip M with a complex linear structure on each tangent space. This mimics the property $i^2 = -1$, so that whenever we have $v \in TM$:

$$(a + ib)v := a + bJ(v) \quad \forall a, b \in \mathbb{R}$$

Note that such a structure can only be defined in an even-dimensional manifold:

$$\det(J)^2 = \det(J^2) = (-1)^{\dim(M)}$$

and the determinant of a vector bundle isomorphism is real whenever the manifold is real. Moreover, if the manifold has a complex structure with local holomorphic coordinates $x^j + iy^j$ then we can define the almost complex structure as a counterclockwise 90 degrees rotation:

$$J\left(\frac{\partial}{\partial x^j}\right) = \frac{\partial}{\partial y^j} \quad \& \quad J\left(\frac{\partial}{\partial y^j}\right) = -\frac{\partial}{\partial x^j}$$

Furthermore, if J is induced by a complex structure, then this one is unique.

The existence of an almost complex structure in a manifold is a well understood topological question: we can formulate it as a reduction of the structure group of the tangent bundle of M from $GL(2n, \mathbb{R})$ to $GL(n, \mathbb{C})$.

But is defining an almost complex structure enough to obtain a complex one? Turns out it is in the case of surfaces but not in the general case. One should look at whether we can define

holomorphic local coordinates as above, which in general we can only do for points but not for neighborhoods of points. If that is the case, we say the almost complex structure J is integrable. We can reduce the matter of integrability to the study of the Nijenhuis tensor associated to J .

Definition 0.4. Given a linear map L on TM , that is, $L \in \Gamma(TM \otimes T^*M)$ is a tensor field of type $(1,1)$, the Nijenhuis tensor N_L is defined as the tensor field of type $(1,2)$ (i.e. $N_L \in \Gamma(TM \otimes T^*M \otimes T^*M)$) such that:

$$N_L(X, Y) = -L^2[X, Y] + L([L(X), Y] + [X, L(Y)]) - [L(X), L(Y)]$$

Theorem 0.5 (Newlander-Nirenberg). An almost complex structure J is integrable iff $N_J = 0$.

In the case of complex dimension 1, the tangent space admits a basis of the form $\{X, J(X)\}$, and we can check the almost complex structure is always integrable:

$$N_J(X, J(X)) = [X, J(X)] + J([J(X), J(X)] + [X, -X]) - [J(X), -X] = 0$$

This means we can either look at almost complex structures or at complex structures in our surface (or conformal classes, which we do not explain here). The space classifying all of them (modulo a certain equivalence relation) is what is called moduli space. To be precise:

Definition 0.6. The **moduli space** of Riemann surface structures over a surface Σ is the quotient of all almost complex structures modulo orientation preserving diffeomorphisms on the surface:

$$\mathcal{M}(\Sigma) = \mathcal{J} / \text{Diff}^+$$

Example 0.7. If we fix our surface Σ to be a torus (closed orientable surface of genus 1), the Riemann surface structures are given by the different choices of lattices Λ such that $\Sigma = \mathbb{C}/\Lambda$, up to scaling of the lattice. With the freedom of rescaling, we can define the basis generating the lattice to be $\{1, \tau\}$ where τ is in the upper half plane $\mathbb{H}^2$ (i.e. $\text{Im}(\tau) > 0$). Notice also that we can translate τ to $\tau + m$ with $m \in \mathbb{Z}$ and obtain the same lattice, so we could assume $\text{Re}(\tau) \in [-\frac{1}{2}, \frac{1}{2}]$. Finally, if we choose the smallest vector in the initial basis to be the one rescaled to 1, then we have that $\|\tau\| \geq 1$. The points in the boundary of this region with positive real part a are identified with the ones with real part $-a$ and same imaginary part.

This region is the fundamental domain of the action of $PSL(2, \mathbb{Z}) = SL(2, \mathbb{Z}) / \{\pm id\}$ on $\mathbb{H}^2$. The resulting space has two singular points: the ones corresponding to the square lattice $\{1, i\}$ and the hexagonal lattice $\{1, \frac{1}{2} + \frac{\sqrt{3}}{2}i\}$ which arise because these lattices have additional symmetries compared to the general ones: the first is stabilised by S (order 2) and the second by ST (order 3), where:

$$PSL(2, \mathbb{Z}) = \langle S, T \mid S^2 = 1, (ST)^3 = 1 \rangle$$

$$S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \& \quad T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

where the brackets represent equivalence classes of matrices. The action is carried by Möbius transformations:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{corresponds to} \quad z \mapsto \frac{az + b}{cz + d}$$

This appearance of singularities is a general phenomenon that we would like to avoid. In particular, we could take the universal covering orbifold of the moduli space (to *unfold the singularities*) and obtain Teichmüller space $\mathcal{T}(\Sigma)$. A different approach yields the same result:

Definition 0.8. For a surface Σ , **Teichmüller space** is the space of all complex structures (or almost complex or conformal) up to the action of diffeomorphisms that are isotopic to the identity:

$$\mathcal{T}(\Sigma) = \mathcal{J} / \text{Diff}_0(\Sigma)$$

Comparing this with our definition of moduli space, we obtain:

Definition 0.9. Suppose we are working with an object X and we have a topology defined in its group of automorphisms. Then we call **mapping class group** $\text{MCG}(X)$ to be the group fitting in the short exact sequence:

$$1 \longrightarrow \text{Aut}_0(X) \longrightarrow \text{Aut}(X) \longrightarrow \text{MCG}(X) \longrightarrow 1$$

In the case of our surface Σ , the mapping class group is the quotient $\text{Diff}^+(\Sigma) / \text{Diff}_0(\Sigma)$, so that $\mathcal{M}(\Sigma) = \mathcal{T}(\Sigma) / \text{MCG}(\Sigma)$.

When talking about Teichmüller space, one generally refers to the case of surfaces of genus $g \geq 2$. This is because they feed on the power of hyperbolic geometry, as a consequence of the following result attributed to Poincaré:

Theorem 0.10 (Uniformisation Theorem). *A simply connected closed orientable Riemann surface is conformally equivalent to one of the following three models:*

- The sphere $\mathbb{S}^2$, with curvature $+1$.
- The Euclidean plane $\mathbb{C}$, with curvature 0 .
- The hyperbolic plane $\mathbb{H}^2$, with curvature -1 .

Consequently, every Riemann surface has one of these models as universal cover, and so we can pushforward the metric via the covering map to obtain a constant curvature metric on the surface. The first case corresponds to $g = 0$, the second to $g = 1$ and the third to $g \geq 2$, as can be seen for example using Gauss-Bonnet.

We will focus from now on on the hyperbolic case. Turns out every complex structure corresponds bijectively to a hyperbolic structure, so here we have a new equivalent point of view for our theory of moduli spaces. In particular, given a complex structure on Σ , we can lift it to its universal cover $\tilde{\Sigma} \cong \mathbb{H}^2$. In it, the fundamental group $\pi_1(\Sigma)$ acts isometrically by deck transformations. We recover the original surface by quotienting by it: $\Sigma \cong \tilde{\Sigma} / \pi_1(\Sigma)$.

Now, the group of orientation-preserving isometries of $\mathbb{H}^2$ is $\text{PSL}(2, \mathbb{R})$, so the action of the fundamental group on hyperbolic space is given by a homomorphism of $\pi_1(\Sigma)$ into $\text{PSL}(2, \mathbb{R})$. This has to be discrete and faithful to obtain a properly discontinuous action (so that the quotient map is a covering map as desired).

Definition 0.11. An injective homomorphism $\pi_1(\Sigma) \longrightarrow \text{PSL}(2, \mathbb{R})$ with discrete image is called **Fuchsian representation**.

We consider two representations to be in the same equivalence class if they differ by a Möbius transformation of $\mathbb{H}^2$, that is, by an element of $\text{PSL}(2, \mathbb{R})$, since they would both give us equivalent complex structures. As a result, one can identify $\mathcal{T}(\Sigma)$ with the connected component of faithful and discrete homomorphisms in the character variety:

$$\text{Rep}(\pi_1(\Sigma), \text{PSL}(2, \mathbb{R})) := \text{Hom}(\pi_1(\Sigma), \text{PSL}(2, \mathbb{R})) / \text{PSL}(2, \mathbb{R})$$

Another equivalent formulation is that of marked hyperbolic structures:

Definition 0.12. *Given a smooth orientable surface Σ , a **marked hyperbolic surface** is a pair (X, f) where X is a hyperbolic surface and $f : \Sigma \rightarrow X$ is a diffeomorphism.*

Teichmüller space is also the space of all marked hyperbolic surfaces modulo the equivalence relation:

$$(X, f) \sim (Y, g) \iff f \circ g^{-1} \text{ is isotopic to an isometry}$$

This is also sometimes called Fricke space, to distinguish it from Teichmüller space depending on if the focus is on Riemann surface or hyperbolic structures.

One can already fathom why Teichmüller theory enjoys such a rich theoretical body, and why one might try to find similar spaces and structures or attempt to replicate its behaviors. Unfortunately, we cannot spend more time here on its intricacies.

Satisfying the above premise, Nigel Hitchin found a similar component in the character varieties associated to other (*semisimple*) Lie groups [Hit92]. We say similar because it is composed solely of discrete and faithful representations (and in some cases, we even know it is a moduli space of certain geometric structures, like Teichmüller space is: [CG93] & [GW07]). Moreover, it contains a copy of Teichmüller space inside it, corresponding to a certain three-dimensional subgroup $\text{PSL}(2, \mathbb{R})$, the *principal TDS* that will appear later.

Definition 0.13. *The **Hitchin component** $\text{Hit}(G)$ associated to a (semisimple) Lie group G , is a connected component of the character variety $\text{Rep}(\pi_1(\Sigma), G)$ that consists of discrete and faithful representations. Their study is named **higher Teichmüller theory**.*

Hitchin showed that such a component is homeomorphic to $\mathbb{R}^{(2g-2)\dim(G)}$, so it is contractible. The Hitchin component in $\text{Rep}(\pi_1(\Sigma), G)$ thus consists of the homomorphisms

$$\pi_1(\Sigma) \rightarrow G$$

which can be continuously deformed to a composition of the form

$$\pi_1(\Sigma) \rightarrow \text{PSL}(2, \mathbb{R}) \rightarrow G$$

where the first is a Fuchsian representation and the second is what is called the *principal map* (we will see what this is when we talk about the *principal TDS*). In the case of $\text{PSL}(n, \mathbb{R})$, the principal map is the unique irreducible representation of $\text{PSL}(2, \mathbb{R})$ of dimension n . The principal map is unique for every G (up to conjugation), and these compositions are what constitute the copy of Teichmüller space inside the Hitchin component.

Definition 0.14. *A **G-Fuchsian representation** is a homomorphism of the above form: a composition of a Fuchsian representation (in $\text{PSL}(2, \mathbb{R})$) and a principal map. The representations in the Hitchin component which are not Fuchsian are sometimes called **G-Hitchin representations**.*

When we say Fuchsian without specifying the group G , we usually refer to $G = \text{PSL}(2, \mathbb{R})$, as 0.11 suggests.

The description given by Hitchin made use of the theory of Higgs bundles via the so called non-abelian Hodge correspondence, which in turn provides a parametrisation of them via equivariant harmonic maps. This perspective has a fruitful connection with physics, e.g. via moduli spaces of flat connections, but as Hitchin wrote in his seminal paper:

Unfortunately, the analytical point of view used for the proofs gives no indication of the geometrical significance of the Teichmüller component. Teichmüller space itself is, as we know, not simply a space of homomorphisms of a fundamental group, but more importantly is the space of conformal structures (or metrics of constant negative curvature) modulo the action of diffeomorphisms homotopic to the identity. This geometrical interpretation gives a natural action of the mapping class group on Teichmüller space. We do have an action in the general case too, but there is no geometrical interpretation to support it. The action on the moduli space of homomorphisms exists via the outer automorphisms of $\pi_1(\Sigma)$, and this at most permutes components: on the other hand since Teichmüller space is embedded in the Teichmüller component, the whole component is preserved by the action. This provokes consideration of the quotient of the Teichmüller component by the mapping class group and possible compactifications, but without further geometrical information it is difficult to proceed.

As a result, effort was made to obtain a different parametrisation that could reveal the underlying geometry in the same fashion as with classical Teichmüller space. One of this new ideas is the one developed by Slegers [Sle21] for $\mathrm{PSL}(n, \mathbb{R})$. He was able to apply the barycenter method by Besson, Courtois and Gallot [BCG95] for such purposes.

Let us denote here $\Gamma := \pi_1(\Sigma)$. The barycenter construction allows us to extend a map between the boundaries of symmetric spaces to a map between the symmetric spaces themselves. To be able to define it, one needs to work within the negative curvature regime. A priori, the barycentric maps on higher rank symmetric spaces are not necessarily well defined due to the existence of flat subspaces, but the special properties of Hitchin representations allow us to carry out its construction. In particular, these representations have a unique associated *Frenet hyperconvex curve*, which can be interpreted precisely as a map between boundaries of symmetric spaces. Moreover, the image of such hyperconvex curve *lands in a region of the boundary away from the flat subspaces*.

Therefore, if we fix a Fuchsian representation

$$\theta : \Gamma \longrightarrow \mathrm{PSL}(2, \mathbb{R})$$

then to each representation in Hitchin component

$$\rho : \Gamma \longrightarrow G$$

we can associate a map between boundaries (here K is a maximal compact subgroup of G , and $X := G/K$ admits then a natural Riemannian metric making it into a symmetric space with isometry group G)

$$\xi_\rho : \partial_\infty \Gamma \cong \partial_\infty \mathbb{H}^2 \longrightarrow \partial_\infty X$$

who in turn gives us via the barycenter method a map between symmetric spaces

$$f_{\theta, \rho} : \mathbb{H}^2 \longrightarrow X$$

This allows us to embed the Hitchin component into the space of continuous functions between two symmetric spaces, up to a certain conjugation:

$$\mathrm{Hit}(G) \longrightarrow \mathcal{C} := C^0(\mathbb{H}^2, X)/G, [\rho] \mapsto [f_{\theta, \rho}]$$

The compact-open topology in $C^0(\mathbb{H}^2, X)$ induces a topology in $\mathcal{C}$ through the quotient map. This parametrisation is similar to the one given by Corlette's theorem [Cor88], via ρ -equivariant harmonic maps (here the maps will also be ρ -equivariant).

Additionally, barycenter maps often admit explicit uniform bounds on its Jacobian [CF02], and thus are said to be (coarsely) area minimising: this is what makes them geometrically relevant. Corlette’s harmonic maps are minimisers of the energy functional, so both parametrisations employ *optimal* maps. We will now explain what all of this means in detail, but first let us establish the basic mathematical framework for the theory.

1 Lie theory

One of the characterisations of Hitchin components is that they only harbor representations which are discrete and faithful. In his seminal paper [Hit92], Hitchin showed that this phenomenon occurs for the *adjoint group of a split real form of a simple complex Lie group* (we will sometimes shorten it by saying *split real group*). We will soon understand exactly what this means. Nevertheless, the important features of these groups are contained in the study of *semisimple* Lie algebras, which we will kick off our discourse with.

In chapter 4, we will mention how discreteness and faithfulness can be captured in the notion of positivity, which has recently lead to the discovery of new Hitchin components in a larger class of groups [GLW21].

1.1 The study of Lie algebras

The main tool in the study of the representation theory of Lie groups is the use of the corresponding *infinitesimal* perspective given by the representation theory of Lie algebras. Let us try to briefly illustrate why semisimple Lie algebras are fundamental in such prospects [FH91].

Given a Lie group representation, we can always obtain a Lie algebra representation by differentiating at the identity (the tangent space at the identity of a Lie group is canonically isomorphic to its associated Lie algebra). The converse is not true, but we have the following result:

Theorem 1.1. *If G is simply connected, every representation of its Lie algebra $\mathfrak{g}$ comes from a representation of G .*

To obtain such a representation, we can use the exponential map (using the usual power series expression). In particular, for general G , every representation in $\mathfrak{g}$ comes from a representation in the universal cover $\tilde{G}$. This motivates the following definitions:

Definition 1.2. *A Lie group map $\phi : H \rightarrow G$ is called an **isogeny** if it is a covering space map of the underlying manifolds. We say two Lie groups G and H are *isogenous* if there exist an isogeny between them (in either direction).*

Every isogeny class has an initial element: the universal cover $\tilde{G}$ of any of them. If the center of the universal cover $Z(\tilde{G})$ is discrete (which happens in the semisimple case which we are interested in), it also has a final object: $\tilde{G}/Z(\tilde{G})$. This is a consequence of the following proposition:

Proposition 1.3. *If H is a Lie group and $\Gamma \subset Z(H)$ a discrete subgroup, then H/Γ admits a unique Lie group structure such that the quotient map is a Lie group map.*

Definition 1.4. *We call the initial element in an isogeny class the **simply connected form**. The final element is denoted **adjoint form** (when it exists).*

In particular, if G is connected then the adjoint form is the **adjoint group**, i.e. the image of G via the adjoint representation:

$$\mathrm{Ad} : G \longrightarrow \mathrm{GL}(\mathfrak{g})$$

Proposition 1.5. • $\mathrm{Ad}(G) \subset \mathrm{Aut}(\mathfrak{g})$.

- The Lie algebra of $\mathrm{Ad}(G)$ is $\mathrm{ad}(\mathfrak{g})$.
- If the ground field has characteristic 0, $\mathfrak{g}$ uniquely determines $\mathrm{Ad}(G)$, called the group of inner automorphisms of $\mathfrak{g}$. If moreover G is semisimple, then $\mathrm{Ad}(G)$ is the connected component of the identity in the group of automorphisms of $\mathfrak{g}$, i.e. $\mathrm{Aut}_0(\mathfrak{g})$.

Once we know why looking at Lie algebra representations might be useful, let us roughly classify them in an elemental manner. Recall first the notion of ideal, analogous to that in rings:

Definition 1.6. A Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$ is an **ideal** if it satisfies

$$[X, Y] \in \mathfrak{h} \quad \text{for all } X \in \mathfrak{h}, Y \in \mathfrak{g}$$

The ideals in Lie algebras correspond to normal subgroups in Lie groups.

Proposition 1.7. Let G be a connected Lie group with a connected subgroup $H \subset G$, with Lie algebras $\mathfrak{h}$ and $\mathfrak{g}$. Then H is a normal subgroup if and only if $\mathfrak{h}$ is an ideal in $\mathfrak{g}$.

In the same way normal subgroups are nice to quotient by, the same happens with ideals:

Proposition 1.8. The bracket operation on $\mathfrak{g}$ induces a bracket on the quotient $\mathfrak{g}/\mathfrak{h}$ if and only if $\mathfrak{h}$ is an ideal in $\mathfrak{g}$.

Now we can introduce different classes of Lie algebras:

Definition 1.9. A Lie algebra is named **simple** if $\dim(\mathfrak{g}) > 1$ and it contains no non-trivial ideals. Equivalently, the adjoint form of $\mathfrak{g}$ has no non-trivial normal Lie subgroups.

Definition 1.10. The **lower central series** of subalgebras $\mathcal{D}_k(\mathfrak{g})$ is defined inductively as:

$$\mathcal{D}_1(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}], \quad \mathcal{D}_k(\mathfrak{g}) = [\mathfrak{g}, \mathcal{D}_{k-1}(\mathfrak{g})]$$

We say $\mathfrak{g}$ is **nilpotent** if $\mathcal{D}_k(\mathfrak{g}) = 0$ for some k .

Definition 1.11. The **derived series** of subalgebras $\mathcal{D}^k(\mathfrak{g})$ is defined inductively as:

$$\mathcal{D}^1(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}], \quad \mathcal{D}^k(\mathfrak{g}) = [\mathcal{D}^{k-1}(\mathfrak{g}), \mathcal{D}^{k-1}(\mathfrak{g})]$$

We say $\mathfrak{g}$ is **solvable** if $\mathcal{D}^k(\mathfrak{g}) = 0$ for some k .

Definition 1.12. A Lie algebra $\mathfrak{g}$ is **semisimple** if it has no non-zero solvable ideals.

Now, for the reason why semisimple Lie algebras are fundamental in the representation theory of general Lie algebras:

Proposition 1.13. The sum of two solvable ideals in a Lie algebra is again solvable. In particular, the sum of all solvable ideals is a maximal solvable ideal $\text{Rad}(\mathfrak{g})$ called the **radical** of $\mathfrak{g}$.

The quotient $\mathfrak{g}/\text{Rad}(\mathfrak{g})$ is semisimple. Thus, for any Lie algebra $\mathfrak{g}$ we have an (splitting) exact sequence:

$$0 \longrightarrow \text{Rad}(\mathfrak{g}) \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\text{Rad}(\mathfrak{g}) \longrightarrow 0$$

where the first algebra is solvable and the last is semisimple.

Example 1.14. The most important examples of nilpotent and solvable Lie algebras are, respectively, the algebras of strictly upper triangular and upper triangular $n \times n$ matrices. In fact, all representations of solvable Lie algebras are upper triangular.

The representation theory of Lie algebras is also somewhat inspired by that of finite groups, although there one enjoys very nice properties. In particular, one tries to decompose the representations in simpler *atomic* components: one can always express a representation as a direct sum of irreducible representations, which are indecomposable themselves. A similar thing happens in nicer cases for Lie groups:

Definition 1.15. An **irreducible representation** $\rho : G \longrightarrow GL(V)$ (where V is a vector space) is one for which the only invariant subspaces are 0 and V . A representation is **completely reducible** if it can be expressed as a direct sum of irreducible representations.

In the case of semisimple Lie groups, every finite dimensional representation is completely reducible. Therefore, the goal of the theory is to classify all finite dimensional irreducible representations. The precise statement in our context is:

Proposition 1.16. If V is a representation of the semisimple Lie algebra $\mathfrak{g}$ (meaning we have a homomorphism $\rho : \mathfrak{g} \longrightarrow \text{End}(V)$) and $W \subset V$ an invariant subspace under the action of $\mathfrak{g}$, then there exists $W^\perp \subset V$ complementary to W and invariant under $\mathfrak{g}$.

Finally, we will see that a semisimple Lie algebra can be defined as a direct sum of simple Lie algebras. And we have a classification for them:

Theorem 1.17. Every simple complex Lie algebra is isomorphic to either:

- $\mathfrak{sl}(n, \mathbb{C})$, $\mathfrak{so}(n, \mathbb{C})$ or $\mathfrak{sp}(2n, \mathbb{C})$ for some n (**classical** Lie algebras)
- $\mathfrak{e}_6$, $\mathfrak{e}_7$, $\mathfrak{e}_8$, $\mathfrak{f}_4$ or $\mathfrak{g}_2$ (**exceptional** Lie algebras)

As for the solvable part of our Lie algebra ($\text{Rad}(\mathfrak{g})$), we can summarise the theory here already. The main points are given by the following theorems:

Theorem 1.18 (Engel's theorem). If $\mathfrak{g} \subset \mathfrak{gl}(V)$ is a Lie subalgebra such that every $X \in \mathfrak{g}$ is a nilpotent endomorphism of V (meaning some power of it is zero), then there exists a non-zero vector $v \in V$ such that $X(v) = 0$ for all $X \in \mathfrak{g}$.

By induction, this means that the action of X on V is strictly upper triangular: inductively, we know $X(v) = 0$, so X acts on $V/\langle v \rangle$ and here we can apply the induction hypothesis to obtain a basis $\bar{v}_i$ for which the action is upper triangular. Lifting each $\bar{v}_i \in V/\langle v \rangle$ to an element $v_i \in V$ an adding v as first vector gives then the desired result.

Theorem 1.19 (Lie's Theorem). Over an algebraically closed field of characteristic zero, if $\pi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is a finite dimensional representation of a solvable Lie algebra, then there's a flag $V = V_0 \supset V_1 \supset \dots \supset V_n = 0$ of invariant subspaces of $\pi(\mathfrak{g})$ with $\text{codim } V_i = i$.

This is a generalisation of the fact that commuting matrices are simultaneously upper triangularizable (they generate an abelian Lie algebra, which is solvable).

Remark 1.20. The statement is equivalent to:

There exists a vector $v \in V$ that is an eigenvector for each linear transformation in $\pi(\mathfrak{g})$.

Indeed, take any non-zero vector spanning V_{n-1} . Conversely, if v is such, take V_{n-1} to be its span and then $\pi(\mathfrak{g})$ admits a common eigenvector in V/V_{n-1} , and we can iterate.

If we take v to be the first vector in the base, the next step in the iteration to be the second, and so on, we will obtain the desired basis in which the matrices are upper triangular as a consequence of the invariance of the flag (the action of the matrix on a vector in the subspace V_i , that is, with the i last components being zero, has to land in V_i , so have i last components zero. Since this has to happen for arbitrary vectors in V_i , the matrix elements from $(n-i+1, n-i)$ to $(n, n-i)$ need to vanish). For $n = 4$, $i = 2$:

$$\begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix} \begin{pmatrix} * \\ * \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} * \\ * \\ 0 \\ 0 \end{pmatrix} \implies \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & * & * \end{pmatrix}$$

In terms of irreducible (finite dimensional) representations, Lie's theorem tells us that for solvable Lie algebras, these have to be one dimensional. So we can narrow ourselves to semisimplicity from now on:

Proposition 1.21. *Every irreducible representation of a complex (more generally for fields of characteristic 0) Lie algebra is of the form $V = A \otimes B$, where A is an irreducible representation of the semisimple part $\mathfrak{g}/\text{Rad}(\mathfrak{g})$ and B is a one dimensional representation on $\text{Rad}(\mathfrak{g})$.*

So, to summarise: we can study representations of Lie groups by looking at the representations of their Lie algebras. In the nicer (completely reducible) finite dimensional case, the building blocks are the irreducible representations. These can be split in a solvable and a semisimple part: the first is one dimensional and the second harbors a profound theory, where our focus will be from now on.

1.2 Semisimplicity

We want to understand the representations of an arbitrary complex semisimple Lie algebra $\mathfrak{g}$. For that, the main step is finding a maximal abelian subalgebra $\mathfrak{h} \subset \mathfrak{g}$ acting diagonally on it: the **Cartan subalgebra**. This acts via eigenvalues on the rest of the space, allowing a decomposition in 1-dimensional spaces $\mathfrak{g}_\alpha$, called **root space decomposition**. In this description, $\mathfrak{h}$ corresponds to $\alpha = 0$ and $\mathfrak{g} = \mathfrak{h} \oplus (\bigoplus \mathfrak{g}_\alpha)$. The visual and intuitive picture will be provided by the introduction of an inner product called the **Killing form**.

Before the representation-theoretical bit, we present the notion of roots abstractly for clarity. It is however difficult to establish the best order for the exposition of these topics, since they are all interrelated and reduce to just different avenues into the same phenomenon. Inevitably, concepts might be mentioned before the appropriate deep analysis on them is done, and it might feel slightly chaotic at times in favor of giving direction and motivation to our discourse.

1.2.1 Root systems, the Weyl group and Dynkin diagrams

The structure of semisimple Lie algebras is closely related to that of a root system. From its definition we will already see that this is a quite geometric notion, which will have a direct translation in the anatomy of the symmetric spaces we will deal with later on. This will be key when applying our barycenter method.

Definition 1.22. A **root system** in an Euclidean inner product vector space $(E, \langle \cdot, \cdot \rangle)$ is a finite set ϕ of non-zero vectors called **roots** such that:

- They span E .
- The only multiples of any $\alpha \in \phi$ are α and $-\alpha$ (we say it is a reduced system).
- ϕ is invariant under reflections through the hyperplanes perpendicular to the roots. In other words:

$$\forall \alpha, \beta \in \phi, \quad w_\alpha(\beta) := \beta - 2 \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \alpha \in \phi$$

- For every $\alpha, \beta \in \phi$, the projection of β onto the line through α is an integer or half-integer multiple of α . This means $2 \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} \in \mathbb{Z}$ (we say it is a crystallographic root system, or it satisfies the integrality condition).

The **rank** of ϕ is $\dim(E)$.

The **root lattice** is the $\mathbb{Z}$ -submodule of E generated by ϕ .

ϕ is said **irreducible** if it cannot be decomposed into $\phi_1 \times \phi_2 \cong \phi$ spanning $E_1 \times E_2 \cong E$.

Finally, we say (E_1, ϕ_1) and (E_2, ϕ_2) are **isomorphic** if and only if there exists an invertible linear transformation taking E_1 to E_2 and ϕ_1 to ϕ_2 and preserving the inner product on the roots.

We can now proceed to show the most elementary examples. For this, it is important to notice:

- β and $w_\alpha(\beta)$ differ by an integer multiple of α , by the third and fourth properties.
- If $S \subset E$ is spanned by $\psi = \phi \cap S$, then ψ is a root system in S . Therefore, the only possible ways in which two roots from a root system of arbitrary rank can be geometrically arranged are given by the root systems of rank 2.

Furthermore, we have the following [Hal03]:

Proposition 1.23. *Let $\alpha, \beta \in \phi$ such that $\langle \alpha, \alpha \rangle \geq \langle \beta, \beta \rangle$ and they are not multiples. Then we have one of these possibilities:*

- $\langle \alpha, \beta \rangle = 0$.
- $\langle \alpha, \alpha \rangle = \langle \beta, \beta \rangle$ and $\angle(\alpha, \beta) = \frac{\pi}{3}$ or $\frac{2\pi}{3}$.
- $\langle \alpha, \alpha \rangle = 2\langle \beta, \beta \rangle$ and $\angle(\alpha, \beta) = \frac{\pi}{4}$ or $\frac{3\pi}{4}$.
- $\langle \alpha, \alpha \rangle = 3\langle \beta, \beta \rangle$ and $\angle(\alpha, \beta) = \frac{\pi}{6}$ or $\frac{5\pi}{6}$.

This is a consequence of the fourth property and the fact that:

$$2 \frac{\langle \alpha, \beta \rangle}{\langle \alpha, \alpha \rangle} 2 \frac{\langle \beta, \alpha \rangle}{\langle \beta, \beta \rangle} = (2 \cos \angle(\alpha, \beta))^2 \in \mathbb{Z}$$

Example 1.24. *The case of rank 1 is trivial: the root system must be $\{\alpha, -\alpha\}$.*

In rank 2, we have 4 possibilities corresponding to $w_\alpha(\beta) = \beta + n\alpha$ for $n = 0, 1, 2, 3$, as a consequence of the previous proposition.

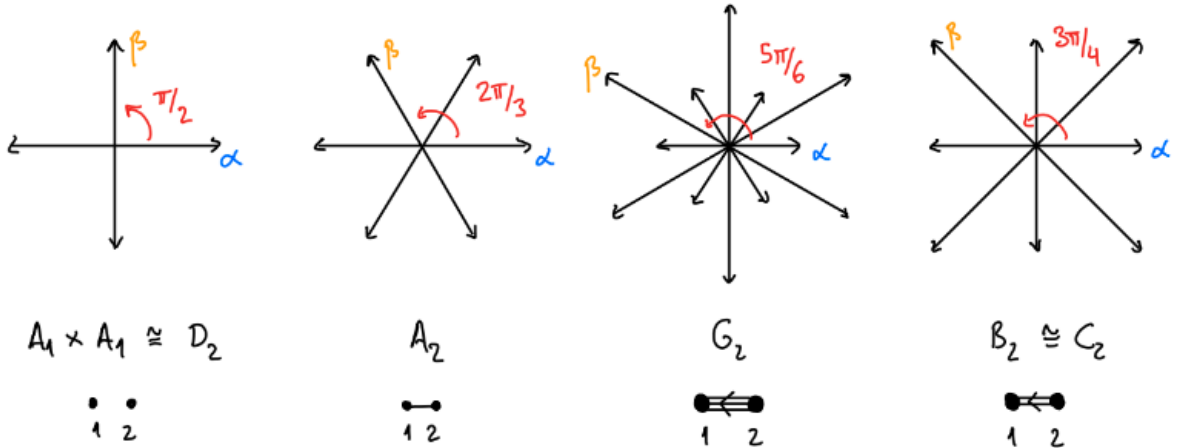


Figure 1: Rank 2 root systems with $\Delta = \{\alpha, \beta\}$ and their corresponding Dynkin diagrams. D_2 and B_2 both generate a square root lattice, while for A_2 and G_2 it is hexagonal.

The reflections along hyperplanes perpendicular to the roots that appear in the definition of root systems form a subgroup of the isometry group of E which will be fundamental in our future developments, because they codify the symmetries present in the root system.

Definition 1.25. The **Weyl group** associated to a root system ϕ is the group $\{w_\alpha \mid \alpha \in \phi\}$ of reflections along hyperplanes perpendicular to the roots. It is a finite reflection group, in particular, a Coxeter group. It acts faithfully on ϕ .

We thus notice that the root systems enjoy very specific symmetries. In particular, we could rotate the picture around in several ways leaving it unchanged. Thus it makes sense to have a way to choose an orientation:

Definition 1.26. A set of **positive roots** ϕ^+ is a subset of ϕ that:

- Contains either α or $-\alpha$ for every $\alpha \in \phi$.
- If $\alpha, \beta \in \phi^+$ and $\alpha + \beta \in \phi$, then $\alpha + \beta \in \phi^+$.

We can think of every set of positive roots to arise by taking a hyperplane not containing any root and choosing one side of it. Any two sets of positive roots differ by an element in the Weyl group.

We can use this to define base elements or simple roots in ϕ as the fundamental components:

Definition 1.27. A **simple root** $\alpha \in \phi$ is a root that cannot be written as the sum of two elements in ϕ^+ . The simple roots form a basis Δ of E such that:

- $\alpha \in \phi \implies \alpha \in \mathbb{Z}[\Delta]$.
- α is expressed as a sum of elements in Δ with all positive or all negative coefficients.

One can check that the Weyl group is generated by the reflections associated to simple roots. There is a notion of duality in root systems:

Definition 1.28. The **dual root system** to ϕ is the set of dual roots $\alpha^\vee := \frac{2}{\langle \alpha, \alpha \rangle} \alpha$ for every $\alpha \in \phi$.

The dual root system shares the Weyl group with the original, the dual of the dual is the original and the dual of the simple roots are the simple roots of the dual. According to the Dynkin diagram classification which we will see soon, B_n is dual to C_n but only isomorphic if $n = 2$, while the rest of the root systems are self-dual.

The geometry of the Weyl group is made explicit through the Weyl chamber structure:

Definition 1.29. Consider the hyperplanes perpendicular to each of the roots $\alpha \in \phi$ that allow us to define the elements w_α in the Weyl group. The complement of these hyperplanes in E is disconnected, and we call each connected component a **Weyl chamber**. We can also define the **fundamental Weyl chamber** associated to a basis of simple roots Δ as:

$$\{v \in E \mid \langle \alpha, v \rangle > 0 \quad \forall \alpha \in \Delta\}$$

The Weyl group acts freely and transitively on Weyl chambers, so its order as a group is equal to the number of Weyl chambers. The fundamental Weyl chamber is obtained by choosing for each hyperplane associated to a simple root the side (half-space) in which the simple root lies and intersecting the resulting half-spaces.

There is a more simplified tool for collecting the information about the different types of root systems: the **Dynkin diagrams**. Their importance goes beyond root systems, semisimple complex Lie algebras or Weyl groups, since they somewhat surprisingly appear in various contexts of mathematics.

A Dynkin diagram is simply a graph with some edges doubled or tripled. The construction of the Dynkin diagram given a root system is as follows: draw one node for each simple root and

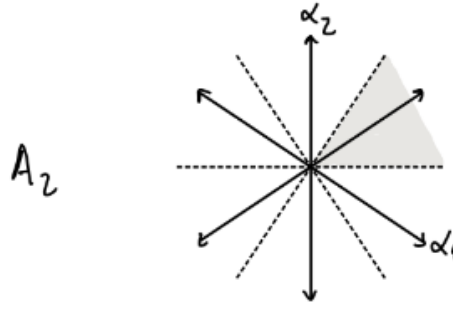


Figure 2: Fundamental Weyl chamber (grey shadow) for the root system A_2 with $\Delta = \{\alpha_1, \alpha_2\}$.

no lines		if $\vartheta = \pi/2$
one line		if $\vartheta = 2\pi/3$
two lines		if $\vartheta = 3\pi/4$
three lines		if $\vartheta = 5\pi/6$.

connect two nodes by a number of lines depending on the angle between the simple roots they represent [FH91]:

The arrow is directed from the longer root to the shorter one, like a *greater than* ($>$) sign. Recall that if the angle is $\frac{2\pi}{3}$, both roots have the same length.

Proposition 1.30. *Dynkin diagrams are connected if and only if the associated root system is irreducible*

This comes as a result of the fact that disconnected roots are perpendicular, so the root system can be decomposed into two, each harboring one of the disjoint roots, and similarly with the subspaces that these roots span.

We will see that the Dynkin diagrams of irreducible root systems give us the classification of semisimple Lie algebras: one can reconstruct the root system from the diagram, and the simple Lie algebra from the root system. Compare the following statement to that of 1.17.

Theorem 1.31. *There is a full classification of the Dynkin diagrams of irreducible root systems [FH91], showed in Figure 3.*

The first four correspond to the classical Lie algebras: $\mathfrak{sl}(n+1, \mathbb{C})$, $\mathfrak{so}(2n+1, \mathbb{C})$, $\mathfrak{sp}(2n, \mathbb{C})$ and $\mathfrak{so}(2n, \mathbb{C})$, respectively.

1.2.2 Cartan subalgebras, root decomposition and the Killing form

A broad overview of the structure and classification of semisimple Lie algebras can be posed in the following manner: we can start by finding a maximal abelian subalgebra $\mathfrak{h} \subset \mathfrak{g}$ acting diagonally. The maximality condition is to ensure that we obtain the most information via this action. This is the Cartan subalgebra (even though in the following development we fix a Cartan subalgebra, it

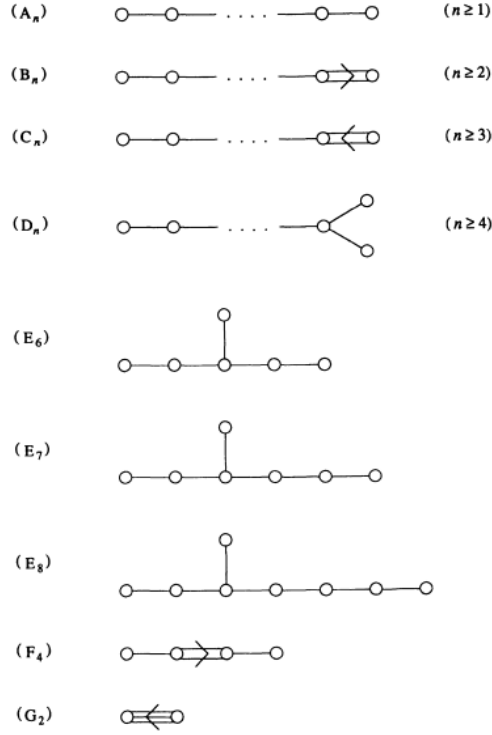


Figure 3: Dynkin diagrams of irreducible root systems [FH91].

is not unique: Cartan subalgebras can be shown to be conjugate by an inner automorphism of the adjoint subgroup of $\text{Aut}(\mathfrak{g})$ [FH91].

Definition 1.32. A *Cartan subalgebra* is a maximal abelian subalgebra of $\mathfrak{g}$.

We can let the elements in the Cartan subalgebra act on $\mathfrak{g}$ via the adjoint representation. The action of $\mathfrak{h}$ in any representation is diagonalisable, so in particular with the adjoint representation we arrive at a **Cartan decomposition**:

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^\vee} \mathfrak{g}_\alpha$$

This direct sum is over a finite set of eigenvalues which are called *weights* of the adjoint representation or *roots* of the Lie algebra $\mathfrak{g}$. Accordingly, the $\mathfrak{g}_\alpha$ are named root spaces. Here, $\mathfrak{h}$ itself corresponds to the *root* $\alpha = 0$, although we do not count this root when drawing analogs to root systems.

The action of $\mathfrak{h}$ preserves each $\mathfrak{g}_\alpha$ and acts on it via scalar multiplication by a linear functional $\alpha \in \mathfrak{h}^\vee$ (i.e. $\alpha : \mathfrak{h} \rightarrow \mathbb{C}$), meaning:

$$\text{ad}(H)(X) = \alpha(H)X \quad \text{where} \quad H \in \mathfrak{h}, X \in \mathfrak{g}_\alpha$$

The process of classification of irreducible representations still requires more steps in which embedded $\mathfrak{sl}(2, \mathbb{C})$ subalgebras and the Weyl group takes part, but we have already presented all the ingredients that will be relevant to us. Except for the fact that we need to introduce a *natural* inner product on $\mathfrak{g}$ (we say natural because it carries nice properties: it will allow us to reduce the information of the root spaces to that of a root system).

We thus need to define this inner product and show some of its characteristics before proceeding with the description. But first, we need to mention one of the properties that makes semisimple Lie algebras *nice*: their representations preserve Jordan decompositions (in particular, the adjoint representation):

Definition 1.33. A linear operator L on a finite dimensional vector space over a field is called **semisimple** if every L -invariant subspace has a complementary L -invariant subspace. A linear operator N is called **nilpotent** if some power N^m of it is the zero operator.

For an underlying algebraically closed field, L semisimple is equivalent to L diagonalizable, i.e. there exists an ordered basis of the vector space consisting of eigenvectors of L .

If L has a matrix representation $A = PDP^{-1}$ with D diagonal, the column vectors of P constitute a basis of eigenvectors of L , explaining the nomenclature.

Definition 1.34. A field $\mathbb{k}$ is called **perfect** if either it has characteristic 0 or, for characteristic p , every element is a p -th power.

Definition 1.35. The **Jordan-Chevalley decomposition** of a linear transformation X of a finite dimensional complex vector space V is given by:

$$X = X_s + X_n \quad (2)$$

where X_s is the semisimple part of X and X_n is the nilpotent part, and both parts commute with each other.

Over a perfect field, such a decomposition exists and is unique, and X_s and X_n can be given by polynomials in X with no constant terms. $\mathbb{C}$ is a perfect field, so for our purposes we have such decomposition.

If X is written in Jordan normal form with respect to some basis, then X_s is the endomorphism whose matrix contains the diagonal elements and X_n the remaining ones.

Example 1.36. An example of invariance of the Jordan decomposition is the following:

For $x \in \mathfrak{gl}(V)$ with Jordan decomposition $x = x_s + x_n$, then $\text{ad}(x) = \text{ad}(x_s) + \text{ad}(x_n)$ is the corresponding Jordan decomposition of $\text{ad}(x) \in \mathfrak{gl}(V)$. Indeed, both commute since $[\text{ad}(x_s), \text{ad}(x_n)] = \text{ad}([x_s, x_n]) = 0$, using $\text{ad}(x) = [x, -]$ and the Jacobi identity. Furthermore, ad preserves semisimplicity and if $x^m = 0$, then $\text{ad}(x)^{2m-1} = 0$. Finally, use uniqueness.

We can now introduce our inner product, which also characterises solvability and semisimplicity of the Lie algebra:

Definition 1.37. The **Killing form** on $\mathfrak{gl}(V)$ is the bilinear form defined by:

$$B_V(X, Y) = \text{Tr}(X \circ Y) \quad (3)$$

From the cyclic property of the trace one can show:

Proposition 1.38.

$$B_V(X, [Y, Z]) = B_V([X, Y], Z) \quad (4)$$

Recall that Lie's theorem 1.19 tells us that if $\mathfrak{g}$ is solvable, its adjoint representation can be put in upper triangular form and $[\mathfrak{g}, \mathfrak{g}]$ acts by strictly upper triangular matrices: indeed, use Jordan decomposition and that taking brackets with something semisimple, i.e. diagonal, is always zero. Thus, for $X \in [\mathfrak{g}, \mathfrak{g}]$ and $Y \in \mathfrak{g}$, $\text{ad}(X) \circ \text{ad}(Y)$ is strictly upper triangular (product of upper triangular with strictly upper triangular is strictly upper triangular). We get more:

Proposition 1.39. $\mathfrak{g}$ solvable $\iff B(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]) = 0$

This is a consequence of:

Theorem 1.40 (Cartan's Criterion). *If $\mathfrak{g}$ is a subalgebra of $\mathfrak{gl}(V)$ and $B_V(X, Y) = 0$ for all $X, Y \in \mathfrak{g}$, then $\mathfrak{g}$ is solvable.*

which builds on Engel's theorem 1.18.

Indeed, if we have $B([\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]) = 0$, the image of $[\mathfrak{g}, \mathfrak{g}]$ by the adjoint representation in $\mathfrak{gl}(\mathfrak{g})$ is solvable, using Cartan's Criterion. Now, the kernel of the adjoint map on $\mathfrak{g}$ is the center of $\mathfrak{g}$, which is abelian, and therefore $[\mathfrak{g}, \mathfrak{g}]$ is solvable (gives a chain for $[\mathfrak{g}, \mathfrak{g}]$ whose successive quotients are abelian: extend such a chain for $\text{ad}([\mathfrak{g}, \mathfrak{g}])$ adding $[\mathfrak{g}, \mathfrak{g}]$ at the start), hence $\mathfrak{g}$ is too (now add $\mathfrak{g}$).

But the Killing form can also say several things about semisimplicity:

Proposition 1.41. *A Lie algebra is semisimple if and only if its Killing form is nondegenerate.*

A semisimple Lie algebra is a product of simple Lie algebras. The only ideals are sum of those factors, so the decomposition is unique.

The adjoint map on a semisimple Lie algebra is an isomorphism into its algebra of derivations.

If the Lie algebra is nilpotent, its Killing form is identically zero, but the converse does not hold, as shown by the next example.

Example 1.42. *Consider the 3-dimensional Lie algebra given by the brackets $[e_1, e_2] = e_2$ and $[e_1, e_3] = \lambda e_3$. Then the Killing form is zero except for $B(e_1, e_1) = 1 + \lambda^2$. If we take $\lambda = \pm i$ and check that this is not nilpotent, we obtain the result.*

Indeed, to calculate the Killing form $B(X, Y)$, we need to look at the trace of $[X, [Y, -]]$, so we plug in the basis vectors for different choices of X and Y (enough to choose among the basis elements by linearity). We only get something non-trivial when looking at $[e_1, [e_1, -]]$, since $[e_1, [e_1, e_2]] = e_2$ and $[e_1, [e_1, e_3]] = \lambda^2 e_3$. We can see it is not nilpotent using a different formulation of Engel's theorem:

A Lie algebra $\mathfrak{g}$ is nilpotent if and only if $\text{ad}(X)$ is a nilpotent endomorphism of $\mathfrak{g}$ for every $X \in \mathfrak{g}$.

Now we look at $\text{ad}(e_1)$, which fixes e_2 . Thus, $\text{ad}^m(e_1) \neq 0$ for any $m \in \mathbb{N}$, so our Lie algebra cannot be nilpotent.

Once the Killing form is introduced, we can turn our attention back to Cartan subalgebras, show some properties and use them to describe root spaces as part of a root system.

Definition 1.43. *An element H of the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ such that $\alpha(H) \neq 0$ for all roots α is called **regular**. Here α denotes either the linear functional in $\mathfrak{h}^\vee$ or the scalar root that produces when it acts on a root space $\mathfrak{g}_\alpha$, i.e. $\text{ad}(H)(X) = \alpha(H)X$, when $X \in \mathfrak{g}_\alpha$.*

Proposition 1.44. *If $H \in \mathfrak{h}$ is regular, then:*

$$\mathfrak{h} = \mathfrak{c}(H) = \{X \in \mathfrak{g} : [H, X] = 0\} \quad (5)$$

The regular elements form a Zariski open subset of $\mathfrak{h}$: the complement of the union of hyperplanes $\{\alpha = 0\}$. Thus, they are dense.

In fact, the regular elements will belong to the Weyl chambers in our cross-comparison with root systems.

All elements of $\mathfrak{h}$ are semisimple (diagonal). Non-regular elements H have associated a $\mathfrak{c}(H)$ larger than $\mathfrak{h}$, since they contain other root spaces. As a consequence of these remarks, we show the existence of Cartan subalgebras first by characterising them as the subalgebras of the form $\mathfrak{c}(H)$ (for some semisimple H) that have minimal dimension.

Definition 1.45. The **rank** of a semisimple Lie algebra is such minimal dimension:

$$\text{rk}(\mathfrak{g}) = \min\{\dim(\mathfrak{c}(H)) \mid H \in \mathfrak{g} \text{ semisimple}\}$$

so regular elements are those whose centraliser $\mathfrak{c}(H)$ has said dimension. A Cartan subalgebra is an abelian subalgebra with all elements semisimple and maximal for that property.

This rank coincides with the dimension of the Euclidean space generated by the associated root system, as expected.

For H semisimple, $\mathfrak{g}$ decomposes into eigenspaces for its adjoint action:

$$\mathfrak{g} = \bigoplus_{\lambda} \mathfrak{g}_{\lambda}(H)$$

where $\mathfrak{g}_{\lambda}(H) = \{X \in \mathfrak{g} : [H, X] = \lambda X\}$ and $\lambda = 0$ corresponds to $\mathfrak{c}(H)$. In the non-semisimple case, we can kill off the nilpotent part eventually by defining the same decomposition with $\mathfrak{g}_{\lambda}(H) = \{X \in \mathfrak{g} : (ad(H) - \lambda I)^k(X) = 0 \text{ for large enough } k\}$.

Proposition 1.46. For any H we have $[\mathfrak{g}_{\lambda}(H), \mathfrak{g}_{\mu}(H)] \subset \mathfrak{g}_{\lambda+\mu}(H)$ via:

$$(ad(H) - (\lambda + \mu)I)^k([X, Y]) = \sum_{j=0}^k \binom{k}{j} [(ad(H) - \lambda I)^j(X), (ad(H) - \mu I)^{k-j}(Y)] \quad (6)$$

(use induction on k and Jacobi identity).

With the above we can follow onto several properties of root spaces:

Proposition 1.47. • If α is a root, then so is $-\alpha$.

- The roots α span $\mathfrak{h}^{\vee}$.
- $[\mathfrak{g}_{\alpha}, \mathfrak{g}_{-\alpha}] \neq 0$ and $[[\mathfrak{g}_{\alpha}, \mathfrak{g}_{-\alpha}], \mathfrak{g}_{\alpha}] \neq 0$.
- $\mathfrak{g}_{\alpha}$ is one dimensional for any root.
- 2α cannot be a root.

We can also define dual roots in this context:

Definition 1.48. Let $\alpha \in \mathfrak{h}^{\vee}$ be a root and define its **dual** $T_{\alpha} \in \mathfrak{h}$ via the pairing B as:

$$B(T_{\alpha}, H) = \alpha(H)$$

On another note, for every root $\alpha \in \mathfrak{h}^{\vee}$ we get a subalgebra isomorphic to $\mathfrak{sl}(2, \mathbb{C})$ (we will later talk more in detail about this especially important Lie algebra):

$$\mathfrak{s}_{\alpha} := \mathfrak{g}_{\alpha} \oplus \mathfrak{g}_{-\alpha} \oplus [\mathfrak{g}_{\alpha}, \mathfrak{g}_{-\alpha}]$$

with basis $X_{\alpha} \in \mathfrak{g}_{\alpha}$, $Y_{\alpha} \in \mathfrak{g}_{-\alpha}$ and $H_{\alpha} \in [\mathfrak{g}_{\alpha}, \mathfrak{g}_{-\alpha}]$ satisfying the commutation relations:

$$[X_{\alpha}, Y_{\alpha}] = H_{\alpha}, \quad [H_{\alpha}, X_{\alpha}] = 2X_{\alpha}, \quad [H_{\alpha}, Y_{\alpha}] = -2Y_{\alpha}$$

meaning H_{α} is determined by being in $[\mathfrak{g}_{\alpha}, \mathfrak{g}_{-\alpha}]$ and $\alpha(H_{\alpha}) = 2$. Thus, we have signaled out a particular regular element for each root.

The previous notions and statements should give us a feeling that we are in fact dealing with a root system within this context. So let us draw the relevant connections:

Start by defining the hyperplane

$$\Omega_\alpha := \{\beta \in \mathfrak{h}^\vee \mid \beta(H_\alpha) = 0\}$$

and the involution W_α on the vector space $\mathfrak{h}^\vee$ with $+1$ -eigenspace this hyperplane and -1 -eigenspace the line spanned by α . These involutions generate the Weyl group associated to the Lie algebra $\mathfrak{g}$. Now, the Killing form B is the unique inner product on $\mathfrak{h}^\vee$ that is preserved by the Weyl group, meaning the Weyl group acts as a group of orthogonal transformations over the inner product space $(\mathfrak{h}^\vee, B)$. Equivalently, it is the unique inner product up to scalars such that the line spanned by a root α is perpendicular to the hyperplane Ω_α . Then W_α is actually the reflection along the hyperplane Ω_α with axis the line spanned by α :

$$W_\alpha(\beta) = \beta - 2\frac{\beta(H_\alpha)}{\alpha(H_\alpha)}\alpha = \beta - \beta(H_\alpha)\alpha$$

As a consequence of the representation theory of $\mathfrak{sl}(2, \mathbb{C})$ that we will formulate later, given a representation of $\mathfrak{g}$ with eigenspace decomposition $V = V_\beta$, the weights can be grouped in equivalence classes modulo α , each of which is invariant under the involution W_α :

$$V_{[\beta]} = \bigoplus_{n \in \mathbb{Z}} V_{\beta+n\alpha}$$

In particular, possibly changing β by $\beta + n\alpha$ for $n \in \mathbb{Z}$, the weights that appear in the previous direct sum are:

$$V_{[\beta]} = \bigoplus_{n=0}^{2\beta(H_\alpha)} V_{\beta-n\alpha}$$

Furthermore:

$$W_\alpha(\beta + n\alpha) = \beta + (-\beta(H_\alpha) - n)\alpha$$

On another note, one can also use $\mathfrak{s}_\alpha$ to show that if α , β and $\alpha + \beta$ are roots, we have $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$. With some more work one may conclude:

Theorem 1.49. *Given a semisimple Lie algebra $\mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$, inducing a root space decomposition*

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^\vee} \mathfrak{g}_\alpha$$

then the non-zero roots form a root system for the inner product vector space $(\mathfrak{h}^\vee, B)$, where B is the Killing form:

$$B(X, Y) = \text{Tr}(ad(X), ad(Y))$$

The Weyl group is generated by the involutions W_α , which are reflections along the hyperplanes Ω_α . These hyperplanes are perpendicular to the lines spanned by the corresponding roots. The dual roots T_α induce an isomorphism:

$$\alpha \mapsto T_\alpha = \frac{2}{B(H_\alpha, H_\alpha)} H_\alpha$$

In particular, this identification allows us to write:

$$2\frac{B(\beta, \alpha)}{B(\alpha, \alpha)} = \beta(H_\alpha) \in \mathbb{Z}$$

Finally, we can transfer to this setting the Weyl chamber geometric picture, the positive roots and the simple roots. In it, the points in a Weyl chamber correspond precisely to regular elements in the Lie algebra.

The point is: from a finite dimensional semisimple Lie algebra, we can construct a root system out of it by looking at the adjoint action of one of its Cartan subalgebras. The root system allows us to classify the Lie algebra, but also provides a visual picture supported by the notions of Weyl group and Weyl chambers. This last point will be exploited in our future study of symmetric spaces.

1.2.3 $\mathfrak{sl}(2, \mathbb{C})$ and the principal TDS

We have already mentioned in several occasions the important role played by this actress in the whole representation theory of Lie algebras. It is thus time we face it fair and square. The crux of this development is perfectly transferable to the real case of $\mathfrak{sl}(2, \mathbb{R})$, in light of the following section on reality. The real case is especially relevant in our setting because the Fuchsian representations in $\mathrm{PSL}(2, \mathbb{R})$ constitute Teichmüller space, and the G -Fuchsian representations factor (by definition) through a representation of $\mathrm{PSL}(2, \mathbb{R})$ into G which we called *principal map*.

The simplest complex semisimple Lie algebra is the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ of 2×2 matrices of zero trace. In fact, any three dimensional complex semisimple Lie algebra $\mathfrak{s}$ is isomorphic to it. If we consider a regular element $H \in \mathfrak{s}$, then the eigenvalues of $\mathrm{ad}(H)$ are going to be a , 0 and $-a$ for some $a \in \mathbb{C} \setminus 0$. We can multiply H by a scalar so that a and $-a$ take the values 1 and -1 . This defines H up to conjugacy. This particular conjugate class of elements are called monosemisimple.

Definition 1.50. A regular element $H \in \mathfrak{s}$ is called **monosemisimple** if the eigenvalues of its adjoint action are 1 , 0 and -1 .

Now we can choose a non-zero eigenvector X for the eigenvalue 1 , which will be unique up to scalars. Now, an eigenvector Y for -1 is uniquely determined by $[X, Y] = H$. So we have the commutation relations for a basis $\{H, X, Y\}$ of $\mathfrak{s}$:

$$[H, X] = X, \quad [H, Y] = -Y, \quad [X, Y] = H$$

In the case of $\mathfrak{sl}(2, \mathbb{C})$, this basis can be realised by the matrices:

$$H = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, \quad X = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

The representation theory of $\mathfrak{sl}(2, \mathbb{C})$ [Kos59] tells us that for every $n \in \mathbb{N}$ there exists only one linear irreducible representation of $\mathfrak{s}$ having that dimension. To describe the representation π_n explicitly, we first define the half integer k via the formula $n := 2k + 1$. We then find a basis of the n -dimensional vector space V satisfying:

$$\pi_n(H)v_j = jv_j \quad \text{for } j \in \mathbb{Z} \cap [-k, k] = k, k-1, \dots, -k$$

This dictates the behavior of X and Y under the representation, which act as the ladder operators for angular momentum in quantum mechanics:

$$\pi_n(X)v_j = v_{j+1}, \quad \pi_n(Y)v_j = v_{j-1}, \quad \text{where } v_{k+1} = v_{-k-1} = 0$$

So v_j are eigenvectors of:

$$\pi_n(X)\pi_n(Y)v_j = \frac{1}{2}(k-j+1)(k+j)v_j \quad \& \quad \pi_n(Y)\pi_n(X)v_j = \frac{1}{2}(k-j)(k+j+1)v_j$$

The point is we obtain, up to scalar multiplication, a realisation of the basis elements as matrices:

$$\pi_n(H) = \begin{pmatrix} k & & & & 0 \\ & k-1 & & & \\ & & \ddots & & \\ & & & \ddots & \\ 0 & & & & -k \end{pmatrix}$$

$$\pi_n(X) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & \sqrt{1(2k)} & & & & & & 0 \\ & 0 & \sqrt{2(2k-1)} & & & & & \\ & & 0 & \sqrt{3(2k-2)} & & & & \\ & & & \ddots & \ddots & & & \\ & & & & \ddots & \ddots & & \\ & & & & & \ddots & \ddots & \\ & & & & & & 0 & \sqrt{(2k)1} \\ 0 & & & & & & & 0 \end{pmatrix}$$

$$\pi_n(Y) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & & & & & & & 0 \\ \sqrt{1(2k)} & 0 & & & & & & \\ & \sqrt{2(2k-1)} & 0 & & & & & \\ & & \sqrt{3(2k-2)} & 0 & & & & \\ & & & \ddots & \ddots & & & \\ & & & & \ddots & \ddots & & \\ & & & & & \ddots & \ddots & \\ & & & & & & 0 & \\ 0 & & & & & & \sqrt{(2k)1} & 0 \end{pmatrix}$$

Note then that the eigenvalues are half integers in even dimension and integers in odd dimensions. They occur with multiplicity 1 and the highest is k , whose (1-dimensional) eigenspace is the kernel of $\pi_n(X)$.

Any other finite dimensional representation π can be decomposed as a direct sum of irreducible ones. To compute such decomposition, one can look at the kernel of $\pi_n(X)$: if n_k denotes the number of components having dimension $2k+1$, then we only need to calculate these n_k to know the representation. From the previous paragraph, the dimension of such kernel is the sum of all n_k , and if the eigenvalues of $\pi(H)$ are k_j , then the dimensions of the irreducible components are $2k_j+1$. As a consequence, the dimension of the eigenspace V_j of $\pi(H)$ for the eigenvalue j has dimension $n_j + n_{j+1} + n_{j+2} + \dots$.

These irreducible representations are fundamental in a different way: their image is a principal three dimensional subalgebra in $\mathfrak{sl}(2, \mathbb{C})$.

Definition 1.51. A *principal three dimensional subalgebra* in a complex simple Lie algebra $\mathfrak{g}$ is a Lie subalgebra spanned by a monosemisimple regular element H and two principal nilpotent elements X and Y , satisfying the previous commutation relations:

$$[H, X] = X, \quad [H, Y] = -Y, \quad [X, Y] = H$$

Here, a *principal nilpotent* element is one that has minimal centraliser, i.e. its kernel under the adjoint action has dimension equal to the rank of $\mathfrak{g}$. This rank is the dimension of the Cartan subalgebra $\mathfrak{h}$, which we recall is the kernel of the regular elements like H .

The principal three dimensional subalgebras are all in the same conjugacy class. Moreover, in them, all regular elements are monosemisimple. They have several more properties which characterise them [Kos59] and make them *optimal* in a certain sense. The generators $\{H, X, Y\}$ are sometimes called principal S -triple.

What's left in this segment on algebra is key to understanding the principal maps that constitute our G -Fuchsian representations. First, we need to get acquainted with the main examples we will treat:

Definition 1.52. *The classical simple Lie algebras are:*

- The special linear group $\mathfrak{sl}(n, \mathbb{C})$: $n \times n$ matrices of trace 0 with complex coefficients.
- The symplectic group $\mathfrak{sp}(2n, \mathbb{C})$: $2n \times 2n$ matrices that preserve a **symplectic form**: this is a non-degenerate skew-symmetric bilinear form. For the sake of simplicity, it usually is chosen to be represented by the block matrix:

$$\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$$

We say a matrix A **preserves** a bilinear form Ω if

$$\Omega A + A^T \Omega = 0$$

- The indefinite orthogonal group $\mathfrak{so}(n, \mathbb{C})$: $n \times n$ skew-symmetric matrices.

One can check that all of these algebras are subalgebras of $\mathfrak{sl}(n, \mathbb{C})$ for a certain n , since they all have trace 0. To see this in the case of the symplectic Lie algebra, we can express a matrix M in block form and compute the defining equation:

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} + \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} = \begin{pmatrix} C - C^T & D + A^T \\ -A - D^T & -B + B^T \end{pmatrix}$$

and thus

$$\mathrm{tr}(M) = \mathrm{tr}(A) + \mathrm{tr}(D) = \mathrm{tr}(A) + \mathrm{tr}(D^T) = \mathrm{tr}(A + D^T) = \mathrm{tr}(0) = 0$$

This could also be shown at the level of Lie groups, with a bit more work.

The following determines the principal map for the largest of these algebras:

Proposition 1.53. *If π_n is the irreducible representation of $\mathfrak{sl}(2, \mathbb{C})$ of dimension n , then*

$$\{\pi_n(H), \pi_n(X), \pi_n(Y)\}$$

is a principal S -triple. Furthermore, these matrices all have trace zero. Therefore, the homomorphism

$$\pi_n : \mathfrak{sl}(2, \mathbb{C}) \longrightarrow \mathfrak{sl}(n, \mathbb{C})$$

is an embedding of $\mathfrak{sl}(2, \mathbb{C})$ as a principal three dimensional subalgebra of the Lie algebra $\mathfrak{sl}(n, \mathbb{C})$.

We need a way of lifting a map from Lie algebras to a map from Lie groups. We mentioned that this was done using exponentiation, which is inspired by the fact that Lie algebras are isomorphic to the tangent space at the identity of a Lie group, when the latter is seen as a manifold. Before that, we shift towards the real case where we actually obtain Hitchin components.

1.3 Reality

There is no such thing as a “real” representation of “reality”. Except perhaps in algebraic formulae.

Claude Simon

In his seminal paper [Hit92], Hitchin shows the existence of Hitchin components for the adjoint groups of the split real forms of complex simple Lie groups. To understand which these are, we need to move now to the realm of real semisimple Lie algebras. We could try to mimic the development done in the complex case, but we can actually deduce all we need from the results obtained there. The point is that for a real semisimple Lie algebra $\mathfrak{g}$, the complexification

$$\mathfrak{g}^{\mathbb{C}} := \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{g} \oplus i \cdot \mathfrak{g}$$

is a complex semisimple Lie algebra. Therefore, we can restrict ourselves to studying real forms of simple Lie algebras:

Definition 1.54. A **real form** $\mathfrak{g}_0$ of a complex semisimple Lie algebra $\mathfrak{g}$ is a real Lie algebra whose complexification is isomorphic to $\mathfrak{g}$.

We can of course extend the notion to Lie groups by our correspondence.

The most important real forms, which we can find for every complex semisimple Lie algebra, are the compact and real form (there are more: we can find 17 simple real Lie algebras associated to the 5 exceptional complex Lie algebras [FH91]). These two differ by their behavior on the root space decomposition of $\mathfrak{g}^{\mathbb{C}}$.

The simplest and most relevant case is that of $\mathfrak{sl}(2, \mathbb{C})$, as expected, where:

- $\mathfrak{sl}(2, \mathbb{R})$ is the split real form: 2×2 matrices of trace 0 with real coefficients.
- $\mathfrak{su}(2)$ is the compact real form: it is generated by the matrices:

$$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \& \quad \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

which are related up to complex scalar multiplication to the Pauli matrices, that represent spin and constitute the basis for the space of 2×2 Hermitian matrices.

Definition 1.55. A real form $\mathfrak{g}$ is **split** if it contains a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ such that its complexification $\mathfrak{h}^{\mathbb{C}} = \mathfrak{h} \otimes \mathbb{C}$ is a Cartan subalgebra of $\mathfrak{g}^{\mathbb{C}}$ and, moreover, $\mathfrak{h}$ acts by the adjoint representation on $\mathfrak{g}$ with real eigenvalues, giving a root space decomposition similar to the case of semisimple Lie algebras:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus \mathfrak{g}_{\alpha}$$

where the root spaces $\mathfrak{g}_{\alpha}$ are the intersection of the complex ones with $\mathfrak{g}$, are 1-dimensional (real dimension) and each pair associated to roots α and $-\alpha$ generates a three dimensional subalgebra isomorphic to $\mathfrak{sl}(2, \mathbb{R})$.

Definition 1.56. A real form $\mathfrak{k}$ is **compact** if its Cartan subalgebra $\mathfrak{h}$ acts by the adjoint decomposition with purely imaginary values, with an associated decomposition:

$$\mathfrak{k} = \mathfrak{h} \oplus \bigoplus \mathfrak{l}_{\alpha}$$

where the root spaces $\mathfrak{l}_{\alpha}$ are 2-dimensional, $\mathfrak{h}$ acts by rotations on them and each generates a subalgebra isomorphic to $\mathfrak{su}(2)$.

One can show that the restriction of the Killing form to the compact form is negative definite and the associated Lie group is compact. In fact, these could serve as equivalent definitions. We will see later that this has an interesting geometric significance.

Both the split and the compact form are unique for every complex semisimple Lie algebra.

We are particularly interested in the split real forms and its adjoint groups. We can use the classification provided by Dynkin diagrams in 1.31 once again and obtain the following complete picture Dynkin diagram - complex semisimple Lie algebra - split real form - adjoint group:

- A_n - $\mathfrak{sl}(n+1, \mathbb{C})$ - $\mathfrak{sl}(n+1, \mathbb{R})$ - $\mathbf{PSL}(n+1, \mathbb{R})$: $(n+1) \times (n+1)$ matrices of determinant one and real coefficients, quotiented by the center $\{\pm I\}$ (that's what P means: projective version of the group)
- B_n - $\mathfrak{so}(2n+1, \mathbb{C})$ - $\mathfrak{so}(n, n+1)$: $(2n+1) \times (2n+1)$ matrices that preserve an indefinite bilinear form of signature $(n, n+1)$:

$$I_{n,n+1}A + A^T I_{n,n+1} = 0, \quad \text{where} \quad I_{p,q} = \text{diag}(\underbrace{1, 1, \dots, 1}_{p\text{-times}}, \underbrace{-1, -1, \dots, -1}_{q\text{-times}})$$

- $\mathbf{SO}(n, n+1)$: $(2n+1) \times (2n+1)$ matrices of determinant one (S), that preserve an indefinite bilinear form of signature $(n, n+1)$:

$$I_{n,n+1}A^T I_{n,n+1} = A^{-1}$$

- C_n - $\mathfrak{sp}(2n, \mathbb{C})$ - $\mathfrak{sp}(2n, \mathbb{R})$ - $\mathbf{PSp}(2n, \mathbb{R})$: matrices preserving a symplectic form, modulo center $\{\pm I\}$.
- D_n - $\mathfrak{so}(2n, \mathbb{C})$ - $\mathfrak{so}(n, n)$ - $\mathbf{PSO}(n, n)$: matrices preserving a bilinear form of signature (n, n) , modulo center $\{\pm I\}$.

In terms of representations, if we have a real semisimple Lie group $\mathfrak{g}$ whose complexification is simple, its irreducible representations on complex vector spaces are exactly the restriction of the irreducible representations of the complexified $\mathfrak{g}^{\mathbb{C}}$. Furthermore, for the split real forms in the classical Lie algebras, every irreducible representation is real, meaning it comes from a representation on a real vector space V by extension of scalars (taking the complexification $V \otimes_{\mathbb{R}} \mathbb{C}$ of that vector space). This is equivalent to it having a conjugate linear involution as endomorphism.

On a related note, the uniqueness of the split and compact forms and their relation can be studied in terms of involutions. First:

$$\sigma : \mathfrak{g}^{\mathbb{C}} \longrightarrow \mathfrak{g}^{\mathbb{C}}, \quad x \otimes z \mapsto x \otimes \bar{z}$$

where $\mathfrak{g}$ is the split real form and $\mathfrak{g}^{\mathbb{C}}$ its complexification. This involution is conjugate linear and preserves Lie brackets. For such an involution, the fixed subalgebra is a real form of $\mathfrak{g}^{\mathbb{C}}$. In particular, σ fixes $\mathfrak{g}$.

For the compact form $\mathfrak{k}$, we have the involution ϕ taking an element in $\mathfrak{h}^{\mathbb{C}}$ to its opposite and X_{α} to Y_{α} (these were defined as $X_{\alpha} \in \mathfrak{g}_{\alpha}$ and $Y_{\alpha} \in \mathfrak{g}_{-\alpha}$ such that $[X_{\alpha}, Y_{\alpha}] = H_{\alpha}$, and $H_{\alpha} = 2T_{\alpha}/B(\alpha, \alpha)$, being T_{α} the dual to α). ϕ commutes with σ .

Finally, the work of Kostant [Kos59] tells us that the formalism of principal TDS has an extension to the real case, and to the adjoint group.

These ingredients already allow us to say everything we want about the principal maps in our G -Fuchsian representations:

Proposition 1.57. *The unique n -dimensional irreducible representation of $\mathfrak{sl}(2, \mathbb{R})$:*

$$\pi_n : \mathfrak{sl}(2, \mathbb{R}) \longrightarrow \mathfrak{sl}(n, \mathbb{R})$$

induces, by taking the exponential map and quotienting by the center, a principal map of the underlying adjoint groups:

$$PSL(2, \mathbb{R}) \longrightarrow PSL(n, \mathbb{R})$$

- *If the dimension $n = 2m$ is even, then we can check that $\pi_n(H)$, $\pi_n(X)$ and $\pi_n(Y)$ preserve a symplectic form, which in their base is written as:*

$$\begin{pmatrix} & & & & & 1 \\ & & & & -1 & \\ & & & \cdot & & \\ & & & \cdot & & \\ & & & \cdot & & \\ & & -1 & & & \\ & 1 & & & & \\ -1 & & & & & \end{pmatrix}$$

Thus, the $2m$ -dimensional irreducible representation factors through the symplectic algebra:

$$\pi_{2m} : \mathfrak{sl}(2, \mathbb{R}) \longrightarrow \mathfrak{sp}(2m, \mathbb{R}) \longrightarrow \mathfrak{sl}(2m, \mathbb{R})$$

and the same happens for the principal map:

$$PSL(2, \mathbb{R}) \longrightarrow PSp(2m, \mathbb{R}) \longrightarrow PSL(2m, \mathbb{R})$$

- *Similarly, if the dimension $n = 2m + 1$ is odd, they preserve an indefinite bilinear form of signature $(m, m + 1)$:*

$$\begin{pmatrix} & & & & & -1 \\ & & & & 1 & \\ & & & -1 & & \\ & & & \cdot & & \\ & & & \cdot & & \\ & & & \cdot & & \\ & & -1 & & & \\ & 1 & & & & \\ -1 & & & & & \end{pmatrix}$$

(check that this diagonalises to $I_{m, m+1}$).

Therefore we have the factoring of the representation and principal map:

$$\pi_{2m+1} : \mathfrak{sl}(2, \mathbb{R}) \longrightarrow \mathfrak{so}(m, m + 1) \longrightarrow \mathfrak{sl}(2m + 1, \mathbb{R})$$

$$PSL(2, \mathbb{R}) \longrightarrow SO(m, m + 1) \longrightarrow PSL(2m + 1, \mathbb{R})$$

The last arrow on the factorings of the representations highlight the fact that these are subalgebras of the special linear algebra, and their adjoint groups are consequently subgroups of the projective special linear group. Note that in every case the map lands in a principal three dimensional subalgebra, or a corresponding principal three dimensional subgroup.

We can arrive to a similar statement for the exceptional G_2 ([FH91] or [Jac22]):

Proposition 1.58. *The unique 7-dimensional irreducible representation of $\mathfrak{sl}(2, \mathbb{R})$ factors in the following way:*

$$\pi_7 : \mathfrak{sl}(2, \mathbb{R}) \longrightarrow \mathfrak{g}_2^{\mathbb{R}} \longrightarrow \mathfrak{so}(3, 4) \longrightarrow \mathfrak{sl}(7, \mathbb{R})$$

consequently establishing the principal map associated to G_2^{split} to be the same as for $PSL(7, \mathbb{R})$:

$$PSL(2, \mathbb{R}) \longrightarrow G_2^{split} \longrightarrow SO(3, 4) \longrightarrow PSL(7, \mathbb{R})$$

Unfortunately, the situation is not so nice for $PSO(n, n)$ ([DZ19]) or the other exceptional groups. The casuistry we have in the former is reduced to:

Proposition 1.59. *The principal map for $PSO(n, n)$ factors through that of $SO(n - 1, n)$:*

$$PSL(2, \mathbb{R}) \longrightarrow SO(n - 1, n) \longrightarrow PSO(n, n)$$

It is thus associated to the $(2n - 1)$ -dimensional irreducible representation of $\mathfrak{sl}(2, \mathbb{R})$ (and therefore does not act irreducibly on $\mathbb{R}^{2n}$).

2 Symmetric spaces

At this point, we should have become more acquainted with our objects of study - surface groups representations over semisimple Lie groups -, and the purpose of the following will be to see how these algebraic ideas interact with the geometry. To every semisimple Lie group G , we will be able to assign a nice, unique manifold: a symmetric space, obtained via quotienting by a maximal compact subgroup $K \subset G$. The manifold G/K is an instance of what is known as *symmetric space*, a most favourable class of manifolds from which we can describe many properties.

The symmetric spaces we obtain have nonpositive sectional curvature, and this is going to allow us to carry key constructions over them, specifically a boundary at infinity encircling them. In a certain way, we will also be able to read the algebraic structure of root system and Weyl chambers within the tangent space of this manifold, and in the construction at infinity (related to the notion of *Tits building*).

These symmetric spaces are relevant for us because we can associate to each representation in Hitchin component a map between them, and this is the crux of our parametrisation. We will proceed to carry the constructions related to symmetric spaces and nonpositive curvature in this section, while in the following ones we will tackle the association of (optimal) maps to representations.

2.1 Nonpositive curvature and convexity

At the heart of our perspective towards nonpositive sectional curvature is the notion of convexity. This notion is actually extendable outside the manifold realm, and so some of the work done here can be subsequently generalised. We start by trying to understand why this is the case.

We normally are presented with the concept of sectional curvature when studying the Gaussian curvature in surfaces. There, our intuition is somewhat extrinsic: as E.F. Beckenbach writes

In the neighborhood of a point where $K < 0$, the surface lies on both sides of the tangent plane, as in the depressions between one's knuckles; accordingly, a surface S with $K \leq 0$ at each of its points is called a saddle surface. In the neighborhood of a point where $K > 0$, the surface lies entirely on one side of the tangent plane, as on the tops of one's knuckles.

Although there is a viewpoint more fruitful for us when we look at the sectional curvature of a Riemannian manifold locally. We choose a point $p \in M$ and consider two geodesic rays starting from it with angle α . We want to compare with the Euclidean case, where geodesics are straight lines. When the curvature is nonnegative, the geodesics tend to come together, converge, compared to the flat case. When it is nonpositive, they tend to diverge faster than the flat case.

We can formalise it in the following way: consider $r > 0$ such that the exponential map $\exp_p : B(0, r) \subset T_p M \rightarrow B(p, r) \subset M$ is a diffeomorphism and $B(p, r)$ is convex. Then we can write the geodesic from p to any $x \in B(p, r)$ using the contracting map:

$$\rho_s : B(p, r) \rightarrow B(p, sr), \quad x \mapsto \exp_p(s \cdot \exp_p^{-1}(x)) \quad \text{for } s \in [0, 1]$$

The geodesic will then be given by $s \mapsto \rho_s(x)$. Now, if we have nonpositive curvature $K \leq 0$, the following holds:

$$d(\rho_s(x), \rho_s(y)) \leq s d(x, y) \quad \text{for all } s \in [0, 1], \quad x, y \in B(p, r)$$

and the opposite in the case of nonnegative curvature $K \geq 0$. This can be used as an alternative definition of nonpositive curvature that can be extended to singular spaces. Notice that in the flat case we obtain equality in the previous expression. It motivates the following:

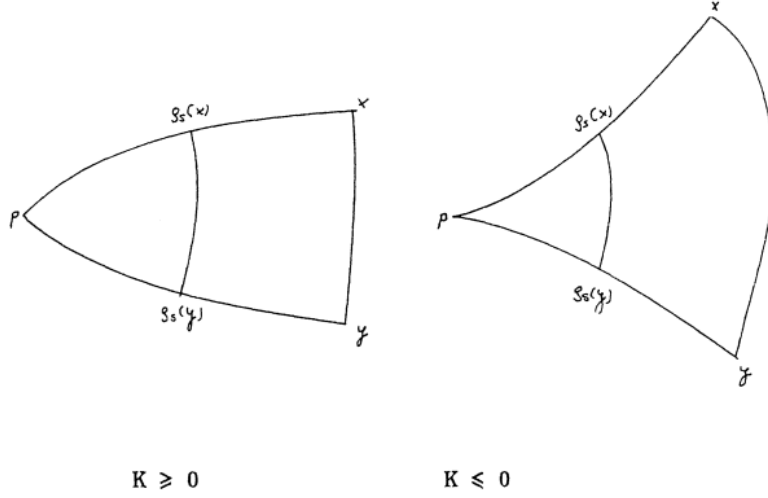


Figure 4: Local behavior of the sign of sectional curvature [BGS85]. Notice how the sum of interior angles is respectively $\leq 2\pi$ or $\geq 2\pi$.

Definition 2.1. We call **model space** for curvature K the unique simply connected surface with constant curvature K . It has infinite diameter if $K \leq 0$; otherwise, it is $\frac{\pi}{\sqrt{K}}$.

Now consider a geodesic metric space X with constant curvature K . Given a triangle $\Delta \subset X$, we call a **comparison triangle** a triangle in the model space for curvature K with the same side lengths.

We say that the geodesic metric space X is a **CAT(K) space** if for every triangle $\Delta \subset X$ (with perimeter less than two times the diameter of the model space), the distance between two points is less or equal than the distance between the corresponding points in the comparison triangle. If a metric space is not geodesic, we still consider it to have curvature $\leq K$ if every point has a geodesically convex CAT(K) neighborhood.

The point of the definition is that we only care about the local picture of geodesics converging or diverging faster compared to a reference space, and for this we do not need a differentiable structure. Nonetheless, this will give us several nice properties. The interesting case here is that of CAT(0) spaces, which will survive most of our following treatment of nonpositive curvature. In them, *distance functions are convex*.

Definition 2.2. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **convex** if for $a < b$ and $s \in (0, 1)$ we have:

$$f(a + s(b - a)) \leq f(a) + s(f(b) - f(a))$$

(**strictly convex** if the inequality is strict).

On a Riemannian manifold, $f : M \rightarrow \mathbb{R}$ is (strictly) convex if for every nontrivial geodesic c , the composition $f \circ c$ is (strictly) convex.

A subset $A \subset M$ of a Riemannian manifold is (geodesically) **convex** if for $a, b \in A$ there is a unique shortest geodesic joining them fully contained in A .

Closed or open convex subsets are submanifolds. If f is a convex function on a convex set, the sublevels $\{f \leq k\}$ or $\{f < k\}$ are convex. Local extrema of f are absolute minima. We will be looking in the future for existence and uniqueness of minima, so we start seeing the relevance of convexity (most of the formalism is build upon some basic slightly technical results, which can be found in [BGS85] and on which we will not dwell). As we mentioned before with more generality:

Proposition 2.3. *Given a convex subset $A \subset M$ of a manifold of nonpositive curvature, the distance function $d : A \times A \rightarrow \mathbb{R}$ is convex.*

Moreover, $K \leq 0$ is equivalent to every $p \in M$ having a convex neighborhood such that distance on it is convex.

Convexity of the distance function can be viewed infinitesimally via convexity of the norm of Jacobi fields.

The last statement means that, in a complete Riemannian manifold M , non-zero Jacobi fields along geodesics do not vanish in the endpoints. Thus there are no conjugate points. The exponential is thus defined on the entire tangent space and its differential is everywhere nonsingular. We can define a new metric on $T_p M$ making $\exp_p$ a local isometry, which is complete by Hopf-Rinow and thus we obtain:

Theorem 2.4 (Cartan-Hadamard). *If M is a complete Riemannian manifold of nonpositive curvature, then for every $p \in M$, $\exp_p : T_p M \rightarrow M$ is a covering map.*

Consequently, the universal cover is contractible, so the fundamental group $\pi_1(M)$ has great relevance from a topological perspective, i.e. M is an Eilenberg-MacLane space of type $K(\pi_1(M), 1)$. As a result, the homology and cohomology of M is given by that of $\pi_1(M)$ as an abstract group, so many geometric properties of M will feed on algebraic properties of $\pi_1(M)$. As an illustration of this:

Proposition 2.5. M is compact $\iff H_{\dim(M)}(\pi_1(M), \mathbb{Z}) \cong \mathbb{Z}$

We can simplify our treatment by looking only at simply connected manifolds and the action of groups on them, since every manifold can then be recovered by quotienting by the action of the fundamental group on a certain contractible space via deck transformations.

Definition 2.6. A **Hadamard manifold** is a connected simply connected complete Riemannian manifold of nonpositive curvature.

2.1.1 Boundary at infinity

When studying conformal models of the hyperbolic plane $\mathbb{H}^2$, one of the constructions carried out in the Poincaré disk model is the attachment of the unit sphere. This is a particular case of a more general construction which is usually named ideal boundary or boundary at infinity, also important in areas such as the study of singularities in general relativity.

In our case, this construction is going to contain all the relevant information we need from the semisimple Lie group we are studying and is going to be the foundation of the barycenter map parametrisation. It can be properly formalised and behaves favourably thanks to the convexity formalism, that is, in the nonpositive curvature regime.

We want to define now a compactification of Hadamard manifolds by attaching a boundary. There are two ways of tackling this:

- For a complete locally compact metric space X , we consider the space of continuous functions on X with the compact-open topology $C^0(X, \mathbb{R})$, which in this case is the topology of uniform convergence on compact sets. We can quotient by constant functions to obtain a space $\mathcal{C}$ with the quotient topology. We consider the embedding:

$$\iota : X \longrightarrow C^0(X, \mathbb{R}) \longrightarrow \mathcal{C}, \quad x \mapsto d_x := d(x, \cdot) \mapsto [d_x]$$

Once we are in the space $\mathcal{C}$, we can take the closure of $\iota(X) \subset \mathcal{C}$, and the boundary will be $X(\infty) := \overline{\iota(X)} \setminus \iota(X)$. This allows us to see points in the boundary (that we have attached by

compactifying) as functions well-defined up to a constant. We will use the notations $X(\infty)$ and $\partial_\infty X$ interchangeably.

Definition 2.7. A **horofunction** is a function $h \in C^0(X, \mathbb{R})$ that projects onto a point $z = [h] \in X(\infty)$. A **open horoball** centered at $z = [h]$ is a sublevel $\{h^{-1}(-\infty, k)\}$ and the corresponding **horosphere** is the level $\{h^{-1}(k)\}$.

If we have a point $x \in X$, the ball $B(x, r)$ is a sublevel of the distance function $\{d_x < r\}$. This analogy inspires us to find an explicit expression for horofunctions in the following way: the function

$$t \mapsto d(x, c(t)) - t$$

(for any unit speed geodesic ray $c : [0, \infty) \rightarrow X$) is bounded from below (by $-d(x, c(0))$):

$$d(x, c(t)) \geq d(c(0), c(t)) - d(x, c(0)) = t - d(x, c(0)) \quad \text{for } t \geq 0$$

and monotone decreasing:

$$d(x, c(t)) \leq d(x, c(s)) + d(c(s), c(t)) = d(x, c(s)) + t - s \quad \text{for } s < t$$

Thus we have a well defined limit:

Definition 2.8. A **Busemann function** is a function of the form:

$$b_\xi : X \rightarrow \mathbb{R}, \quad x \mapsto \lim_{t \rightarrow \infty} (d(x, c(t)) - t)$$

where $c : [0, \infty) \rightarrow X$ is the unit speed geodesic ray obtained by exponentiating $\xi := c'(0) \in T_{c(0)}^1 X$

We first note that this is in fact a horofunction: it is a limit of the (equivalence classes of) distance functions $[d_{c(t)}] = [d_{c(t)} - t]$ (this is one instance that explains why we quotiented by constant functions before; the next will be that being a Busemann function is invariant under addition of constants; finally, the second description of the boundary will consider as equivalent the geodesic rays which are at bounded distance, that is, in some sense differing by a constant is being equivalent in terms of *having the same endpoint at infinity*). Moreover, a Busemann function is not in $\iota(X)$ because it does not assume the minimum (the minimum would be *in the boundary*: we could show that $[d_{x_i}] \rightarrow [d_x] \iff x_i \rightarrow x$ and similarly for $[h] \in X(\infty)$ we find $x_i \in X$ such that $[d_{x_i}] \rightarrow [h]$)

Being a limit, some key properties of distance functions are preserved: in particular, we point out that d_x is convex, 1-Lipschitz and for points $y \in X$ and $x \notin B(y, r)$, there exist points $w, z \in \partial B(y, r)$ such that $|d_x(w) - d_x(z)| = 2r$. The same will be satisfied by any horofunction h , and moreover we have the following result [BGS85]:

Proposition 2.9. Given $h \in C^0(X, \mathbb{R})$, the following are equivalent:

- h is a Busemann function
- h is a horofunction
- h is convex, 1-Lipschitz and for every $x \in X$ and $r > 0$, there exist $x_1, x_2 \in \partial B(x, r)$ with $|h(x_1) - h(x_2)| = 2r$
- h is a convex C^1 function with $\|\text{grad}(h)\| = 1$ (Busemann functions are actually C^2)

We can deduce from this that if h is Busemann, so is $h + k$ for any constant $k \in \mathbb{R}$. Also, $[h_1] = [h_2] \iff \text{grad}(h_1)_x = \text{grad}(h_2)_x$ for an arbitrary $x \in X$. We will see that this is equivalent to saying that if h_1 and h_2 are defined using the geodesic rays c and $\tilde{c}$, then their *endpoints* have to coincide: $[h_1] = [h_2] \iff c(\infty) = \tilde{c}(\infty)$.

This description of Busemann functions, that were explicitly defined using geodesic rays or their corresponding unit tangent vector at 0, suggest a more geometric approach to the definition of boundary at infinity, apt only in the case of Hadamard manifolds, that turns out to be equivalent to this one.

- Now we are going to look at equivalence classes of geodesic rays, saying that they *converge to/have the same endpoint at infinity* whenever their distance is (uniformly) bounded:

Definition 2.10. *Given two unit speed geodesic rays $c_i : [0, \infty) \rightarrow X$, we say they are **asymptotic** if there is a constant $k \in \mathbb{R}$ such that $d(c_1(t), c_2(t)) \leq k$ for all $t \geq 0$. The equivalence class of a geodesic ray is denoted by $c(\infty)$.*

We now say that the points at infinity in $X(\infty)$ are precisely the equivalence classes of asymptotic geodesics. We can give it a topology as follows. Given $z \in X(\infty)$ and $x \in X$, we have a unique (unit speed) geodesic ray such that $c(0) = x$ and $c(\infty) = z$. Since we are in a Hadamard manifold, we can define:

Definition 2.11. *The **angle** at $x \in X$ between two points $y, z \in \overline{X} := X \cup X(\infty)$ is:*

$$\angle_x(y, z) = \angle(c'_1(0), c'_2(0))$$

where c_1 and c_2 are, respectively, geodesics from x to y and from x to z , and the angle is measured within the tangent space at x .

For $\epsilon > 0$ and $x \in X$, $z \in X(\infty)$, consider the *cone* $\{y \in \overline{X} \mid y \neq x \text{ and } \angle_x(z, y) < \epsilon\}$. Then the *cone topology* in $\overline{X}$ given by the open sets of X and these cones induces a topology in $X(\infty)$ which is named *sphere topology* due to its resemblance to the usual sphere topology associated to the angle as metric.

Recall again that every horofunction was a Busemann function and they were defined by the unit tangent vector at 0 of the corresponding geodesic ray. The point is that, for any $x \in X$, we obtain a bijection $T_x^1 X \cong X(\infty)$ by associating $\xi \mapsto b_\xi$ or equivalently $\xi = c'(0) \mapsto c(\infty)$. With the sphere topology on the boundary and the angle topology on the unit tangent space, this is a homeomorphism.

Now we can upgrade the boundary by defining a metric on it, which in the particular case at hand (symmetric spaces associated to semisimple Lie groups) will reflect the algebraic properties of the group.

2.1.2 Tits metric

We have again two ways of coming up with a relevant metric on $X(\infty)$.

For starters, we can recall the angle at $x \in X$ between two points $w, z \in X(\infty)$ as we defined above, and define a global angle as:

$$\angle(z, w) = \sup_{x \in X} \angle_x(z, w)$$

This is a metric on $X(\infty)$, but there is a different approach. Consider unit speed geodesic rays c_1 and c_2 from x to w and from x to z , respectively. The function:

$$t \mapsto \frac{d(c_1(t), c_2(t))}{t}$$

is monotone increasing and bounded (by convexity and the triangle inequality), so we have the limit:

$$l(z, w) := \lim_{t \rightarrow \infty} \frac{d(c_1(t), c_2(t))}{t}$$

which is also a metric (independent of the choice of $x \in X$).

Actually, we can show:

Proposition 2.12. *If $c : [0, \infty) \rightarrow X$ is a geodesic ray with $c(\infty) = z$, then:*

$$\angle(z, w) = \lim_{t \rightarrow \infty} \angle_{c(t)}(z, w)$$

On a different note, if we find a point $x \in X$ such that $\angle(z, w) = \angle_x(z, w) < \pi$, then the geodesics c_1 and c_2 bound a totally geodesic Euclidean sector.

Example 2.13. *This is what happens when $X = \mathbb{R}^n$ (except for antipodal points, at angle π): there, the metric space $(X(\infty), \angle)$ is the sphere $\mathbb{S}^{n-1}$ with the usual angle metric. On the other hand, $(X(\infty), l)$ is also the unit sphere but with the metric $l(z, w) = \|z - w\|$ (distance between two points is given by the straight line joining them). These two metrics then relate by the expression:*

$$l(z, w) = 2 \sin \frac{\angle(z, w)}{2}$$

so they are equivalent and correspond to the usual sphere.

This interesting relation might make us question if we can in general relate both metrics. The answer is positive:

Proposition 2.14. *In an arbitrary Hadamard manifold X , for $z, w \in X(\infty)$ we have:*

$$l(z, w) = 2 \sin \frac{\angle(z, w)}{2}$$

Moreover, the metric spaces $(X(\infty), \angle)$ and $(X(\infty), l)$ are complete.

The second statement can be shown with the angle metric, using that being Cauchy for $\angle$ implies being Cauchy for $\angle_x$ for every $x \in X$.

It is important to note that the topology induced by this metric(s) is not necessarily equivalent to the *cone topology* we mentioned above. The extreme cases are $\mathbb{R}^n$ (that we saw above) and $\mathbb{H}^n$:

Example 2.15. *In the hyperbolic spaces $\mathbb{H}^n$, one can show that any two points z, w in the boundary can be joined by a geodesic c inside the space. Thus, the metric reads $\angle(z, w) \geq \angle_{c(t)}(z, w) = \pi$ (we can show too $\angle \leq \pi$ always, so equality is reached). Therefore, all the points in the boundary are isolated, i.e. the metric space $(H^n(\infty), \angle)$ (same for l) is the set $\mathbb{S}^{n-1}$ with the discrete topology.*

In several applications, it is interesting to consider the length spaces associated to these metric spaces, since in general they contain more geometric information. Nonetheless, in some cases the metric described above is already named *Tits metric*, and for our purposes we could ignore the following and still be able to develop the specific theory we are concerned with. For completeness, though, I will choose to briefly go for the length space formalism.

On a (nice) metric space it is natural a priori to expect that we might be able to get from one point to another following a curve whose length is as close as we want to the distance between both points given by the metric. This does not necessarily mean that we have a geodesic joining the points: for example, picture a space with a *hole*, e.g. $\mathbb{R}^n \setminus 0$ and try to join two points belonging to a line that crosses 0 by a minimal curve. The correct notion we want to use is that of length space, which is weaker than that of geodesic metric space.

Definition 2.16. Consider a metric space (X, d) and define the length of a curve $\gamma : [0, 1] \rightarrow X$ by $L(\gamma) = \sup \sum_{i=0}^{k-1} d(\gamma(t_i), \gamma(t_{i+1}))$, where the supremum is taken over partitions of $[0, 1]$. We can define the **intrinsic metric** $d_I(x, y) := \inf L(\gamma)$ on X , where the infimum is over all continuous curves γ that join x and y , or ∞ if there is no such curve. The resulting space (X, d_I) is called **length space** associated to (X, d) .

Note that $d_I \geq d$ and that $(d_I)_I = d_I$ (if $d = d_I$ we say the starting metric space was already a length space).

Definition 2.17. The **Tits metric** on $X(\infty)$ is the intrinsic metric Td associated to the metric space $(X(\infty), \angle)$. In particular, $Td = \angle_I$ and we can show $Td = l_I$.

The name comes from its relation to the Tits building in the case of symmetric spaces of non-compact type (precisely the ones we are interested in).

The Tits metric does not change our picture too much, the difference is thus subtle: it induces the same topology on $X(\infty)$ as $\angle$, so the discrepancy is purely geometric. We will call this topology *Tits topology* to distinguish it from the *cone topology* (sphere topology) that we first mentioned. This is because $\angle(z, w) = \min(Td(z, w), \pi)$, so it only affects points at infinity that can be joined by a geodesic in X . However, we can have the limiting behavior $Td(z, w) = \pi$ and the points not joined by a geodesic in X .

What can we say about the relation between the sphere topology and the Tits topology?

On the extreme cases $\mathbb{R}^n$ and $\mathbb{H}^n$, we have exactly the same picture as before (although now the distance between points in the boundary of hyperbolic space is ∞). So let us see what happens when we mix both regimes:

Example 2.18. Consider the Riemannian manifold $\mathbb{H}^2 \times \mathbb{R}$, where the boundary at infinity is a sphere: the poles are the endpoints of a geodesic $\{p\} \times \mathbb{R}$ and the equator is the boundary of a copy of hyperbolic plane $\mathbb{H}^2 \times \{q\}$.

Thus, two points z, w on the equator can be joined by a geodesic c , and $\{c\} \times \mathbb{R}$ is a flat plane, so z, w are antipodal in this plane, their distance is π and the minimal curve joining them goes through one pole. The metric of an hemisphere is then that of a disc with the French railway metric, i.e. the faster way to get from one point to another is always crossing through the center (corresponding to the pole). Thus, at the pole the distance balls are not compact, i.e. the Tits metric is not necessarily locally compact, as opposed to the sphere metric.

But we have some similarities:

Definition 2.19. We say a function $f : X \rightarrow \overline{\mathbb{R}} = [-\infty, \infty]$ is **lower semicontinuous** at $x_0 \in X$ if:

$$f(x_0) \leq \liminf_{x \rightarrow x_0} (f(x))$$

or equivalently, for every $y < f(x_0)$, there exists a neighborhood V of x_0 such that $y < f(x)$ for all $x \in V$. In other words, $f^{-1}(y, \infty)$ is open for every $y \in \mathbb{R}$.

Lower semicontinuity is relevant because it is a sufficient condition for guaranteeing the existence of minima in compact spaces.

Proposition 2.20. *Convergence on the Tits metric implies convergence on the sphere metric. Moreover, the function $Td_w : X(\infty) \rightarrow \overline{\mathbb{R}}$ is lower semicontinuous in the sphere topology for all $w \in X(\infty)$.*

Of course, much more can be said, and there are several technical results [BGS85] that show what global geometric properties are revealed by the Tits metric. These are mostly stated in the form of comparison results: when the distance is $< \infty$, points can be joined by minimal (i.e. realising their distance) geodesics at infinity (so when points cannot be joined by a geodesic in X , they can be joined by a geodesic at infinity); when the distance is $< \pi$, the minimal geodesic at infinity is unique; if the distance is π , then the geodesic in X joining the points bounds a flat half plane, and so on. The key is that the metric space description does not distinguish between the different subbehaviors associated to points z, w joined by a geodesic in X , grouping all of them as $\angle = \pi$, while Tits metric is more fine and can give us valuable insights.

There is a degenerate case, which is that of *visibility manifolds*, where every pair of points at infinity can be joined by a geodesic in X (so they can be *seen* by following the geodesic; although perhaps this might be motivated rather by general relativity and light following geodesics and therefore carrying through them the visual information that an outside observer could perceive).

One can moreover show that for any Hadamard manifold, $X(\infty)$ has curvature ≤ 1 in the sense of CAT(1) spaces, i.e. in terms of comparison triangles.

2.2 Symmetric spaces

To each semisimple real Lie group we study, we will be able to associate a particular Riemannian manifold that reflects its algebraic structure: this will be a *symmetric space of non-compact type*.

When looking at manifolds, we might be interested in which properties are satisfied by its isometry group and what structure it can be given. A *big enough* isometry group would perhaps give us information about the manifold itself. In particular, this consideration allows us to outline two favourable classes of manifolds: *homogeneous spaces* and their subclass *symmetric spaces*. We are interested in the latter.

Definition 2.21. *A **symmetric space** is a simply connected complete Riemannian manifold X whose isometry group $\text{Iso}(X)$ contains an inversion about every point. An isometry $f : X \rightarrow X$ is an **inversion** about $p \in X$ if it fixes the point ($f(p) = p$) and its differential at p acts as minus the identity, reversing the geodesics through p ($df_p = -Id_{T_p M}$).*

The connected component of the identity of the isometry group $G := \text{Iso}_0(X)$, with the compact-open topology, admits a Lie group structure, while the isotropy group of a point $p \in X$ is a compact subgroup $K := \text{Stab}_G(p)$. G acts transitively on X , so we can express the symmetric space as the quotient $X = G/K$ (in the spirit of homogeneous spaces).

Recall the involutions we talked about in the section about real semisimple Lie groups. In this case, the inversion f about the point p allows us to define the automorphism:

$$\Sigma_p : G \rightarrow G, \quad g \mapsto fgf$$

By the usual identification of the Lie algebra as the infinitesimal version of the Lie group (the tangent space at the identity), we obtain a Lie algebra isomorphism $\sigma_p : \mathfrak{g} \rightarrow \mathfrak{g}$ defined as $\sigma_p = (d\Sigma_p)_I$. It is an involution, so it is diagonalisable with eigenvalues ± 1 (since its square is the identity), and decomposes the Lie algebra in two eigenspaces:

$$\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{k}$$

where $\mathfrak{p}$ corresponds to eigenvalue -1 and to the tangent space at a point, and $\mathfrak{k}$ corresponds to eigenvalue $+1$ and to the Lie algebra of the compact subgroup K . In the noncompact case that we will be focusing on in a moment, this Lie algebra involution is of a special type.

Definition 2.22. A *Cartan involution* σ on a real semisimple Lie algebra $\mathfrak{g}$ with Killing form B is an involution for which $B(X, \sigma Y)$ is a positive definite bilinear form.

Any real semisimple Lie algebra has a Cartan involution, and they are related to our discussion about involutions in the Lie theory section.

Finally, we have the *Cartan relations*:

$$[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k}, \quad [\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p} \quad \& \quad [\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$$

A local characterisation of a symmetric space is that its curvature tensor is invariant under parallel transport (idea behind the so called *locally symmetric spaces*, which appear basically as quotients of symmetric spaces by discrete groups of isometries without fixed points), so this is another perspective through which this class of manifolds is optimal. We can also use the holonomy to classify them:

Definition 2.23. Given a vector bundle over a smooth manifold $E \rightarrow X$ with connection ∇ , we define the *holonomy group* of the connection at $x \in X$ to be:

$$\text{Hol}_x(\nabla) = \{P_\gamma \mid \gamma \text{ is a loop based at } x\}$$

where $P_\gamma : E_x \rightarrow E_x$ is the (linear, invertible) parallel transport map around the loop γ .

If X is connected, the choice of point $x \in X$ only affects the holonomy up to conjugation by P_γ , where γ is a path from the original point to the new one (similar as what happens when defining the fundamental group).

In case X is a Riemannian manifold, one naturally chooses the tangent bundle and the Levi-Civita connection and talks about the holonomy group of X .

Holonomy is the tool that allows us to know whether a Riemannian manifold can be expressed as a product of Riemannian manifolds. It also informs about other useful topological and geometrical properties.

Theorem 2.24 (De Rham decomposition). *Every simply connected complete Riemannian manifold X can be decomposed (uniquely up to permutation) as a product of simply connected complete Riemannian manifolds X_i whose holonomy representation is irreducible, times an Euclidean factor E :*

$$X \cong E \times X_1 \times X_2 \times \cdots \times X_n$$

Moreover, the holonomy representation of the original is the product of the holonomy representations of the factors (holonomy is trivial on Euclidean factors).

If we apply this result to symmetric spaces we get the following classification:

- Non-compact type: $X = G/K$ has no Euclidean factors and nonpositive sectional curvature; equivalently, G is semisimple and σ_p is a Cartan involution (X is not compact).
- Compact type: X has no Euclidean factors and nonnegative sectional curvature; equivalently, G is semisimple and compact.
- X is a direct product of irreducible symmetric spaces of compact or non-compact type, and with an Euclidean factor.

This means we have a decomposition of symmetric spaces according to curvature. We will now focus ourselves on the non-compact type, since it is the case for G real semisimple and split, for which we have Hitchin components.

From the algebraic point of view, our construction is the following. Given our adjoint group G of a split real form of a complex simple Lie group, we consider a maximal compact subgroup $K \subset G$. Now, the quotient $X = G/K$ can be given a symmetric space structure, where G will correspond to the connected component of the identity of the isometry group of X and K to the isotropy group of a point $x \in X$ (the choice of point only affects the group K up to conjugation).

Since $\mathfrak{g}$ is semisimple, the Killing form is nondegenerate and we can consider $\mathfrak{p}$ to be the orthogonal complement to the Lie algebra $\mathfrak{k}$ of K , i.e. $\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{k}$. We have a Cartan involution and the Cartan relations, so we obtain a positive definite quadratic form, which is moreover invariant under the adjoint representation of K . Consequently, we can define a G -invariant metric such that $G \rightarrow G/K$ is a Riemannian submersion, for which the curvature satisfies:

$$K(Y, Z) = -\|[Y, Z]\|^2 \quad \text{for } Y, Z \in \mathfrak{p} \quad \text{such that } \|Y\| = \|Z\| = 1$$

We will see in a second how the existence of flat submanifolds can hinder our application of the barycenter method, so our parametrisation depends on whether we can control where these lie. Thus, this formula is very revealing, since flats are going to be associated to commutativity in the Lie algebra picture, so to abelian subalgebras: the Cartan subalgebras, that already proved quite important.

Definition 2.25. *A **flat** in X is a totally geodesic complete Euclidean (flat) submanifold of maximal dimension. The **rank** of X is the dimension of any flat.*

If the rank is 1, then the flats are just geodesics, so the manifold has curvature strictly negative. The higher rank cases are, of course, richer. By the previous remark, we can note that the tangent space to a flat has to be a maximal abelian subalgebra, and thus the flat corresponds to the exponential of the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{p} \subset \mathfrak{g}$. Now, since all Cartan subalgebras are conjugate, for every pair of flats F and F' containing x we have an isometry $g \in K$ fixing x such that $gF = F'$.

Now, we can put into use the algebraic machinery: recall the Cartan subalgebra induced a root space decomposition, and the roots formed a root system within $\mathfrak{h}^\vee$. Moreover, since the Killing form is nondegenerate by semisimplicity, we can translate the root system picture from $\mathfrak{h}^\vee$ to $\mathfrak{h}$. This means that we have a Weyl chamber decomposition in the tangent space to each flat.

The interior of the Weyl chamber corresponded to regular elements $H \in \mathfrak{h}$, so we say these tangent vectors spawn regular geodesics. On the contrary, the walls of the Weyl chambers corresponded to elements $H \in \mathfrak{h}$ that commuted with elements outside $\mathfrak{h}$: we call them singular, they spawn singular geodesics (belonging to more than one flat) and the flats they belong to are generated by the elements with which they commute. We know that the singular elements corresponded to finitely many hyperplanes within the Weyl chamber picture, associated to the reflections that produce the Weyl group. In particular, a singular subspace of codimension n corresponds to the intersection of $n + 1$ flats containing x .

Notice how we defined the rank of $\mathfrak{g}$ to be the dimension of $\mathfrak{h}$, and this is the dimension of the tangent space to a flat F , so its equal to the rank of $X = G/K$. Finally, it is key to note that every element in the tangent space $\mathfrak{p}$ is semisimple (diagonalisable) and thus corresponds to a Cartan subalgebra (elements that are also diagonal after diagonalising the original), so the union of all Weyl chambers and walls is the whole tangent space at a point.

We will now transport this Weyl chamber structure into the boundary at infinity and read off the geometric and algebraic information of the resulting construction. The examples of symmetric spaces with nonpositive curvature will come later, but we already mention that they all can be

seeing effectively as totally geodesic submanifolds of the irreducible $\mathrm{SL}(n, \mathbb{R})/\mathrm{SO}(n)$. This comes from the fact that simply connected symmetric spaces of nonpositive curvature are totally geodesic submanifolds of $\mathrm{GL}(n, \mathbb{R})/\mathrm{O}(n) \cong \mathrm{SL}(n, \mathbb{R})/\mathrm{SO}(n) \times \mathbb{R}$ (since $\mathrm{SL}(n, \mathbb{R}) \times \mathbb{R}^\times \cong \mathrm{GL}(n, \mathbb{R})$ and the connected component of the identity of $\mathrm{O}(n)$ is $\mathrm{SO}(n)$; the isometry is given by $(t, x) \mapsto e^{t/\sqrt{n}}x$). Again, the special linear group is the starting point and the key to the theory, evidencing the connection between such symmetric spaces and the classification of semisimple Lie groups.

2.3 Metric, building and flags at infinity

The Weyl chamber decomposition associated to the different flats in the tangent space of a point respects the incidence relations that describe the boundary at infinity (from the point of view of asymptotic geodesic rays). This is because we defined flats to be totally geodesic submanifolds, so their geodesic behavior is favourable. In particular, we can use the homeomorphism of the boundary at infinity with the unit tangent space at a point to intersect our Weyl chamber decomposition of tangent space with the unit tangent ball, obtaining a Weyl chamber decomposition at infinity. Its properties are going to be those of a *Tits building*, and they will connect to the Tits metric we defined above.

If we find ourselves in a symmetric space K of rank n , the Weyl chambers have dimension n , since they are open dense subsets of the tangent space at a flat, i.e. a Cartan subalgebra. When intersected with the unit sphere $\mathbb{S}^{n-1}$, they will become $(n-1)$ -dimensional Weyl chambers at infinity. Similarly, the walls lose 1 dimension when intersected at infinity.

The Weyl group acts simply transitively on Weyl chambers, and therefore the order of the group is the cardinality of the Weyl chambers in the flat. A choice of positive roots $\{\alpha_1, \dots, \alpha_n\}$ correspond to a choice of fundamental domain for the action of the Weyl group, i.e. to a special choice of Weyl chamber. This is the subset:

$$\mathfrak{a}^+ := \{H \in \mathfrak{h} \mid \alpha_i(H) > 0 \quad \forall i \in \{1, \dots, n\}\}$$

The hyperplanes that bound said chamber are then given by:

$$\{H \in \mathfrak{h} \mid \alpha_i(H) = 0\}$$

for each i . A generic wall can be written as:

$$\{H \in \mathfrak{h} \mid \alpha_{i_m}(H) > 0 \quad \text{if } i_m \in \{1, \dots, r\} \quad \& \quad \alpha_{i_m}(H) = 0 \quad \text{if } i_m \in \{r+1, \dots, n\}\}$$

so we get exactly $\binom{n}{r}$ walls of dimension r .

We are a bit careless with the notation and talk about Weyl chamber decomposition to refer both to the Weyl chamber $\mathfrak{a} \subset \mathfrak{h}$ described above, corresponding to the tangent space to a flat F containing the point $x \in X$, as well as to the set $\exp(\mathfrak{a})x \in F$ (similar for walls).

It will be useful to *parametrise* the set of Weyl chambers in some manner, mainly to construct bijections to it and confer it a structure. For this, we use that G acts transitively on the set CX of all Weyl chambers in all flats, i.e. any Weyl chamber in the decomposition is obtained by acting on $\mathfrak{a}^+$ via an element in G . The stabiliser of $\mathfrak{a}^+$ under this action is the Borel subgroup B , so we obtain $CX \cong G/B$ [BGS85].

Definition 2.26. A **Borel subgroup** of an algebraic group G is a maximal Zariski closed, connected solvable algebraic subgroup.

Subgroups $B \subset P \subset G$ between the Borel subgroup and G are called **parabolic** and are characterised by G/P being an algebraic variety which is complete (generalisation of compact space).

In the adjoint groups of classical Lie algebras, the Borel subgroup corresponds to upper triangular matrices.

Definition 2.27. A **flag** F in a vector space V is a strictly increasing nested sequence of vector subspaces:

$$\{0\} = F^0 \subset F^1 \subset \dots \subset F^k = V$$

where $k \leq n = \dim(V)$. In particular, if $k = n$ we say it is a **full flag**, so $\dim(F^i) = i$. Otherwise, we talk about **partial flags**.

When G is semisimple, the quotient G/P is also called a (generalised) flag variety, since it can be seen as a space of (perhaps partial) flags (there is a correspondence between parabolic subgroups and homogeneous spaces whose points are flags in a certain finite dimensional vector space).

In particular, we see that the space of Weyl chambers CX is a generalised flag variety, so admits the structure of a homogeneous space.

We can look at this generalised flag variety as a subset of the boundary at infinity in the following way: consider the Weyl chamber decomposition at infinity. Associate to each Weyl chamber its barycenter: if the Weyl chamber $\mathfrak{a}$ corresponds to a choice of positive roots $\alpha_i \in \mathfrak{h}^\vee$, whose duals via the Killing form are the $H_i \in \mathfrak{h}$, the barycenter is $b := \sum_i H_i \in \mathfrak{a}$ (every Weyl chamber can be made to correspond to a choice of positive roots, so the procedure is general).

If necessary, we can rescale the barycenter so that it has norm 1 and therefore belongs to the unit tangent space. This allows us to see it as a point at infinity. Since the space of all Weyl chambers was isomorphic to G/B , the same occurs to the space of all barycenters, which is sometimes named *Furstenberg boundary* and denoted $\partial_F X = G/B \subset X(\infty)$ (the Furstenberg boundary is a more general construction that appears in seemingly unrelated contexts, but we only care about this specific instance for symmetric spaces of non-compact type). Notice that the space of barycenters is the orbit of any one of them under G : $G \cdot b \cong G/B$ (in general, Furstenberg boundary could be identified with the orbit of any regular point in the boundary, i.e. the endpoint of a geodesic tangent to a regular vector).

The boundary at infinity can also be seen as a space of flags for the symmetric space associated to the adjoint groups of the split real forms of classical simple Lie algebras, but we delay its description to the next section.

We would like to show that distinct Weyl chambers at infinity do not intersect, and they cover the whole boundary at infinity, since the Weyl chamber decomposition covers the whole tangent space. We say that they define a *tesselation* of $X(\infty)$, where Weyl chambers are like disjoint pieces (simplices) that assembled together via the walls reconstruct the boundary at infinity. This already hints at the fact that it will be enough to know this incidence relations between chambers to build up $X(\infty)$, and this is the algebraic idea of a *building*.

We need the following tool:

Definition 2.28. The **Iwasawa decomposition** (sometimes called **KAN decomposition**) of a semisimple Lie group G is the product $G = KAN$, where K is a maximal compact subgroup associated to the Lie algebra $\mathfrak{k}$ of eigenvalue $+1$ via the Cartan decomposition, A is a flat obtained by exponentiating a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{p} \subset \mathfrak{g}$, and N is the nilpotent subgroup corresponding to exponentiation of the sum of the root spaces for a choice of positive roots.

This product decomposition consists explicitly of an analytic diffeomorphism:

$$K \times A \times N \longrightarrow G, \quad (k, a, n) \mapsto kan$$

As a consequence of this decomposition we obtain:

Proposition 2.29. *If $\mathfrak{a}$ and $\mathfrak{b}$ are Weyl chambers in $X(\infty)$ and $\mathfrak{a} \cap \mathfrak{b} \neq \emptyset$, then $\mathfrak{a} = \mathfrak{b}$. We have the same behavior at the level of walls.*

Proof. • The group AN acts transitively on X .

- For a geodesic ray $c(t) = e^{tH}x$ (for $H \in \mathfrak{a}^+$ a Weyl chamber corresponding to a choice of positive roots) and $n = e^Y \in N$:

$$\begin{aligned} d(c(t), nc(t)) &= d(e^{tH}x, e^Y e^{tH}x) = d(x, e^{-tH} e^Y e^{tH}x) = \\ &= d(x, \text{Ad}_{e^{-tH}}(Y)) = d(x, e^{-\text{ad}_{tH}}(Y)x) \end{aligned}$$

where the exponential means the usual power series decomposition and we use the relation between the adjoint representations at the level of groups or algebras.

Since $n \in N$, $Y = \sum_i Y_i$ where the sum is over the root spaces corresponding to a choice of positive roots $\{\alpha_i\}$. Thus:

$$\begin{aligned} (\text{ad}_{tH})^n(Y) &= (\text{ad}_{tH})^{n-1}([tH, Y]) = \sum_i (\text{ad}_{tH})^{n-1}(\alpha_i(tH)Y_i) = \sum_i \alpha_i(tH)^n Y_i \\ \implies e^{-\text{ad}_{tH}}(Y) &= \sum_i e^{-t\alpha_i(H)} Y_i \end{aligned}$$

with $\alpha_i(H) > 0$ for every i (since α_i positive root and $H \in \mathfrak{a}^+$).

- Furthermore, if we take $a \in A$ it is going to commute with $c(t)$, since it belongs to the exponential of the Cartan subalgebra (it is diagonal). So:

$$d(anc(t), c(t)) = d(ne^{tH}x, e^{tH}a^{-1}x) \xrightarrow{t \rightarrow \infty} d(x, a^{-1}x) = d(ax, x)$$

by the previous item, which shows that $t \mapsto d(anc(t), c(t))$ is decreasing from $d(ancx, x)$ to $d(ax, x)$. In particular, anc and c are asymptotic.

- Choose $c(\infty) \in \mathfrak{a} \cap \mathfrak{b}$. Without loss of generality, $\mathfrak{a} = (A^+x)(\infty)$ belongs to a flat F based at x (meaning we exponentiate a Cartan subalgebra from x to obtain the flat), and $\mathfrak{b} \subset F'$ based at x' , with c' such that $c'(0) = x'$ and $c'(\infty) = c(\infty)$. The Iwasawa decomposition says that AN acts transitively on X , so $x' = anx$ for $an \in AN$. But an leaves $\mathfrak{a}$ fixed by the calculation above, so $c' = anc$.

Both c' and c are then regular geodesics, so an maps the interior of one Weyl chamber to the other (actually, one flat to the other): $\mathfrak{b} = ana = \mathfrak{a}$. This is because since $an \in AN \in G$ is an isometry, it preserves the curvature, so if it maps a regular geodesic to a regular geodesic, it has to map a neighborhood of that regular geodesic to a neighborhood of that regular geodesic (and these have to be neighborhoods within the same flat, since the isotropy group of a point was K and our isometry is in AN according to the KAN decomposition), because the only intersections with other flats happen at singular geodesics.

□

We can look at this tessellation from a purely combinatorial point of view in four ways:

- We can consider chambers and walls at infinity as a partially ordered set: $a \leq b$ means a is a wall of b . We add the 0 element as wall of every other element.

- This is equivalent to considering the chambers and walls of all flats through a point $x \in X$, and identifying x to the element 0. This works because every chamber or wall at infinity has a unique representative in a flat through x .
- Another option is to consider chambers and walls through all points in X , and say that two of them are equivalent if they are at finite Hausdorff distance (this is a generalisation to being asymptotic). Then, $a \leq b$ if a is contained in a neighborhood of b .
- Finally, there is a perspective in terms of parabolic groups. Parabolic subgroups are precisely the isotropy groups of a point at infinity. We will denote by G_z the subgroup of G fixing $z \in X(\infty)$. We will show that if z, w belong to the same chamber or wall at infinity, then $G_z = G_w$. If z lies in a wall of w , then $G_w \subset G_z$. Thus, we can define an order on parabolic subgroups given by $G_z \leq G_w$ if $G_w \subset G_z$, and the 0-element is G .

Proof. Let us set the notation as before: we have $z = c(\infty)$ with $c(0) = x$ and c belonging to the flat Ax , so $c(t) = e^{tH}x$ for $H \in \mathfrak{b} \subset \overline{\mathfrak{a}^+} \subset \mathfrak{a}$. If $G = KAN$, we first get $AN \subset G_z$ as a consequence of the previous proof, so we write $G_z = (G_z \cap K)AN$.

Choose $g = e^Y \in K$ for $Y \in \mathfrak{k}$, with $Y = Y_0 + \sum_{\alpha} Y_{\alpha}$ its full root decomposition (the root 0 corresponds to the Cartan subalgebra). A similar computation as before gives:

$$d(gc(t), c(t)) = d(e^Y e^{tH}x, e^{tH}x) = d(e^{-\text{ad}_t H}(Y)x, x)$$

Again, $e^{-\text{ad}_t H}(Y) = \sum_{\alpha} e^{-t\alpha(H)}Y_{\alpha} + Y_0$. We will have that $g \in G_z$ whenever the previous distance calculation decays exponentially (so that we are at bounded distance and gc and c identify at infinity). For this to happen, we need that whenever $Y_{\alpha} \neq 0$, $\alpha(H) = 0$. But this is equivalent to $Y \in (\mathfrak{h} \oplus \bigoplus_{\alpha \in \Lambda^H} \mathfrak{g}_{\alpha}) \cap \mathfrak{k}$ where Λ^H is the set of roots for which $\alpha(H) \geq 0$ (that $Y \in \mathfrak{k}$ we knew beforehand).

Now, if $\mathfrak{b}$ is a Weyl chamber, it corresponds to a choice of positive roots, so all the elements belonging to it are positive under the action of the same roots, i.e. for any pair $H, H' \in \mathfrak{b}$, $\Lambda^H = \Lambda^{H'}$. If $\mathfrak{b}$ is a wall, then it is a wall for a certain Weyl chamber associated to a choice of positive roots, but it is also a wall for another Weyl chamber in another flat, so we need to add new elements in Λ^H , the number of which depends on the codimension of the wall. This reasoning shows that if H, H' belong to the same chamber or wall, then $\Lambda^H = \Lambda^{H'}$. If the inclusion is strict, i.e. $H \in \mathfrak{b} \subset \overline{\mathfrak{b}'}$ and $H' \in \mathfrak{b}'$ ($\mathfrak{b}$ wall of $\mathfrak{b}'$), then $\Lambda^{H'} \subset \Lambda^H$.

For a bigger Λ^H , we obtain more possibilities for what Y can be, so we get a bigger G_z . In the end, for $z = c(\infty)$, $c(t) = e^{tH}x$ and $w = c'(\infty)$, $c'(t) = e^{tH'}x$, we get $\Lambda^H \subset \Lambda^{H'} \iff G_z \subset G_w$. It makes sense that fixing a chamber is done by less elements than the ones that fix one of their walls, and also that fixing a chamber means fixing its walls.

□

The last ingredient in our *building* will be the apartments, that correspond to the boundary at infinity of a flat. Each Weyl chamber will belong to infinitely many apartments (flats) parametrised by N .

Definition 2.30. An n -dimensional (Tits) **building** is an abstract simplicial complex Δ formed by the union of subcomplexes called **apartments** such that:

- Δ is thick, meaning every k -subsimplex of Δ belongs to at least three n -simplices if $k < n$ (think about walls belonging to the closure of two Weyl chambers in a flat, but also of Weyl chambers of other flats).

- Every apartment A is thin, i.e. $(n - 1)$ -simplices inside an apartment belong to exactly two adjacent n -simplices (in a flat, a codimension 1 wall is shared by exactly two chambers, which the wall separates).
- For every two simplices, we can always find an apartment containing them.
- If two simplices belong simultaneously to two apartments A and B , then we have a simplicial isomorphism $A \rightarrow B$ fixing both apartments and their walls.

We call a sequence $\{\mathfrak{a}_1, \dots, \mathfrak{a}_n\}$ a **gallery** if $\mathfrak{a}_i$ is adjacent to $\mathfrak{a}_{i+1}$, that is, they share a codimension 1 wall. Such a gallery is said to have length n .

The distance between chambers (n -simplices from a building perspective) can be taken to be the length of the shortest gallery joining them. An apartment containing both chambers also contains a shortest gallery between them.

We close this section by noticing the similarity of the Tits metric and the Tits building.

Proposition 2.31. *In a symmetric space X , the Tits metric on $X(\infty)$ determines the Tits building and viceversa. One says that $(X(\infty), \text{Td})$ is a geometric realisation of the abstract simplicial complex that is the Tits building.*

We start by noticing that minimal geodesics in the Tits metric are piecewise contained in Weyl chambers. Indeed, for $z, w \in X(\infty)$, take a flat F such that $z, w \in F(\infty)$. Take $x \in F$ and thus the distance is realised by the angle in the unit tangent: $\text{Td}(z, w) = \angle_x(z, w)$. If the angle is less than π (they are not opposite points), then they are joined by a minimal geodesic in $F(\infty)$, which is then piecewise contained in Weyl chambers if we look at the boundary as the unit tangent space at x .

This allows us to decide whether a point is in a chamber or a wall. A point z is in a chamber if there is an $\epsilon > 0$ such that the ball $B(z, \epsilon)$ in the Tits metric is isometric to an ϵ -ball in the standard sphere (geodesics only move through Weyl chambers until they reach a wall). Otherwise, the point z is in a wall. We can get rid of the set of all points contained in chambers and iterate the process to detect walls of dimension $n - 1$, and so on. Thus we obtain the incidence relation needed to set up the Tits building from the Tits metric.

Conversely, if we know the Tits building structure, we assign the length to a Weyl chamber to be 2π divided by the number of Weyl chambers in an apartment. And the incidence relations govern the gluing. For example, points in the center of two Weyl chambers are at distance equal to the length of a Weyl chamber times the length of the shortest gallery joining those chambers.

These will all become clearer in the next section, when we introduce the relevant examples in these constructions that actually turn out to (to some extent) cover all the casuistry.

2.4 Hierarchy in the non-compact regime and examples

Doubtless we cannot see that other higher Spaceland now, because we have no eye in our stomachs.

FLATLAND, Edwin A. Abbott

It is time that we finally address the concrete symmetric spaces that we want to deal with for our application within higher Teichmüller theory. These are going to correspond to the Lie groups for which Hitchin showed the existence of Hitchin components: the adjoint groups of split real forms of complex simple Lie groups. In particular, as we mentioned, we will see that their

symmetric spaces are all totally geodesic submanifolds of the one associated to $\mathrm{PSL}(n, \mathbb{R})$ (which is the same as the one associated to $\mathrm{SL}(n, \mathbb{R})$). The structures at infinity will also fit nicely, so that we will be able to reduce the problem of constructing a parametrisation of Hitchin components for several groups to just checking that it works in $\mathrm{PSL}(n, \mathbb{R})$.

In the same spirit of the previous section, where we explicitly proved some claims basing ourselves on [BGS85], because they illustrated applications of the general theory and they were not as omnipresent in the literature as most of the other ideas exposed, here we will draw upon [BH09] and modify the statements (and notation!) there to our needs, and so adapt or complete the proofs as required. By introducing explicit examples, we also intend to shed some light on the precedent analysis.

Recall that we are interested in symmetric spaces of non-compact type, that are simply connected and nonpositively curved, with no Euclidean de Rham factor. Then, the connected component of the identity of the isometry group is a semisimple Lie group with trivial centre and no compact factors (i.e. it is the case of our adjoint groups of split real forms). So to a symmetric space X corresponds such a group $G = \mathrm{Iso}_0(X)$, so that $X = G/K$ for a maximal compact K , endowed with a G -invariant Riemannian metric.

We are particularly interested in the case of rank at least 2, where the space $P(n, \mathbb{R})$ of symmetric, positive definite, real $n \times n$ matrices of determinant 1 is universal in the sense that any simply connected symmetric space of non-compact type is isometric to a totally geodesic submanifold of $P(n, \mathbb{R})$.

Proposition 2.32. *$P(n, \mathbb{R})$ is the symmetric space $\mathrm{PSL}(n, \mathbb{R})/\mathrm{PSO}(n)$*

Proof. • Reminder from linear algebra: a matrix A is symmetric iff for any vectors $u, v \in \mathbb{R}^n$ we have $(Au, v) = (v, Aw)$. We can always find an orthonormal basis of eigenvectors of A : there exist a matrix $O \in \mathrm{SO}(n)$ such that $O^T A O$ is diagonal. It is positive definite if $(Av, v) > 0$ for every $v \neq 0$. In this case, the entries of the diagonal version are all positive.

- The space of positive definite symmetric matrices is an open cone in the space of symmetric matrices, which has dimension $n(n+1)/2$. Thus, $P(n, \mathbb{R})$ is a differentiable manifold of dimension $n(n+1)/2 - 1$, as it is the kernel of the determinant map into $\mathbb{R}^\times$, which has constant rank 1. The tangent space then consists of the symmetric matrices (if we ignore determinant one, we are open in the manifold of symmetric matrices, so we inherit its tangent space; the space of symmetric matrices is a vector subspace of the space of $n \times n$ real matrices, so its tangent space is isomorphic to itself) which have trace 0 (condition of determinant 1 is added, and in infinitesimal terms it corresponds to the Lie algebra of the special linear group). The scalar product on $T_p P(n, \mathbb{R})$ is given by:

$$(X, Y)_p = \mathrm{Tr}(p^{-1} X p^{-1} Y)$$

which defines a Riemannian metric.

- $\mathrm{SL}(n, \mathbb{R})$ acts on $P(n, \mathbb{R})$ via:

$$g.A = g A g^T$$

This action is transitive: given $p \in P(n, \mathbb{R})$, we have $O \in \mathrm{SO}(n)$ such that $D = O^T p O$ is diagonal with positive entries and determinant 1: $\det(D) = \det(O)\det(p)\det(O)$. Define $p^{1/2} = O D^{1/2} O^T$, where $D^{1/2}$ has diagonal entries the square root of the diagonal entries of D . Then, $p^{1/2}.I = O D^{1/2} O^T I O (D^{1/2})^T O^T = O D^{1/2} D^{1/2} O^T = p$.

This action is by Riemannian isometries: for $X, Y \in T_p P(n, \mathbb{R})$, the derivative of $g \in \mathrm{SL}(n, \mathbb{R})$ at p maps X to $g X g^T \in T_{g.p} P(n, \mathbb{R})$, so:

$$(g.X, g.Y)_{g.p} = \mathrm{Tr}((g^{-T} p^{-1} g^{-1})(g X g^T)(g^{-T} p^{-1} g^{-1})(g Y g^T)) = \mathrm{Tr}(p^{-1} X p^{-1} Y) = (X, Y)_p$$

by the cyclic property of the trace.

The stabiliser of the identity consists of the matrices $g \in \mathrm{SL}(n, \mathbb{R})$ such that $gg^T = I$, so it is $\mathrm{SO}(n)$. Moreover, $-I$ acts trivially: $(-I)A(-I) = A$, so the effective action is carried by $\mathrm{PSL}(n, \mathbb{R})$.

- Thus, $P(n, \mathbb{R})$ corresponds to the homogeneous space $\mathrm{PSL}(n, \mathbb{R})/\mathrm{PSO}(n)$, and we just need to check we get an inversion about every point in the group of isometries. Indeed, for $p \in P(n, \mathbb{R})$ choose σ_p that sends $q \mapsto pq^{-1}p$. Then, it is an involution: $q \mapsto pq^{-1}p \mapsto p(p^{-1}qp^{-1})p = q$.

It is an isometry since it is the precomposition of the previous isometry $q \mapsto pqp^T = pqp$ with $q \mapsto q^{-1}$. The latter is also an isometry: its derivative sends X , initial vector of $t \mapsto q + tX$ to $q^{-1}Xq^{-1}$, to $q^{-1}Xq^{-1}$ initial vector of $t \mapsto (q + tX)^{-1}$, so:

$$(q^{-1}Xq^{-1}, q^{-1}Yq^{-1})_{q^{-1}} = \mathrm{Tr}(qq^{-1}Xq^{-1}qq^{-1}Yq^{-1}) = \mathrm{Tr}(q^{-1}Xq^{-1}Y) = (X, Y)_q$$

Its differential is multiplication by -1 , since that is the differential of the map $q \mapsto q^{-1}$ at I and σ_p is obtained from this by conjugation by $q \mapsto p^{1/2}.q$, by above.

□

Once we have this identification, we can use our previous knowledge: it is a symmetric space of non-compact type, and we have a Weyl chamber decomposition on the flats ($n!$ Weyl chamber in each flat), and the rank (dimension of flats) is equal to the dimension of the Cartan subalgebra, i.e. $n - 1$ (think about diagonal matrices of trace 0). The KAN decomposition corresponds to $K = \mathrm{SO}(n)$, A diagonal matrices with positive entries and determinant 1 and N upper triangular matrices with diagonal entries 1.

If we choose I as basepoint for a flat, every Weyl chamber corresponds to a permutation $\sigma \in \mathcal{S}_n$: it is an open convex cone with elements

$$\left(\begin{matrix} t_1 & & \\ & \ddots & \\ & & t_n \end{matrix} \right) \mid \sum_i t_i = 0 \quad \& \quad t_{\sigma(1)} > \cdots > t_{\sigma(n)} \}$$

The walls of codimension k are obtained by replacing k of the inequalities $t_{\sigma(1)} > \cdots > t_{\sigma(n)}$ by equalities. Two chambers are adjacent whenever $\exists i$ such that $\sigma'(i) = \sigma(i + 1)$ and $\sigma'(i + 1) = \sigma(i)$, while $\sigma'(j) = \sigma(j)$ for $j \neq i$.

The Weyl group is $\mathcal{S}_n$ and acts simply transitively on the set of Weyl chambers by permutation of the diagonal entries.

The root system associated to $\mathfrak{h}$ is given by the roots $\{H_{ij}\}_{i \neq j}$, where $H_{ij} = \frac{1}{\sqrt{2}}(e_i - e_j)$ and e_i is a matrix with a 1 on the diagonal element (i, i) and 0 elsewhere (we chose the normalised version of the roots to then look at them as points at infinity). Therefore, the barycenter of a Weyl chamber is given by the same of those corresponding to a positive choice of roots, so as above:

$$b = \sum_{i < j} H_{ij} = \frac{1}{\sqrt{2}} \begin{pmatrix} n-1 & & & \\ & \ddots & & \\ & & (n-k) \cdot 1 + (k-1) \cdot (-1) & \\ & & & \ddots & \\ & & & & 1-n \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} n-1 & & & \\ & n-3 & & \\ & & \ddots & \\ & & & 3-n & \\ & & & & 1-n \end{pmatrix}$$

This can also be normalised to be seen as a barycenter at infinity, i.e. the natural representative of its Weyl chamber at infinity:

$$\frac{1}{\sqrt{\prod_{k=1}^n (n - 2k + 1)}} \begin{pmatrix} n-1 & & & \\ & n-3 & & \\ & & \ddots & \\ & & & 3-n & \\ & & & & 1-n \end{pmatrix}$$

The group $\mathrm{PSL}(n, \mathbb{R})$ corresponds to the Dynkin diagram A_{n-1} (recall 1.31), which is the simplest case. Since the angles between walls correspond to angles between roots by the symmetry of the Weyl decomposition, this tells us that adjacent walls will be at an angle of $\pi/3$ (connection by a single vertex in the Dynkin diagram) and non-adjacent walls will be at angles $\pi/2$ (lack of vertex in the Dynkin diagram indicates orthogonality). This is confirmed using the scalar product: look again at our preferred Weyl chamber. The codimension 1 walls are given by $t_i = t_{i+1}$. We can associate to such a wall the vector $v_i := H_{i,i+1}$. Then (v_i, v_j) is $-1/2$ if $i = j - 1$ and 0 otherwise.

We also have the Tits metric and building at infinity and now we can proceed to describe its boundary as a space of flags.

Choose $z \in P(n, \mathbb{R})$, which we identify with $Y \in T_I^1 P(n, \mathbb{R}) \subset \mathfrak{p}$, so $\|Y\| = 1$ and Y is a symmetric matrix. In particular, $z = c(\infty)$ for $c(t) = e^{tY} \cdot \mathrm{SO}(n)$. Moreover, it can be diagonalised: the matrices that become diagonal after this are the ones that belong to the tangent space $\mathfrak{h}$ of the flat based at I to which Y belongs.

Now Y is diagonal with eigenvalues $\lambda_1 > \dots > \lambda_k$ and corresponding eigenspaces $V_i \subset \mathbb{R}^n$ ($2 \leq k \leq n$). Set $\lambda(z) = (\lambda_1, \dots, \lambda_k)$ and $F^i(z) = V_1 \oplus \dots \oplus V_i$.

Proposition 2.33. *We have a bijection between $X(\infty)$ and the pairs $(F(z), \lambda(z))$ that satisfy:*

- $\lambda_1 > \dots > \lambda_k$ (diagonalisation eigenvalues thanks to symmetry)
- $\sum_{i=1}^k m_i \lambda_i = 0$ (null trace), where $m_i = \dim(F^i(z)) - \dim F^{i-1}(z)$
- $\sum_{i=1}^k m_i \lambda_i^2 = 1$ (norm 1 tangent vector)

The bijection is, of course, $z \mapsto (F(z), \lambda(z))$ and the inverse is given by (F, λ) mapping to the matrix which has eigenvalues λ_i and corresponding eigenspaces $V_i = F^i - F^{i-1}$. By the above conditions, it belongs to the unit tangent space (this is not lost by the diagonalising process, since we conjugate then by a member of $\mathrm{SO}(n)$) Note that if we have previously fixed a basis of $\mathbb{R}^n$, it might then very well not be diagonal as an image of (F, λ) already.

We can furthermore describe the action of G on $X(\infty)$ as an action on flats: $g.F = (g(F^1), \dots, g(F^k))$ and the stabiliser of full flags ($k = n$) corresponds to a conjugate of the Borel subgroup (the upper triangular matrices in a certain basis). This is because if we choose a basis $\{e_1, \dots, e_n\}$ so that F^i is spanned by $\{e_1, \dots, e_i\}$, then the flag is preserved by the upper triangular matrices (the matrix computations follow the philosophy of the calculation previous to 1.21). This choice of basis is what gives the specific conjugation.

As a result, full flags $\mathrm{Flag}(\mathbb{R}^n)$ correspond to the Furstenberg boundary via $\mathrm{Flag}(\mathbb{R}^n) \cong G/B \cong \partial_F X \subset X(\infty)$. This makes sense with our Weyl chamber description, where tangent vectors have distinct eigenvalues, so distinct 1-dimensional eigenspaces, so full flags. In particular, the inclusion $\mathrm{Flag}(\mathbb{R}^n) \subset X(\infty)$ corresponds to $F \mapsto (F, \lambda_b)$ where the barycenter eigenvalues λ_b are the eigenvalues obtained after normalising our preceeding calculation of the barycenter.

Proposition 2.34. *Two points $z, w \in \partial_F X$ can be connected by a geodesic in X if and only if the flags $F(z)$ and $F(w)$ are transverse, i.e. $F^i(z) \oplus F^{n-i}(w) = \mathbb{R}^n$ for all $i \in \{1, \dots, n\}$.*

Proof. Points z, w in the Furstenberg boundary can be connected by a geodesic within X whenever they belong to a flat in which they are opposing points. This is because for every pair of Weyl chambers we can find a flat containing them, and so the same happens for the barycenters, and the geodesic joining them is in X when the Tits metric is at least π , so they have to be antipodal. Then the geodesic joining them is like a diameter in the Weyl chamber picture of tangent space.

Now, because they are opposing, and adjacent Weyl chambers correspond to a transposition $\sigma \in \mathcal{S}_n$ of the diagonal entries, then we would have inverted all the entries at the moment we get

to the antipodal point, advancing through a gallery within the flat. Or seen in another way, since they are antipodal matrices, if we choose z to be the barycenter in the Weyl chamber associated to our choice of positive roots, then it has the expression above. w will be the opposite of this matrix, so it will have precisely the entries of z in reversed order. $\square$

Since the codimension k walls of Weyl chambers relate to k equalities appearing among diagonal entries, they correspond to partial flags. In particular, the vertices have maximum degeneracy, i.e. they correspond to $(\tilde{F}, \tilde{\lambda})$ with $k = 1$, so by an argument similar to the ones above using roots H_{ij} :

$$\tilde{\lambda} = \frac{1}{\sqrt{i(n-i)^2 + (n-i)i^2}} \underbrace{(n-i, \dots, n-i)}_{i \text{ times}} \underbrace{(-i, \dots, -i)}_{n-i \text{ times}}$$

where we properly normalised (norm is $\sqrt{\sum_i m_i \lambda_i^2}$). The only non-trivial element in the flag is $\tilde{F}^1 = V_i$ (where V_i was the subspace corresponding to F^i in the flag of the Weyl chamber whose vertex this is: we have $n-1$ vertices in the closure of every Weyl chamber, corresponding to $F^1, \dots, F^{n-1}$). This means vertices at infinity are in one to one correspondence with proper subspaces of $\mathbb{R}^n$, so the set of vertices is the disjoint union of Grassmanians $\bigsqcup_{i=1}^{n-1} \text{Gr}_i(\mathbb{R}^n)$.

Ok, so we already got a glimpse at our main example. In the beginning, we mentioned how every other relevant case was a totally geodesic submanifold of it with nice embedding of structures at infinity. We will conclude the section by showing this and looking at an example. We need two notions:

Definition 2.35. A subgroup $G \subset GL(n, \mathbb{R})$ is called **reductive** if it is closed and it is stable under transposition. More generally, a Lie group is reductive if its Lie algebra is, i.e. if its radical is equal to its center.

A group is called **algebraic** if it admits the structure of an algebraic variety, i.e. it corresponds to the subset of zeros of a finite collection of polynomials.

The point is we have ([FH91]):

Proposition 2.36. If G is semisimple, then it is algebraic (as we can check on our classification). In terms of $\mathfrak{g}$ being reductive, the following are equivalent:

- $[\mathfrak{g}, \mathfrak{g}]$ is semisimple.
- The adjoint representation of $\mathfrak{g}$ is semisimple.
- $\mathfrak{g}$ is a product of a semisimple and an abelian Lie algebra (if we take the abelian part to be trivial, we see semisimple $\implies$ reductive).
- $\mathfrak{g}$ has a finite dimensional faithful semisimple representation.

It turns out reductive algebraic groups satisfy the following properties that are useful for us (this statement can be generalised, but this is enough for us):

Proposition 2.37. Let $G \subset PSL(n, \mathbb{R})$ algebraic subgroup. Then for $Y \in T_I P(n, \mathbb{R})$ such that $e^Y \in G$, we also get $e^{tY} \in G$ for all $t \in \mathbb{R}$.

Moreover, if G is reductive, set $K = G \cap PSO(n)$ and $X = G \cap P(n, \mathbb{R})$. Then:

- X is the G -orbit of I and a totally geodesic submanifold of $P(n, \mathbb{R})$ diffeomorphic to the symmetric space G/K , non-positively curved and with K as the maximal compact subgroup (up to conjugation).

- For any totally geodesic submanifold M of $P(n, \mathbb{R})$ the group $G = \{g \in \mathrm{PSL}(n, \mathbb{R}) \mid g.M = M\}$ is reductive and $M = G \cap P(n, \mathbb{R})$.

We only need the first two claims, so we will show those.

Proof. • First note that a necessary condition for $G.I \subset P(n, \mathbb{R})$ to be totally geodesic is precisely that for $Y \in T_I P(n, \mathbb{R})$ and $e^Y \in G$, the whole geodesic ray $e^{tY} \in G$.

Now, we want to reduce the statement (which can be generalised to algebraic subgroups of the general linear group) to showing that if a polynomial on matrix entries (defining the set of zeros for which the group is algebraic) vanishes at e^{kY} for some Y symmetric (with trace 0) and all $k \in \mathbb{Z}$, then it vanishes at e^{tY} for all $t \in \mathbb{R}$. We have the premise because we can iteratively apply the exponential on e^Y , then on e^{2Y} and so on, and if $e^{kY} \in G$, then we have its inverse e^{-kY} .

Start by diagonalising Y ; then we now how the exponential looks like:

$$Y = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_{n-1} & \\ & & & & \lambda_n \end{pmatrix} \implies e^{tY} = \begin{pmatrix} e^{t\lambda_1} & & & \\ & e^{t\lambda_2} & & \\ & & \ddots & \\ & & & e^{t\lambda_{n-1}} & \\ & & & & e^{t\lambda_n} \end{pmatrix}$$

So if the polynomial $P(x_1, \dots, x_n) = \sum c_{i_1 \dots i_n} x_1^{i_1} \dots x_n^{i_n}$ of degree m satisfies $P(e^{k\lambda_1}, \dots, e^{k\lambda_n}) = 0$ for all $k \in \mathbb{Z}$, then $P(e^{t\lambda_1}, \dots, e^{t\lambda_n}) = 0$ for all $t \in \mathbb{R}$.

Consider the set of tuples $(i_1, \dots, i_n) \in \mathbb{N}^n$ such that $i_1 + \dots + i_n \leq m$, and denote the different values taken by the product of monomials $e^{i_1\lambda_1} \dots e^{i_n\lambda_n}$ on each of those tuples by $s_1, \dots, s_l$. We have

$$P(e^{t\lambda_1}, \dots, e^{t\lambda_n}) = \sum_{j=1}^l a_j s_j^t \quad \text{for} \quad a_j = \sum_{(i_1, \dots, i_n) \text{ s.t. } e^{i_1\lambda_1} \dots e^{i_n\lambda_n} = s_j} c_{i_1 \dots i_n}$$

The system of l equations $\{\sum_j a_j s_j^k = 0 \mid k \in \{0, 1, \dots, l-1\}\}$ on the variables $a_1, \dots, a_l$ has a non-zero Vandermonde polynomial as determinant of the coefficient matrix: $\prod_{j > j'} (s_j - s_{j'})$. Therefore, since $\sum_j a_j s_j^k = 0$ for $k \in \mathbb{Z}$, then $a_j = 0$ (recall that the non-vanishing of the Vandermonde polynomial implies that there is a unique solution; we can reach that solution for $a_j = 0$, so this has to be the only option).

- If $Y \in T_I P(n, \mathbb{R})$ and $p = e^Y \in G$, then $p = e^{Y/2}.I$. So $p \in X \iff e^X \in G \iff e^{X/2} \in G \iff p \in G.I$, so X is the G -orbit of I .

G is a Lie group and its action differentiable, so $G.I$ is a differentiable submanifold of $P(n, \mathbb{R})$. Since the stabiliser of $\mathrm{PSL}(n, \mathbb{R})$ is $\mathrm{PSO}(n)$, X is diffeomorphic to G/K . The unique geodesic ray containing I and $p \in X$ is the image of $t \mapsto e^{tY}$, and the ray is contained entirely in G by above, so X is totally geodesic.

Being totally geodesic implies that the inversions about points in $P(n, \mathbb{R})$ leave X invariant, so it is a symmetric space. It inherits the curvature from $P(n, \mathbb{R})$, so it is nonpositively curved.

K stabilises I in G ; it is compact because G is closed in $\mathrm{GL}(n, \mathbb{R})$ (reductive) and $\mathrm{PSO}(n)$ is compact. It is maximal because every compact subgroup of $\mathrm{PSL}(n, \mathbb{R})$ is conjugate to a subgroup of $\mathrm{PSO}(n)$ (standard proof of this fact uses Haar measures).

□

We actually improve on the result by the following consideration: if K acts on $T_I X$ irreducibly, then the totally geodesic submanifold X obtained is an isometrical embedding of G/K into $P(n, \mathbb{R})$. The precise statement is [Ebe97]:

Proposition 2.38. *Let X be a symmetric space of noncompact type, $p \in X$, $G = \text{Iso}_0(X)$ and with positive inner product $\phi_p(Y, Z) = -B(\sigma(Y), Z)$, where σ is the Cartan involution. identify $(\mathfrak{g}, \phi_p)$ with $\mathbb{R}^n$ with its usual scalar product by a fixed linear isometry ($n = \dim(G)$). Then:*

- *$\text{Ad} : G \longrightarrow GL(n, \mathbb{R})$ is a faithful representation with image a self-adjoint subgroup of $SL(n, \mathbb{R})$.*
- *The map $F_p : X \longrightarrow P(n, \mathbb{R})$ given by $F_p(g(p)) = \text{Ad}(g) \cdot SO(n)$ is a diffeomorphism into a complete totally geodesic submanifold, whose metric is a multiple with some positive constant of the original metric.*

In any case, most of the important ideas are contained in the proof above. We can distill what we will use later:

Corollary 2.39. *Every adjoint group G of a split real form of a simple complex Lie group has an associated symmetric space $X = G/K$ isometric to a totally geodesic submanifold of $P(n, \mathbb{R}) = SL(n, \mathbb{R})/SO(n)$, for suitable $n \in \mathbb{N}$.*

Furthermore, the structures at infinity embed: $\partial_F X \subset \partial_F P(n, \mathbb{R})$, $\partial_\infty X \subset \partial_\infty P(n, \mathbb{R})$

Proof. G is semisimple, so algebraic and reductive; it is also a subgroup of $SL(n, \mathbb{R})$. Therefore, we have the embedding as a totally geodesic submanifold. Since the boundary at infinity is defined in Hadamard manifolds by geodesic rays, being totally geodesic ensures the embedding happens nicely. For the Furstenberg boundary, we take the perspective through which we see it as the G -orbit of a barycenter of a Weyl chamber at infinity. Just note that, for $H \subset G$, the H -orbit of a barycenter embeds in its G -orbit, and Weyl chambers in G/K are intersections of Weyl chambers in $P(n, \mathbb{R})$ with G/K [BH09]. $\square$

This corollary is fundamental, because it will be a key step in allowing us to directly translate the parametrisation constructed in [Sle21] for the Hitchin component of $SL(n, \mathbb{R})$ to the Hitchin component of other adjoint groups of split real forms.

Finally, let us look at the only non-trivial example we can draw:

Example 2.40 ($P(3, \mathbb{R})$). *The easiest example of symmetric space could be $\mathbb{H}^2 = PSL(2, \mathbb{R})/PSO(2)$, which is well known. But here the flats are 1-dimensional, so we do not really get any insight into the Weyl chamber and flag structures at infinity in the general regime. That is why we choose the higher rank case corresponding to $X = P(3, \mathbb{R}) = PSL(3, \mathbb{R})/PSO(3)$ (rank 2).*

Choose as basepoint the identity I . Every $z \in T_I^1 X$ is a 3×3 symmetric matrix with trace 0 and norm 1. Again, finding a flat containing it is done by diagonalising the matrix: we obtain a matrix with eigenvalues $a \geq b \geq c$ and eigenvectors $\{v_a, v_b, v_c\}$, such that:

- $a + b + c = 0$ (null trace)
- $a^2 + b^2 + c^2 = 1$ (unit norm)

Thus, all the information can be gathered up in a ratio constant r defined as:

$$r := \frac{b - c}{a - b}$$

Note that $a > 0$ and $c < 0$, so $r \in [0, \infty]$: $r = 0$ if $b = c$ and $r = \infty$ if $a = b$.

We associate to this element $z \in X(\infty)$ a strip of the form $\mathbb{R}v_1 + [-r, r]v_2$. It will degenerate into a plane if $a = b$ and into a line if $b = c$. If it does not degenerate into those, i.e. $a > b > c$, we associate to it the flag $\{0\} \subset F^1(z) = \mathbb{R}v_a \subset F^2(z) = \mathbb{R}v_a + \mathbb{R}v_b \subset F^3(z) = \mathbb{R}^3$ (this flag is the same for the whole Weyl chamber, in particular, we get the bijection between full flags in $\mathbb{R}^3$ and barycenters of Weyl chambers $\text{Flag}(\mathbb{R}^3) \cong CX = \text{PSL}(3, \mathbb{R}).b$). For $a = b$ we will have the flag given by $\{0\} \subset F^2 \subset \mathbb{R}^3$ and for $b = c$ the one given by $\{0\} \subset F^1 \subset \mathbb{R}^3$. This agrees with our description of vertices as flags with only one non-trivial subspace.

Accounting for the action of the Weyl group via permutation of the diagonal entries, we can draw the whole picture of a Weyl chamber at infinity within the perspective of unit tangent space:

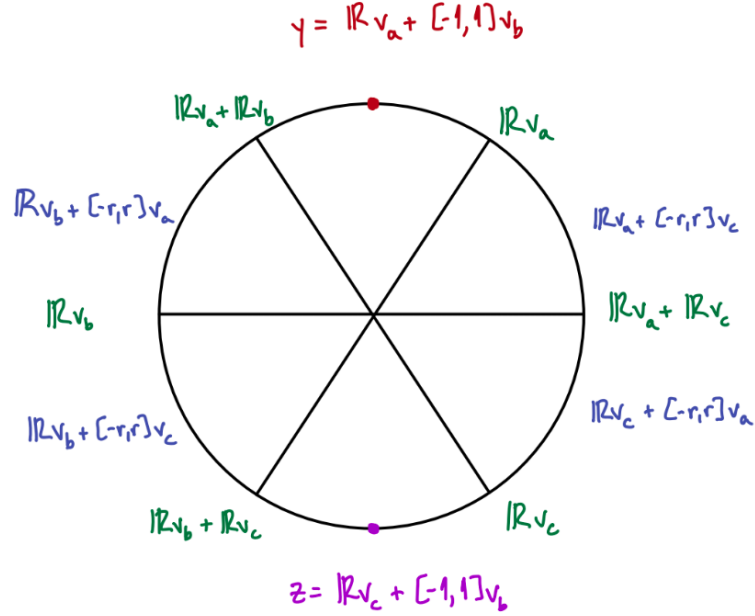


Figure 5: Flag structure of a flat at infinity in $P(3, \mathbb{R})$, expressed in terms of strips. y and z are barycenters of opposite Weyl chambers, so they can be joined by a geodesic in $P(3, \mathbb{R})$ and their flags are transverse. Weyl chambers at infinity are 1-dimensional and correspond to $0 < r < \infty$, and vertices to $r \in \{0, \infty\}$.

3 The barycenter method

With the previous sections we have settled most of the matters regarding *foundational* formalisms present in our theory. In turn, now we will treat more specific tools that originate in this context.

The nub of the barycenter method consists of taking a map between the boundaries of two symmetric spaces Y and X and obtaining a map between the symmetric spaces themselves:

$$\text{Given } F : \partial_\infty Y \longrightarrow \partial_\infty X, \quad \text{construct } f : Y \longrightarrow X$$

This map is moreover optimal (in a coarse area minimising sense [Sle21]) and we have estimates of bounds for its Jacobian.

This is to be compared with a parametrisation using harmonic maps (optimal in the sense of minimising energy), also between symmetric spaces [Cor88]. We expect the barycenter map to be more relevant in geometric terms due to the knowledge we extract about its Jacobian.

To arrive at such a map, we will need to consider measures on the boundary at infinity of our symmetric spaces and then show that a certain function constructed using these measures has a unique minimum (this is where convexity plays an important role), which we will call *barycenter of the measure* (in some manner, its *center of mass*). The map between the boundaries at infinity will be given by the Hitchin representation itself, and we will take care of it in the next section.

Denoting $\mathcal{M}(\partial_\infty X)$ the space of Borel measures in the boundary at infinity of X with finite mass equipped with the weak $*$ topology, the whole process will consist of defining the map:

$$f : Y \longrightarrow \mathcal{M}(\partial_\infty Y) \longrightarrow \mathcal{M}(\partial_\infty X) \longrightarrow X$$

where the first arrow is a natural embedding using visual measures, the second corresponds to a pushforward of measures coming from the map between boundaries of symmetric spaces provided by and unique to every Hitchin representation, and the third is the minimisation step: finding the barycenter of the measure as unique minimum of a certain natural convex function, the so called *weighted Busemann function*.

We divide the section in two parts: on the one hand, the construction of barycenter maps through understanding the first and third arrows above, and on the other hand sketching the reasons why barycenter maps are relevant.

3.1 How to: barycenter maps

Given a function, when searching for a point in which it reaches a minimum, and whether this minimum is unique, one usually looks to convex analysis. If the convex function is defined in the extended real line, one tends to avoid the value $-\infty$ or that it is constant $+\infty$ in its whole domain, since in those examples the result is clear. This motivates:

Definition 3.1. *A convex function $f : X \longrightarrow [-\infty, \infty]$ is called **proper** if:*

- $f(x) > -\infty$ for all $x \in X$
- $\exists x_0 \in X$ such that $f(x_0) < +\infty$

Now, the ideal setup to obtain a unique minimum is to have a strictly convex proper function that is bounded from below.

We consider a measure $\mu \in \mathcal{M}(\partial_\infty X)$ and we want to associate to it a *minimum* that we will call barycenter of the measure. For this, we look for a natural convex function associated to it: our formalism of boundary at infinity described in terms of Busemann functions suggests the following:

Definition 3.2. A *weighted Busemann function* associated to a measure $\mu \in \mathcal{M}(\partial_\infty X)$ is defined as:

$$B_\mu : X \longrightarrow \mathbb{R}, \quad x \mapsto \int_{\partial_\infty X} b_\eta(x) d\mu(\eta)$$

where b_ν is the Busemann function of 2.8. If B_μ is bounded from below, proper and strictly convex, we define the **barycenter** of μ , denoted $\text{bar}(\mu)$, as the unique minimiser of B_μ .

Question: in which instances do we obtain such a unique minimiser?

Intuitively, the idea is that in flat spaces, a unique minimiser does not exist, so we cannot carry out the barycenter method. But we do achieve it in strictly negatively curved spaces. Therefore, in our spaces, which are non-positively curved, we will try to avoid the flatness corresponding to singular geodesics associated to walls of Weyl chambers in the flat, where several flats intersect at once.

Turns out Busemann functions are asymptotically linear along geodesic rays, so the same happens for weighted Busemann functions. The positivity of the *asymptotic slope* (the slope associated to its asymptotic linear form) will tell us whether the measure admits a barycenter or not. Let us break this down step by step.

Definition 3.3. The *asymptotic slope* of a Lipschitz continuous convex function $f : X \longrightarrow \mathbb{R}$ at a point $\xi \in \partial_\infty X$ is:

$$\text{slope}_f(\xi) = \lim_{t \rightarrow \infty} \frac{f(c(t))}{t} = \lim_{t \rightarrow \infty} \frac{d}{dt}(f \circ c)(t)$$

where c is any unit speed geodesic ray such that $c(\infty) = \xi$.

Note that if at infinity $f \circ c \sim at + b$, then $\text{slope}_f(\xi) = a$, so the nomenclature is adequate. The Lipschitz assumption is elected to ensure the independence on the choice of geodesic ray c . The existence of the limit follows by boundedness via the Lipschitz condition and monotonicity (non-decreasing) via convexity.

Proposition 3.4. The slope of a weighted Busemann function B_μ at a point $\xi \in \partial_\infty X$ is given by:

$$\text{slope}_\mu(\xi) := \text{slope}_{B_\mu}(\xi) = \int_{\partial_\infty X} -\cos \angle(\xi, \eta) d\mu(\eta)$$

Proof. • Notice that B_μ is $\|\mu\|$ -Lipschitz and convex, since Busemann functions are 1-Lipschitz and convex. It is also well-defined up to an additive constant, as Busemann functions. It is common practice to normalise Busemann functions at the tip $p \in X$ (basepoint chosen such that $\partial_\infty X \cong T_p^1 X$): we set $b_\xi(p) = 0$; this yields continuity of the maps:

$$\partial_\infty X \times X \longrightarrow \mathbb{R}, \quad (\xi, x) \mapsto b_\xi(x) \quad \& \quad \mathcal{M}(\partial_\infty X) \times X \longrightarrow \mathbb{R}, \quad (\mu, x) \mapsto b_\mu(x)$$

- In any proof of 2.9 (3) $\implies$ (4) (e.g. [BGS85]), one shows that at a point $x \in X$, $\nabla b_\xi := \text{grad}(b_\xi) = -\dot{c}$, where c is the geodesic ray such that $c(0) = x$ and $c(\infty) = \xi$. We also calculated in 2.11 the angle between points at infinity $\xi, \eta \in \partial_\infty X$ by looking at their representatives in the unit tangent space $\dot{c}(0)$ and $\dot{c}'(0)$ for the unit speed geodesics c and c' , respectively, and setting $\angle_x(\xi, \eta) = \angle(\dot{c}(0), \dot{c}'(0))$.

Thus, for a unit speed geodesic ray ρ with $\rho(\infty) = \xi$, choose for every t unit speed geodesics $c_t(s) = \rho(s + t)$ and c'_t representing ξ and η , respectively, with $c_t(0) = c'_t(0) = \rho(t)$. We can compute the slope from:

$$\begin{aligned}\frac{d}{dt}(b_\eta \circ \rho)(t) &= \langle \nabla b_\eta(\rho(t)), \rho'(t) \rangle = \langle -\dot{c}'_t(0), \dot{c}_t(0) \rangle = \\ &= -\cos \angle_{\rho(t)}(\eta, \xi) \|\dot{c}'_t(0)\| \|\dot{c}_t(0)\| = -\cos(\angle_{\rho(t)}(\eta, \xi))\end{aligned}$$

Using 2.12, we get $\lim_{t \rightarrow \infty} \angle_{\rho(t)}(\eta, \xi) = \angle(\eta, \xi)$, so:

$$\text{slope}_{b_\eta}(\xi) = \lim_{t \rightarrow \infty} \frac{d}{dt}(b_\eta \circ \rho)(t) = -\cos \lim_{t \rightarrow \infty} \angle_{\rho(t)}(\eta, \xi) = -\cos \angle(\eta, \xi)$$

- In the case of weighted Busemann functions, the differential is computed to be:

$$(dB_\mu)_x(\xi) = - \int_{\partial_\infty X} \cos \angle_x(\xi, \eta) d\mu(\eta)$$

and by the integral monotone convergence theorem:

$$\text{slope}_\mu(\xi) = - \int_{\partial_\infty X} \cos \angle(\xi, \eta) d\mu(\eta)$$

□

The way we study the existence and uniqueness of minima for measures is through this asymptotic slope:

Definition 3.5. We say the measure $\mu \in \mathcal{M}(\partial_\infty X)$ is **stable** if $\text{slope}_\mu > 0$.

Proposition 3.6. μ is stable $\iff B_\mu$ is proper and bounded from below. If that is the case, B_μ attains a minimum, so $\text{bar}(\mu)$ exists.

Proof. • The set of nonpositive slope of μ corresponds to the boundary at infinity of the sublevel sets of B_μ :

$$\partial_\infty \{B_\mu \leq k\} = \{\text{slope}_\mu \leq 0\} \quad \text{for any } k \in \mathbb{R}$$

Indeed, $\{B_\mu \leq k\}$ is non-empty and convex, so choose any point $x \in \{B_\mu \leq k\}$ and $\xi \in \partial_\infty \{B_\mu \leq k\}$. By convexity, the geodesic ray c from x to ξ has to be contained in $x \in \{B_\mu \leq k\}$, so B_μ is non-increasing along it, otherwise it would get out of the sublevel set, so $\text{slope}_\mu(\xi) \leq 0$:

$$\text{slope}_\mu(c(\infty)) = \lim_{t \rightarrow \infty} \frac{B_\mu(c(t))}{t} \leq \lim_{t \rightarrow \infty} \frac{k}{t} = 0$$

Conversely, if $\text{slope}_\mu(\xi) \leq 0$, then B_μ is non-increasing along the geodesic ray from x to ξ , so this ray is contained in a sublevel set, and so ξ belong to the boundary at infinity of that sublevel set.

- A symmetric space X is a Riemannian manifold, so locally compact. In such context, proper and bounded from below is equivalent to strictly positive slope on the boundary of X .

Suppose $\text{slope}_\mu(\xi) \leq 0$, then B_μ is non-increasing along rays asymptotic to ξ . In particular, the boundary at infinity of sublevel sets is non-empty, so sublevel sets are non-empty for all $k \in \mathbb{R}$. As a result, B_μ cannot be proper.

Conversely, sublevel sets being compact imply the function is proper. Which means that if B_μ was not proper, then sublevel sets would be noncompact. Thus, they are unbounded, so they must contain a geodesic ray, by local compactness of X . This ray defines an ideal point with non-positive asymptotic slope, by the previous item.

- Thus, μ stable means B_μ proper, convex and bounded from below. Its set of minima has to be compact, non-empty and totally geodesic, so it has to be a point [KLM03].

One can alternatively show that properness implies strict convexity of B_μ , noticing that a function f is (strictly) convex $\iff$ its Hessian ∇f is (strictly) positive [Sle21]. $\square$

Example 3.7. Thanks to the expression for the slope in 3.4, we can use our previous angle calculations from 2.13 and 2.15 to give an idea of why we want to avoid flatness.

In the case of hyperbolic space $\mathbb{H}^n$, the angle between any two points is π , so the slope reads:

$$\text{slope}_\mu(\xi) = \int_{\partial_\infty \mathbb{H}^n} -\cos(\pi) d\mu(\eta) = \mu(\partial_\infty \mathbb{H}^n \setminus \{\xi\}) - \mu(\{\xi\})$$

and this is positive for all ξ as long as the measure is not atomic or the atoms of the measure have weight less than half. Our particular application concerns non-atomic measures, thus the measure is stable and we get a barycenter.

Notice that if the measure is a Dirac measure (support on only one point) or a sum of two Dirac measures (two points), then translations along geodesics with these atoms as endpoints leave the weighted Busemann functions unchanged, so the barycenter is also fixed by those translations, thus cannot be unique. Nonetheless, if you have three points and you translate along a geodesic connecting two of them, the third would move (when extending the map to the boundary), so we do not get such invariability, and therefore a measure with only three atoms in its support can have a barycenter if it satisfies the above condition.

When looking at Euclidean space $\mathbb{R}^n$, the angle is the usual sphere angle. In fact, it corresponds to angles measured in the unit tangent space at any point: $\angle(\xi, \eta) = \angle_x(\xi, \eta)$ for all $\xi, \eta \in \partial_\infty \mathbb{R}^n$ and all $x \in \mathbb{R}^n$. The fact that every single point has this same behavior should already convince us that there cannot be a special unique minimiser, a barycenter. Formally, we can see that in the sphere, the angle function between two vectors is continuous, so the same holds for:

$$\partial_\infty \mathbb{R}^n \times \partial_\infty \mathbb{R}^n \longrightarrow \mathbb{R}, \quad (\xi, \eta) \mapsto \angle(\xi, \eta)$$

and the same is true for $(\xi, \eta) \mapsto \langle \xi, \eta \rangle = \cos \angle(\xi, \eta)$, so slope_μ is continuous for every $\mu \in \mathcal{M}(\partial_\infty \mathbb{R}^n)$. Moreover, $\cos(x) = -\cos(-x)$, so $\cos \angle(\xi, \eta) = -\cos \angle(-\xi, \eta)$. This implies $\text{slope}_\mu(\xi) = -\text{slope}_\mu(-\xi)$, and by continuity there must exist $\xi_0 \in \partial_\infty \mathbb{R}^n$ such that the slope reaches the intermediate null value: $\text{slope}_\mu(\xi_0) = 0$.

In particular, if μ is uniform, then the slope is always zero, since the integral of the cosine of the angle is over the standard volume element on the sphere, so vanishes by the symmetry of the cosine function (it is even).

Ultimately, this means that no measure on the boundary at infinity of Euclidean space has a barycenter.

So now the plan to learn about which measures admit a barycenter is to check whether slopes of weighted Busemann functions are strictly positive in our boundary at infinity. We can save time via the following technical detail, based on the simplicial structure of a Weyl chamber at infinity and the asymptotic linearity of Busemann functions:

Proposition 3.8. For a Weyl chamber $\Delta \in \partial_\infty X$, if the slope is positive on all vertices of Δ , then it is positive in the whole of Δ .

Proof. Denote the vertices $\{x_1, \dots, x_n\}$. The key is that we can express $\xi \in \Delta$ as a linear combination of them by seeing them as unit tangent vectors:

$$\xi = \frac{\sum_{i=1}^{n-1} \alpha_i x_i}{\|\sum_{i=1}^{n-1} \alpha_i x_i\|}$$

This expression is independent of the choice of flat or basepoint and depends only on the eigenvalues of ξ (recall how vertices correspond to a single eigenvalue from the Weyl chamber they are delimiting), the same way the boundary at infinity is.

Now, for a generic $\eta \in \partial_\infty X$, there is a flat F containing Δ and η at infinity, and for every $z \in F(\infty)$, $\angle(z, \eta) = \angle_x(z, \eta)$ is reached when measuring from the unit tangent space of any basepoint $x \in F$, in the spirit of 2.12. Thus:

$$\begin{aligned} \text{slope}_{B_\eta}(z) = -\cos \angle(z, \eta) &\implies \text{slope}_{B_\eta}(\xi) = \frac{\sum_{i=1}^{n-1} \alpha_i \text{slope}_{B_\eta}(x_i)}{\|\sum_{i=1}^{n-1} \alpha_i x_i\|} \implies \\ \text{slope}_\mu(\xi) &= \int_{\partial_\infty X} \text{slope}_{B_\eta}(\xi) d\mu(\eta) = \frac{\sum_{i=1}^{n-1} \alpha_i \int_{\partial_\infty X} \text{slope}_{B_\eta}(x_i) d\mu(\eta)}{\|\sum_{i=1}^{n-1} \alpha_i x_i\|} = \frac{\sum_{i=1}^{n-1} \alpha_i \text{slope}_\mu(x_i)}{\|\sum_{i=1}^{n-1} \alpha_i x_i\|} \end{aligned}$$

Which is greater than 0 provided $\text{slope}_\mu(x_i) > 0$ for every $i \in \{1, \dots, n-1\}$, our hypothesis. $\square$

This discussion concerned the third arrow:

$$\text{bar} : \mathcal{M}(\partial_\infty X) \longrightarrow X, \mu \mapsto \min_{x \in X} B_\mu(x)$$

In the next sections we will study the slope of vertices at infinity.

Now, we turn towards the first arrow:

$$Y \longrightarrow \mathcal{M}(\partial_\infty Y)$$

We choose the measures at the boundary that are appropriate for our purposes: suppose we have a map $F : \partial_\infty \mathbb{H}^2 \longrightarrow \partial_F X \subset \partial_\infty X$, where $\mathbb{H}^2$ is the hyperbolic plane and X is a symmetric space. For $z \in \mathbb{H}^2$, consider the diffeomorphism $E_z : T_z^1 \mathbb{H}^2 \longrightarrow \partial_\infty \mathbb{H}^2$ that identifies the boundary at infinity with the unit tangent space at a point. Let $\mathcal{L}$ denote the Lebesgue measure on $T_z^1 \mathbb{H}^2$. Define the family of **visual measures** $\{\nu_z\}_{z \in \mathbb{H}^2}$ on $\partial_\infty \mathbb{H}^2$ by:

$$\nu_z = (E_z)_* \left(\frac{\mathcal{L}}{2\pi} \right) \quad \text{where} \quad f_* \mu(x) = \mu(f^{-1}(x)).$$

Thinking about $\mathbb{H}^2$ from the Poincaré disk model, $\nu_0 = \frac{\mathcal{L}}{2\pi}$ is a multiple of the standard Lebesgue measure on the circle, and these are the unique $\text{PSL}(2, \mathbb{R})$ -equivariant measures on $\partial_\infty \mathbb{H}^2$. We can pushforward the measures by F_* , and obtain our first two arrows (for $Y = \mathbb{H}^2$):

$$\mathbb{H}^2 \longrightarrow \mathcal{M}(\partial_\infty \mathbb{H}^2) \longrightarrow \mathcal{M}(\partial_\infty X), z \mapsto \nu_z \mapsto F_*(\nu_z)$$

We still need to say where the map F comes from or if it exists at all, and this is going to be given by the algebraic properties of Hitchin representations, investigated in the next section.

3.2 Why: Jacobian bounds

Our construction is a particular instance of a more general method that turns out to be useful to solve a myriad of problems in differential geometry (or at least give insight and partial results on ongoing conjectures). Because of this applicability, the explicit knowledge of Jacobian bounds and its *coarse optimality* within its homotopy class (this optimality relates to minimising a functional dependent on volume and volume entropy), we consider it relevant to illustrate briefly the general procedure. That is what we do now ([CF02]).

The main timeline of the barycenter method is roughly as follows:

- Douady-Earle introduce the idea of a conformal construction for the hyperbolic plane [DE86].
- Besson-Courtois-Gallot extend it to negatively curved targets [BCG95].
- Connell-Farb treat the case of symmetric spaces of non-compact type in [CF01] and then follow with nonpositively curved targets with negative Ricci curvature in [CF02].

The set up is given by the following diagram:

$$\begin{array}{ccccc}
& & \mathcal{M}(Y) & \xrightarrow{\tilde{f}_*} & \mathcal{M}(X) & \xrightarrow{* \nu_z} & \mathcal{M}(\partial_\infty X) \\
& & \uparrow \mu^s & & & & \downarrow \text{bar} \\
Y & \xrightarrow{\tilde{f}} & X & & Y & \xrightarrow{\tilde{F}_s} & X \\
\pi_N \downarrow & & \downarrow \pi_M & & \pi_N \downarrow & & \downarrow \pi_M \\
N & \xrightarrow{f} & M & & N & \xrightarrow{F_s} & M
\end{array}$$

We need two ingredients before we can explain what the steps above mean.

Definition 3.9. For a Riemannian manifold (N, g) , we define the **volume entropy** to be:

$$h(g) := \lim_{R \rightarrow \infty} \frac{1}{R} \log \text{vol}(B(y, R))$$

where the ball $B(y, R)$ is in the universal cover Y of N (and the limit is independent of the choice of $y \in Y$).

It can be normalised to be scale invariant:

$$\text{ent}(g) := h(g)^n \text{Vol}(N, g)$$

We also need to introduce a class of measures, named after Patterson and Sullivan, since they correspond to a generalisation of their work produced by several authors. The idea is that given a discrete group of isometries Γ acting on a symmetric space of non-compact type X (such as the action of the fundamental group via deck transformations), the measures encode the density at infinity of Γ -orbits in X measured with the Riemannian metric.

Definition 3.10. Given a discrete subgroup of isometries Γ acting on a Hadamard manifold X with no Euclidean factors, and which is the universal cover of a Riemannian manifold (M, g) , the **Patterson-Sullivan measures** $\{\nu_x\}_{x \in X}$ are a family of measures on $\partial_\infty X$ satisfying the following properties:

- They have no atoms (points with non-zero mass).
- They are equivariant with respect to Γ : $\nu_{\gamma x} = \gamma_* \nu_x$ for all $\gamma \in \Gamma$.
- For all $x, y \in X$, the measure ν_y is absolutely continuous with respect to ν_x , with Radon-Nikodym derivative:

$$\frac{d\nu_x}{d\nu_y}(\xi) = e^{h(g)b_\xi(y)}$$

where the Busemann function is normalised at x : $b_\xi(x) = 0$, i.e. is defined with a geodesic ray c such that $c(0) = x$ and $c(\infty) = \xi$.

- Each ν_x is supported in a specific orbit of Γ in $\partial_\infty X$.

These measures are constructed in [Alb99]. In the setting of symmetric spaces of non-compact type, each ν_x is supported in a copy of the Furstenberg boundary $\partial_F X$, giving the matter an algebraic flavor, as we will see.

In the barycenter method, given $s > h(g)$, we start with a map $f : N \rightarrow M$ and want to improve it to a map $F_s : N \rightarrow M$ which is *optimal* (in a sense related to the *ent* functional) in its homotopy class.

We take the universal cover of both N and M and the corresponding lift of f , which we denote $\tilde{f} : Y \rightarrow X$. We want to intuitively *average* this map to obtain another map $\tilde{F}_s : Y \rightarrow X$. We intend to do this in an equivariant manner, so that this map descends to $F_s : N \rightarrow M$.

Denote the metric on Y by g . Then the first inclusion $\mu^s : y \mapsto \mu_y^s$ comes from localising and normalising this metric to make it into a probability measure in the Lebesgue class:

$$\frac{d\mu_y^s}{dg}(z) = \frac{e^{-sd(y,z)}}{\int_Y e^{-sd(y,z)} dg}$$

$s > h(g)$ was in fact chosen so that the denominator would not diverge.

The pushforward $\tilde{f}_*$ is clear. After that, we convolve with the Patterson-Sullivan measures $\{\nu_z\}_{z \in X}$ associated to the discrete subgroup $\pi_1(M) < \text{Iso}(X)$; since they are probability measures, and the same holds for μ_y^s , we obtain the probability measures:

$$\sigma_y^s(U) = \int_X \nu_x(U) d(\tilde{f}_* \mu_y^s)(x) \quad \text{for Borel sets } U \in \partial_\infty X$$

We conclude by finding the barycenter of such measures, as before (we need to ensure it is well-defined). The process yields:

$$\tilde{F}_s : Y \rightarrow X, \quad y \mapsto \mu_y^s \mapsto \tilde{f}_*(\mu_y^s) \mapsto \tilde{f}_*(\mu_y^s) * \nu_x \mapsto \text{bar}(\tilde{f}_*(\mu_y^s) * \nu_x)$$

and each step is equivariant, so it descends to the desired map $F_s : N \rightarrow M$. The homotopy with f is given by:

$$H_s : N \times [0, 1] \rightarrow M, \quad (n, t) \mapsto F_{s + \frac{t}{1-t}}(n)$$

where $t = 0$ corresponds to F_s and $t = 1$ to f . This comes as a consequence of $\sigma_y^s \rightarrow \nu_{\tilde{f}(y)}$ and $\tilde{F}_s(y) \rightarrow \tilde{f}(y)$ as $s \rightarrow \infty$, so $F_\infty = f$.

Since taking the barycenter consisted in minimising the Busemann functional:

$$B_{\sigma_y^s}(x) = \int_{\partial_\infty X} b_\eta(x) d\sigma_y^s(\eta)$$

and the minimum is reached for $x = \tilde{F}_s(y)$, we have the implicit equation (critical point of a functional):

$$\int_{\partial_\infty X} (db_\eta)_{\tilde{F}_s(y)}(\cdot) d\sigma_y^s(\eta) = 0$$

We can just differentiate this expression via the implicit function theorem and obtain a formula for the Jacobian. After the usage of several estimates (e.g. Hölder inequality) one arrives at ([CF01]):

$$|\text{Jac}(\tilde{F}_s)| \leq \left(\frac{s}{\sqrt{n}} \right)^n \frac{\det \left(\int_{\partial_\infty X} (db_\eta^2)_{\tilde{F}_s(y)}(\cdot) d\sigma_y^s(\eta) \right)^{1/2}}{\det \left(\int_{\partial_\infty X} D(db_\eta)_{\tilde{F}_s(y)}(\cdot, \cdot) d\sigma_y^s(\eta) \right)}$$

where $n = \dim(Y) = \dim(X)$.

When M is locally symmetric and non-positively curved, Patterson-Sullivan measures are supported on the Furstenberg boundary and we can use eigenvalue estimates of Ddb and other algebraic perks to bound:

$$|\text{Jac}(\tilde{F}_s)| \leq C \frac{\left(\det \int_K A_\eta \begin{pmatrix} 1 & 0_{1 \times (n-1)} \\ 0_{(n-1) \times 1} & 0_{(n-1) \times (n-1)} \end{pmatrix} A_\eta^* d\sigma_y^s(\eta) \right)^{1/2}}{\det \int_K A_\eta \begin{pmatrix} 0 & 0 \\ 0 & I_{(n-\text{rank}(X))} \end{pmatrix} A_\eta^* d\sigma_y^s(\eta)}$$

where $K \subset G = \text{Iso}(X)$ is the maximal compact subgroup, $A_\eta \in K$ and C depends only on dimensions and the smallest Ricci curvatures of the manifolds involved. This can be improved [CF01] to:

$$|\text{Jac}(F_s)| \leq C \left(\frac{s}{h(g_1)h(g_2) \cdots h(g_k)} \right)^n$$

where (X, g_0) is decomposed in the irreducible symmetric factors (X_i, g_i) for $i = 1, \dots, k$, and C depends only on the dimension of X . Moreover, for $s = h(g)$ the entropy can be bounded by Ricci curvatures via the Bishop-Gromov comparison theorem (which roughly states that under a lower bound in Ricci curvature $\geq (n-1)\kappa$, we get an upper bound on volume of R -balls on our manifold by volume of R -balls in the space form of constant curvature κ). $h(g_i)$ can also be bounded by Ricci curvatures. We say this gives us a *universal bound* of the Jacobian for F_s , solely dependant on dimension and smallest Ricci curvatures of M and N .

In more general cases, so restrictive bounds have not been found, and we have some knowledge about cases in which they are non-existent. But this and the applications of the method ([CF02]) should give a glimpse of how much we can infer about the geometry of the spaces using this construction, and why we hope they could prove useful as a parametrisation of Hitchin components, in order to approach Hitchin's considerations.

4 Hitchin components

In this section, which again will marginally resemble a survey, we will run through the research around Hitchin components to contextualise the main problem treated by this work and distill the technical properties that are to be employed in our construction. More concretely, we will show which properties make our Hitchin representations naturally produce measures for which there exists a barycenter.

Higher Teichmüller theory consists of studying connected components of discrete and faithful representations in character varieties. It is motivated by the fact that Teichmüller space itself is such a connected component of the representation variety $\text{Rep}(\pi_1(\Sigma), \text{PSL}(2, \mathbb{R}))$.

Definition 4.1. *A **higher Teichmüller space** is a subset of $\text{Hom}(\pi_1(\Sigma), G)/G$ composed of the union of connected components that consist fully of discrete and faithful representations.*

This definition raises some questions:

- Which groups G admit higher Teichmüller spaces in their character varieties?
- Can we also generalise by taking Σ not to be a surface and still obtain subsets of discrete and faithful representations (perhaps by relaxing the condition of the subset being a union of connected components)?
- In which cases do these spaces contain relevant geometric information, for example by being moduli of certain geometric structures?
- How do higher Teichmüller spaces relate between each other and which similarities and differences can be identified with classical Teichmüller space and among each other?

Not every group necessarily has higher Teichmüller spaces: when G is a simply connected complex Lie group, the representation variety is irreducible as an algebraic variety, hence connected, and not all representations are discrete and faithful [Wie19]. Of course, there exist groups G for which we find higher Teichmüller spaces, otherwise we would not be having this discussion. The first and main examples are spaces of maximal representations and our beloved Hitchin components. They lie in the intersection of many areas of mathematics and also were discovered through distinct perspectives and methods.

- When G is a Hermitian Lie group, the quotient G/K admits a G -invariant complex structure, so it is a Kähler manifold (Riemannian, complex and symplectic structures) with Kähler 2-form ω (this can in fact be the definition for Hermitian Lie group). We have a complete classification of Hermitian Lie groups, two of them exceptional.

In this setting, we can define the *Toledo invariant* of a representation, which is a notion of volume of the representation and can be treated in terms of integrals of Kähler forms and related to bounded cohomology and simplicial volume [KM08]. It is constant on connected components of the character variety and bounded from above by the Euler characteristic of Σ and the real rank of G .

Definition 4.2. *The representations for which the Toledo invariant attains its maximum value are called **maximal representations**.*

Turns out Teichmüller space corresponds to maximal representations $\pi_1(\Sigma) \rightarrow \text{PSL}(2, \mathbb{R})$, and, in general, maximal representations are discrete and faithful: they constitute higher Teichmüller spaces.

- As we discussed through in our own exposition, when G is split real, we have a principal embedding $\mathrm{PSL}(2, \mathbb{R}) \rightarrow G$ (up to conjugation), and a connected component containing G -Fuchsian representations $\pi_1(\Sigma) \rightarrow \mathrm{PSL}(2, \mathbb{R}) \rightarrow G$ is composed by discrete and faithful representations. These are then higher Teichmüller spaces.

A remark is in order here: we first defined **Hitchin components** the way we have just defined higher Teichmüller spaces, since we assumed we were not yet familiar with the notion of split real Lie groups and wanted to give just a broad idea in the introduction. The accurate way would be to define higher Teichmüller space as a space of discrete and faithful representations, and Hitchin component as the higher Teichmüller space obtained in the split real regime by looking at components with G -Fuchsian representations.

Hitchin first proved the existence of these components (hence its nomenclature) using Higgs bundles theory, through the so called non-abelian Hodge correspondence. Let us briefly sketch the main idea (for more information, see [Tho22]). We skip most details since this deviates from our main goals, but the hope is to motivate the reader to research further or just highlight the relevant characters in the theory.

The underlying principle is to use PDE techniques to construct representations of fundamental groups in complex geometry. To one of our representations of $\pi_1(\Sigma)$ we can associate a vector bundle $E \rightarrow \Sigma$ with a connection ∇ whose curvature F_∇ (measuring its non-commutativity) vanishes identically:

$$0 = F_\nabla(X, Y) := [\nabla_X, \nabla_Y] - \nabla_{[X, Y]}$$

Such connection is called flat and this expression yields a differential equation. The quotient by conjugation in the representations corresponding to the character variety will amount to an equivalence of flat connections modulo gauge, i.e. modulo bundle automorphisms. This is contained in:

Theorem 4.3 (Riemann-Hilbert correspondence).

$$\mathrm{Rep}(\pi_1(\Sigma), G) \cong \{\text{flat } G\text{-connections over } \Sigma\} / \text{gauge}$$

In each gauge class we want to find and select a harmonic representative, again given by an equation: for a bundle $\nabla = d + \psi$ we require $d^*\psi = 0$. For this we need to restrict our scope of representations and focus on those which are *completely reducible* (invariant subspaces admit an invariant complement):

Theorem 4.4 (Corlette-Donaldson).

$$\mathrm{Rep}^{c.r.}(\pi_1(\Sigma), G) \cong \{\text{flat } G\text{-bundles}\} / \text{gauge} \cong \{\text{harmonic flat } G\text{-bundles}\} / \text{constant}$$

On the other hand, we have Higgs bundles, a purely holomorphic object in contrast with the topological/representation-theoretical nature of the elements of the character variety:

Definition 4.5. A **Higgs bundle** is a pair (E, ϕ) consisting of a holomorphic vector bundle $E \rightarrow \Sigma$ and a **Higgs field** ϕ , i.e. an $\mathrm{End}(E)$ -valued holomorphic $(1, 0)$ -form such that $\phi \wedge \phi = 0$.

We say it is **stable** if for any ϕ -invariant holomorphic subbundle $F \subset E$ we have:

$$\mu(F) < \mu(E), \quad \text{where} \quad \mu(E) = \frac{\deg(E)}{\mathrm{rk}(E)}$$

and **polystable** if it is a direct sum of stable Higgs bundles with the same quotient as above. The degree in the quotient is given by the first Chern class.

We also want to consider the moduli space of Higgs bundles modulo the gauge group of G -valued functions $\mathcal{G} = \Omega^0(\Sigma, G)$ and find a harmonic representative. To every Higgs bundle one can associated a flat connection ∇_A , the Chern connection related to a hermitian metric on E . Then a harmonic Higgs bundle (E, ϕ) is one equipped with a hermitian metric such that the **Hitchin equation**:

$$F_{\nabla_A} + [\phi, \phi^*] = 0$$

holds (this is our main PDE, which is linked with the Yang-Mills equation). One can show that there is a unique representative in a gauge orbit for the polystable case of degree 0:

Theorem 4.6 (Hitchin-Simpson).

$$\{\text{polystable Higgs bundles of degree 0}\}/\text{gauge} \cong \{\text{harmonic Higgs bundles}\}/\text{unitary gauge}$$

And now the trick is to relate harmonic representatives in both regimes:

Theorem 4.7 (non-abelian Hodge correspondence). *The moduli space of polystable Higgs bundles of degree 0 is diffeomorphic to the character variety of completely reducible representations $\pi_1(\Sigma) \longrightarrow GL(n, \mathbb{C})$.*

For d_A the Chern connection, the correspondence (flat-Higgs, Higgs-flat) is:

$$(E, \nabla = d_A + \psi, h) \mapsto (E, \bar{\partial}_E = d_A^{0,1}, \phi = \psi^{1,0}, h), \quad (E, \bar{\partial}_E, \phi, h) \mapsto (E, \nabla = d_A + \phi + \phi^{*h}, h)$$

The name tries to generalise a correspondence of cohomology theories given by the Hodge correspondence: Dolbeault, de Rham and topological (simplicial-singular-cellular) cohomologies. A similar picture can be drawn in our case: Higgs bundles, flat connections and representations of the fundamental group, respectively.

Finally, this paradigm brings us to a parametrisation of Hitchin components, since we can show that the Higgs bundles of the form:

$$\phi = \begin{pmatrix} 0 & t_2 & t_3 & \cdots & t_{n-1} & t_n \\ 1 & 0 & t_2 & \cdots & t_{n-2} & t_{n-1} \\ 0 & 1 & 0 & \cdots & t_{n-3} & t_{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & t_2 \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

correspond to representations satisfying a certain reality condition and ultimately landing on $\mathrm{PSL}(n, \mathbb{R})$, so constituting a subset of the complex linear character variety that identifies with the Hitchin component of the linear split real groups (and moreover this procedure generalises to the other adjoint groups of split real forms of simple complex Lie algebras).

The t_j are certain holomorphic sections and can be interpreted as elements of $H^0(\Sigma, (T^*\Sigma)^j)$, and so the parametrisation for $\mathrm{PSL}(n, \mathbb{R})$ is via $\mathrm{Hit}_n(\Sigma) \cong \bigoplus_{j=2}^n H^0(\Sigma, (T^*\Sigma)^j)$.

Parametrisations are then associated to harmonic maps which are solutions of differential equations, from which we cannot extract much information. Nonetheless, as shown in [Cor88], they have the same form as the ones we will construct with the barycenter method: we will embed the Hitchin component of G into a space of maps between hyperbolic plane and the symmetric space G/K modulo isometries. Corlette's description uses harmonic maps as optimal maps, but we will use the barycenter maps that are also in some sense optimal within their class.

It is interesting to know that for groups that are simultaneously split and Hermitian, the Hitchin component is contained in the space of maximal representations (i.e. Hitchin representations have maximal Toledo invariant, but not the other way around). Teichmüller space is both a Hitchin component and a space of maximal representations.

The work of Hitchin did not establish Hitchin components as higher Teichmüller spaces in the discrete-faithful sense: this development had to wait until the work in Anosov representations of Labourie [Lab04] and in positive representations of Fock-Goncharov [FG06].

Anosov representations were the conducting thread that brought together the worlds of maximal representations and Hitchin components; they are defined in terms of a specific dynamical behavior. They allow us to construct unique continuous boundary maps:

- If $\rho : \pi_1(\Sigma) \longrightarrow G$ is Hitchin, we have a ρ -equivariant boundary map $F_\rho : \partial_\infty \mathbb{H}^2 \cong \mathbb{S}^1 \longrightarrow G/B \cong \partial_F X$ into the generalised flag variety with special properties (transversality, as in 2.34), where $X = G/K$.
- If $\rho : \pi_1(\Sigma) \longrightarrow G$ is maximal, the situation is similar but the target is G/P_S , where P_S is a maximal parabolic subgroup fixing a point in the Shilov boundary S of X (so still into a generalised flag variety and with transversality properties). The idea behind the Shilov boundary is to make X correspond to a bounded symmetric domain D (*Harish-Chandra embedding*), and defining S as the points in ∂D for which for all holomorphic functions f on D , continuous on $\overline{D}$, we get $|f(x)| \leq \max_{y \in S} |f(y)|$ for all $x \in D$.

The notion of positive representations was a step forward in generalising the theory and allowing extensions of it to other classes of groups. The latest advances in this direction are θ -Anosov and θ -positive representations, which ensure the existence of higher Teichmüller spaces for groups admitting a positive structure [GLW21]. This also includes new groups such as the $\mathrm{SO}(p, q)$ for $p \neq q$, not belonging to the families above.

Positivity is actually defined in terms of boundary maps in flag varieties as above, and how they transform triples and quadruples of points (*they send positive triples into positive triples*). The motivating notion for considering such boundary maps appears in [Lab04] and is based on what's called Frenet hyperconvex curves.

In [CG93], we had a explicit description of Hitchin components for $\mathrm{PSL}(3, \mathbb{R})$ as holonomies of convex real projective structures on Σ , which basically means that for such a representation ρ , we have an open set $A \subset \mathbb{RP}^2$ such that $A/\rho(\pi_1(\Sigma)) \cong \Sigma$. As a result, ρ preserves the (C^1 convex) boundary curve of A . For $\mathrm{PSL}(n, \mathbb{R})$ representations, this motivates:

Definition 4.8. A curve $F : \mathbb{S}^1 \longrightarrow \mathbb{RP}^n$ is called **Frenet hyperconvex** if for $n_1, \dots, n_k \in \mathbb{N}$ and distinct points $x_1, \dots, x_k \in \mathbb{S}^1$ such that $\sum_i n_i \leq n + 1$, there exists a family of maps $(F^1, \dots, F^n)$ with $F^j \subset F^{j+1}$, called **osculating flag**, satisfying:

- F^j takes values on $\mathrm{Gr}_k(\mathbb{R}^n)$ and $F^1 = F$.
- *Transversality:* the sum $\bigoplus_j F^{n_j}(x_j)$ is direct.
- *Continuity:* $\lim_{x_j \rightarrow x} \bigoplus_j F^{n_j}(x_j) = F^{\sum_j n_j}(x)$ for x_j distinct.

F^1 determines F^j by the definition, to the point that if F is smooth, these are generated by the derivatives of F at x up to order $j - 1$. In general, it is not smooth, but its image is a C^1 submanifold. In the case of $\mathrm{PSL}(n, \mathbb{R})$ -Fuchsian representations, this curve corresponds to the Veronese embedding $\mathbb{RP}^1 \longrightarrow \mathbb{RP}^n$, which is $\mathrm{PSL}(2, \mathbb{R})$ -equivariant.

The link to our problem comes through:

Definition 4.9. We say a representation $\rho : \pi_1(\Sigma) \longrightarrow PSL(n, \mathbb{R})$ is **B-Anosov** (B referring to the Borel subgroup of $PSL(n, \mathbb{R})$) if there are constants $C, D > 0$ such that:

$$\frac{\lambda_{k+1}(\rho(\gamma))}{\lambda_k(\rho(\gamma))} \leq Ce^{-D|\gamma|} \quad \text{for all } \gamma \in \pi_1(\Sigma), \quad 1 \leq k \leq n-1$$

where $\lambda_1(\rho(\gamma)) \geq \dots \geq \lambda_n(\rho(\gamma))$ are the ordered eigenvalues of $\rho(\gamma)$.

Such a representation admits an equivariant flag curve with transversality for pairs of points. Moreover:

Definition 4.10. A representation $\rho : \pi_1(\Sigma) \longrightarrow PSL(n, \mathbb{R})$ is called **hyperconvex** if it is B-Anosov and its (unique) associated flag curve $F_\rho : \partial_\infty \pi_1(\Sigma) \longrightarrow \text{Flag}(\mathbb{R}^n)$ is ρ -equivariant and is the osculating flag of a Frenet hyperconvex curve:

- *Equivariance:* $F_\rho(\gamma t) = \rho(\gamma)F_\rho(t)$ for all $\gamma \in \pi_1(\Sigma)$ and $t \in \partial_\infty \pi_1(\Sigma)$.
- *Transversality:* for $t_i \in \partial_\infty \pi_1(\Sigma) \cong \mathbb{S}^1$, $n_i \in \mathbb{N}$ with $\sum_i n_i \leq n$ we have $\bigoplus_j F^{n_i}(t_i)$ is a direct sum (in particular, if $\sum_i n_i = n$, then the direct sum equals $\mathbb{R}^n$).
- *Continuity:* for $t_i \neq t \in \partial_\infty \pi_1(\Sigma)$, $n_i \in \mathbb{N}$ with $\sum_i n_i \leq n$ we have $\lim_{t_i \rightarrow t} \bigoplus_i F^{n_i}(t_i) = F^{\sum_i n_i}(t)$.

The crux of this definition lies in the following proposition, implied by [FG06] and [Lab04]:

Proposition 4.11. Every representation in the Hitchin component of $PSL(n, \mathbb{R})$ is hyperconvex. It moreover acts irreducibly on $\mathbb{R}^n$.

Remark 4.12. The Milnor-Schwarz lemma tells us that a group acting nicely and discretely by isometries on a metric space is quasi-isometric to that space. Being quasi-isometric then implies that the boundaries at infinity are isomorphic.

It is the case that, given a Fuchsian representation $\theta : \pi_1(\Sigma) \longrightarrow PSL(2, \mathbb{R})$, the orbit map $\gamma \in \pi_1(\Sigma) \longrightarrow \theta(\gamma)z \in \mathbb{H}^2$ for any $z \in \mathbb{H}^2$ is a quasi-isometry between $\pi_1(\Sigma)$ and $\mathbb{H}^2$, so $\partial_\infty \pi_1(\Sigma) \cong \partial_\infty \mathbb{H}^2 \cong \mathbb{S}^1$, illustrating how F_ρ can be seen as a Frenet hyperconvex curve.

Besides, recall that for $G = PSL(n, \mathbb{R})$ we get $X = G/K = P(n, \mathbb{R})$ and the space of flags is identified with the Furstenberg boundary: $\text{Flag}(\mathbb{R}^n) \cong G/B \cong \partial_F P(n, \mathbb{R}) \subset \partial_\infty P(n, \mathbb{R})$

Thus, the flag curve of a hyperconvex representation can be seen as a map between the boundaries of two symmetric spaces: $F_{\theta, \rho} : \partial_\infty \mathbb{H}^2 = \partial_\infty P(2, \mathbb{R}) \longrightarrow \partial_\infty X = \partial_\infty P(n, \mathbb{R})$.

As we mentioned in the previous section, this is the map that we were missing in the application of the barycenter method to our problem, the map that would allow us to push forward the measures at infinity.

In the next section we will use these properties of $PSL(n, \mathbb{R})$ -Hitchin representations to deduce more about other split real groups. In particular, recall how these were subgroups of it and the boundaries and flags at infinity embedded nicely (2.39). The same way we can describe the Furstenberg boundary of $P(n, \mathbb{R})$ as the space of full flags, we obtain other characterisations of flag spaces, and so there is an analogous description of hyperconvexity of representations as flag curves with target G/B . The extension for any group would be to consider a map into the generalised flag variety G/P_θ , for P_θ parabolic (recall B is minimal parabolic, so fits in the description), and this leads us back to θ -positive representations [GLW21].

- As we have seen, for $G = PSL(n, \mathbb{R})$ the Borel subgroup $B \subset G$ is a stabiliser of a full flag in $\mathbb{R}^n$.

- For $G' = \mathrm{SO}(n, n+1)$, the Borel subgroup $B' = B \cap G' \subset G'$ corresponds again to its upper triangular elements. Denote by $\mathbb{R}^{p,q}$ the vector space $\mathbb{R}^{p+q}$ together with the standard indefinite symmetric bilinear form of signature (p, q) , i.e. $I_{p,q}$. The space G'/B' is:

$$\{F = (F^1, \dots, F^{2n+1}) \mid F^k \in \mathrm{Gr}_k(\mathbb{R}^{n,n+1}), \quad F^k \subset F^{k+1}, \quad F^{2n+1-k} = (F^k)^\perp\}$$

These are the full flags in $\mathbb{R}^{2n+1}$ such that the subspaces of dimension less than half are *isotropic* (meaning the bilinear form vanishes on them) and the ones greater than half are their orthogonals with respect to the bilinear form, so determined by the first ones. This is a homogeneous space because G' acts transitively on $\mathrm{Gr}_k(\mathbb{R}^{n,n+1})$.

- For $G'' = \mathrm{PSp}(2n, \mathbb{R})$, we consider $\mathbb{R}^{2n}$ with the standard symplectic form ω , and now we have a transitive action of G'' on its Grassmanians, resulting in the flag variety G''/B'' for $B'' = B \cap G'' \subset G''$ given by the full flags in $\mathbb{R}^{2n}$ which are isotropic with respect to ω for dimension less than n , and orthogonal to those for dimension greater than n (again $F^{2n-k} = (F^k)^\perp_\omega$).
- For $G''' = \mathrm{PSO}(n, n)$ we could expect a similar picture for $\mathbb{R}^{n,n}$, but here the action on $\mathrm{Gr}_n(\mathbb{R}^{n,n})$ has two orbits: positive and negative isotropic n -planes. Choose a basis $\{e_1, \dots, e_{2n}\}$ and an n -plane $P \in \mathrm{Gr}_n(\mathbb{R}^{n,n})$, with basis $\{u_1, \dots, u_n\}$. This extends to a basis $\{v_1, \dots, v_n\}$, which spans another n -plane. We compute:

$$\frac{u_1 \wedge \dots \wedge u_n \wedge v_1 \wedge \dots \wedge v_n}{e_1 \wedge \dots \wedge e_{2n}}$$

If it is $+1$, we say P is positive. If it is -1 , we say P is negative. We denote these components by $\mathrm{Gr}_n^+(\mathbb{R}^{n,n})$ and $\mathrm{Gr}_n^-(\mathbb{R}^{n,n})$. Now, G'''/B''' consists of flags of the form:

$$F^1 \subset \dots \subset F_+^n, F_-^n \subset \dots \subset F^{2n-1}, \quad \text{so that}$$

$$F^k \in \mathrm{Gr}_k(\mathbb{R}^{n,n}) \quad \& \quad F^{2n-k} = (F^k)^\perp \quad \text{for } k \neq n; \quad F_\pm^n \in \mathrm{Gr}_n^\pm(\mathbb{R}^{n,n})$$

Here $F^1 \subset \dots \subset F^{n-1}$ already determine the whole flag F .

One can try to investigate many angles into higher Teichmüller spaces: number of components, geometric structures that they might parametrise, quotients by the action of the mapping class group and its relation to the moduli space of Riemann surfaces, dynamical properties, entropy, compactifications, sets of coordinates (e.g. generalising Fenchel-Nielsen coordinates on finite type Teichmüller spaces), the principles behind the existence of components of discrete and faithful representations when we change Σ by a higher dimensional manifold (perhaps asking for compactness, hyperbolicity, and so on), etc.

In conclusion, higher Teichmüller theory is a rich subject, still mostly uncharted, that builds bridges between many areas and also full of recent ideas and extensive collaboration. There are many fronts that can be explored, and the developments suggest that they would have important implications in relevant topics in mathematics and theoretical physics (via mathematical physics, e.g. constructing non-perturbative quantum theories).

5 Geometric parametrisation

There is no real ending. It's just the place where you stop the story.

FRANK HERBERT

The stage is set, the curtains up and the lights on. It is time we present our core statements. We will start by showing the procedure for the barycentric parametrisation in $\mathrm{PSL}(n, \mathbb{R})$ in the style of [Sle21], to then broaden the scope of this result to other split real groups. Afterwards, we will comment on possible lines of further study.

5.1 Slegers parametrisation and its extensions

Fix a Fuchsian representation $\theta : \pi_1(\Sigma) \rightarrow \mathrm{PSL}(2, \mathbb{R})$. This induces a θ -equivariant identification $\partial_\infty \pi_1(\Sigma) \cong \partial_\infty \mathbb{H}^2$ as in 4.12 (this actually comes from the osculating flag $\partial_\infty \pi_1(\Sigma) \rightarrow \mathrm{Flag}(\mathbb{R}^2) \cong \mathbb{S}^1$).

Choose a B -Anosov representation $\rho : \pi_1(\Sigma) \rightarrow \mathrm{PSL}(n, \mathbb{R})$ with flag curve $F_{\theta, \rho} : \partial_\infty \pi_1(\Sigma) \cong \partial_\infty \mathbb{H}^2 \rightarrow \mathrm{Flag}(\mathbb{R}^n) \cong \partial_F X \subset \partial_\infty X$, with $X = G/K = P(n, \mathbb{R})$. The dependence on θ comes from the identification of boundaries at infinity in the source, so as to obtain a map between boundaries of symmetric spaces.

Consider the visual measures $\{\nu_z\}_{z \in \mathbb{H}^2} \subset \mathcal{M}(\partial_\infty \mathbb{H}^2)$, and push them forward using $F_{\theta, \rho}$, defining $\mu_z^{\theta, \rho} := (F_{\theta, \rho})_* \nu_z \in \mathcal{M}(\partial_\infty X)$.

The key statement is:

Theorem 5.1 (Slegers). *If $\rho : \pi_1(\Sigma) \rightarrow \mathrm{PSL}(n, \mathbb{R})$ is Hitchin, then the measure $\mu_z^{\theta, \rho}$ is stable for all $z \in \mathbb{H}^2$.*

Being in the Hitchin component is a necessary condition:

Example 5.2. *Consider the representation $\rho : \pi_1(\Sigma) \rightarrow \mathrm{PSL}(2, \mathbb{R}) \times \mathbb{R} \rightarrow \mathrm{PSL}(3, \mathbb{R})$, where the first map is given by θ on the first factor and constant equal to zero in the second, and the second map is the inclusion:*

$$(A, t) \mapsto \begin{pmatrix} e^t \cdot A & 0 \\ 0 & e^{-2t} \end{pmatrix}$$

Such a map satisfies the eigenvalue B -Anosov definition, but does not yield a stable measure (since it is not irreducible, so not Hitchin): the boundary map associated to the first arrow has to land in $\partial_\infty \mathbb{H}^2 \subset \partial_\infty (\mathbb{H}^2 \times \mathbb{R})$, so it is fixed by translations $(z, t) \mapsto (z, t + s)$. This means the weighted Busemann function for a pushforwarded measure is also invariant under these translations and thus its minimiser cannot be unique.

To prove 5.1, we need the notion of intersection type of a subspace and a flag:

Definition 5.3. *The **intersection type** of a k -plane $U \in \mathrm{Gr}_k(\mathbb{R}^n)$ and a flag $F \in \mathrm{Flag}(\mathbb{R}^n)$ is defined as the n -tuple:*

$$\mathrm{type}_k(U, F) = (s_0, \dots, s_n)$$

where $s_i = \dim(U \cap F^i)$.

Note that a tuple is an intersection type if and only if $s_0 = 0$, $s_n = k$ and $s_{i+1} = s_i$ or $s_{i+1} = s_i + 1$.

We use this concept to reduce the proof to ruling out the existence of a *bad vertex* at infinity. Let us analyse the formula for the slope 3.4 (denote $F_{\theta,\rho}(t) = \xi(t)$ for $t \in \partial_\infty \mathbb{H}^2$):

$$\text{slope}_{\mu_z^{\theta,\rho}}(\eta) = \int_{\partial_\infty X} -\cos \angle(\eta, \xi(t)) d\mu_z^{\theta,\rho}(\xi(t)) = \int_{\partial_\infty \mathbb{H}^2} -\cos \angle(\eta, \xi(t)) d\nu_z(t)$$

Transversality from the flag curve means that any two $\xi(t)$ and $\xi(s)$ can be joined by a geodesic in X (see 2.34), so $\angle(\xi(t), \xi(s)) = \pi$ for all $t \neq s \in \partial_\infty \mathbb{H}^2$.

If we had $\angle(\eta, \xi(t)) < \pi/2$, then for all $s \neq t$, the triangle inequality would mean:

$$\angle(\eta, \xi(s)) \geq \angle(\xi(t), \xi(s)) - \angle(\eta, \xi(t)) > \pi - \pi/2 = \pi/2$$

so at most for one $t \in \partial_\infty \mathbb{H}^2$ the inequality $\angle(\eta, \xi(t)) < \pi/2$ could be verified. The argument of the integral is thus non-negative (a cosine between $\pi/2$ and π and ν_z has no atoms), so the slope is non-negative.

What's more, ξ is continuous and $\angle$ is lower semicontinuous, so $t \mapsto -\cos \angle(\eta, \xi(t))$ is lower semicontinuous (similar argument as for 2.20). Since visual measures share class with Lebesgue measures, slope can only be 0 precisely if $\angle(\eta, \xi(t)) = \pi/2$ for all $t \in \partial_\infty \mathbb{H}^2$. We want to avoid then the η that are at angle $\pi/2$ with all the $\xi(t)$.

Furthermore, 3.8 suggests that we just check this for $\eta \in \partial_\infty X$ that are vertices of Weyl chambers. In conclusion: we have to show that there does not exist any vertex η such that $\angle(\eta, \xi(t)) = \pi/2$ for all $t \in \mathbb{H}^2$.

By our previous discussion, vertices correspond to k -planes and barycenters of Weyl chambers to full flags. Thus:

Proposition 5.4. *If the vertex $\eta \in \partial_\infty X$ corresponds to a k -plane $U \in \text{Gr}_k(\mathbb{R}^n)$ and $\xi \in G \cdot b$ is the barycenter of a Weyl chamber corresponding to a flag $F \in \text{Flag}(\mathbb{R}^n)$, then $\angle(\eta, \xi) = \pi/2$ if and only if:*

$$\sum_{i=1}^n s_i = \frac{(n+1)k}{2}, \quad \text{where} \quad \text{type}_k(U, F) = (s_0, \dots, s_n)$$

Proof. Find a flat F containing both η and the Weyl chamber of ξ at infinity. Without loss of generality (angles at infinity and intersection types are invariant under isometries of X), suppose $I \in F$, so the angle is realised as $\angle(\eta, \xi) = \angle_I(\eta, \xi)$ (similar to 2.12).

We denote $T_I F = \mathfrak{h}$, the Cartan subalgebra of $\text{PSL}(n, \mathbb{R})$, i.e. trace 0 diagonal matrices, so identified with $\{(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n \mid \sum_i \lambda_i = 0\}$. For the standard basis $(e_i)_{i=1}^n$ of $\mathbb{R}^n$, the vector $X_i = (1, \dots, 1, \underbrace{n-1}_{i\text{-th place}}, 1, \dots, 1)$ can be normalised to represent the vertex at infinity associated to

$\langle e_i \rangle \in \text{Gr}_1(\mathbb{R}^n)$. Moreover, for the k -multi-index $I = \{i_1, \dots, i_k\} \subset \{1, \dots, n\}$, the vector $X_I = \sum_{j=1}^k X_{i_j}$ normalises to a vertex corresponding to $\langle e_{i_1}, \dots, e_{i_k} \rangle \in \text{Gr}_k(\mathbb{R}^n)$ (recall the discussion after 2.34).

The scaling of the inner product does not affect the angle, so we use these vectors directly as if they had norm 1. Moreover, the inner product is defined in terms of the Killing form and is invariant under the Weyl group, so $\langle X_i, X_j \rangle = \langle X_1, X_2 \rangle$ for all $i \neq j$. $\sum_{i=1}^n X_i = 0$ because it is invariant under the Weyl group. All together yields:

$$0 = \langle \sum_i X_i, \sum_i X_i \rangle = n\|X_1\|^2 + n(n-1) \langle X_1, X_2 \rangle \implies \langle X_i, X_j \rangle = \frac{-1}{n-1}, \quad i \neq j$$

$$\implies \langle X_I, X_J \rangle = |I||J| \frac{-1}{n-1} + |I \cap J| \left(1 + \frac{1}{n-1}\right) = \frac{n|I \cap J| - k \cdot l}{n-1}$$

(for I a k -multi-index and J an l -multi-index) by counting all pairs and taking into account that if $X_{i_a} = X_{j_b}$, then we do not count it as a $\langle X_1, X_2 \rangle$ but as a $\|X_1\| = 1$.

As before 2.33, the barycenter can be written as $b = (n-1, n-3, \dots, 3-n, 1-n)$ up to scaling, and acting by an element of the Weyl group we can assume $\xi = b$. The vertex η is associated to a k -plane U spanned by vectors $(e_i)_{i \in I}$ for a k -multi-index I , so $\eta = X_I$. Thus, $\angle(\eta, \xi) = \angle(b, X_I)$.

For the calculation, use $b = \frac{2}{n} \sum_{i=1}^{n-1} (n-i)X_i$, since analysing component-wise:

$$\begin{aligned} \left(\sum_{i=1}^{n-1} (n-i)X_i \right)_k &= \underbrace{(n-1)(n-k)}_{\text{contribution from the } X_k} + \underbrace{(-1) \sum_{i \neq k} (n-i)}_{\text{contribution from the } X_{i \neq k}} = \\ &= (n-1)(n-k) + (n-k) - \sum_i (n-i) = n(n-k) - \frac{n(n-1)}{2} = \frac{n(n-2k+1)}{2} = \frac{n}{2} b_k \end{aligned}$$

Now, $s_i = \dim(U \cap F^i) = |I \cap \{1, \dots, i\}|$, so using the above:

$$\begin{aligned} \angle(\xi, \eta) = \angle(b, X_I) = \frac{\pi}{2} &\iff 0 = \frac{n}{2} \langle b, X_I \rangle = \sum_{i=1}^{n-1} \langle X_1 + \dots + X_i, X_I \rangle = \\ &= \sum_{i=1}^{n-1} \frac{ns_i - ki}{n-1} = \frac{n}{n-1} \sum_{i=1}^{n-1} s_i - \frac{kn}{2} \iff \sum_{i=1}^{n-1} s_i = \frac{(n-1)k}{2} \end{aligned}$$

We just need to add $s_n = k$ to culminate the proof. $\square$

With this, the worst case scenario translates to:

Proposition 5.5. *With the above setup, if $\angle(\eta, \xi(t)) = \pi/2$ for all $t \in \mathbb{S}^1$, then $\text{type}_k(U, F(t))$ is constant in t .*

Proof. Consider the projection onto the orthogonal complement $\pi : \mathbb{R}^n \longrightarrow U^\perp$. For a continuous family of p -planes $\{V_t\}_{t \in \mathbb{S}^1}$ in $\mathbb{R}^n$, we can choose around $t \in \mathbb{S}^1$ a continuous family of linear maps $A_t : \mathbb{R}^p \longrightarrow \mathbb{R}^n$ such that $V_t = \text{im}(A_t)$. The rank of a continuous family of maps is lower semicontinuous (if we have a point $t \in \text{rk}^{-1}(c, \infty)$ and the rank of the map parametrised by t is $k > c$, then it has a $k \times k$ submatrix with non-zero determinant; continuity of the determinant gives us a neighborhood U containing t such that for $s \in U$ the corresponding $k \times k$ submatrix for s has determinant non-zero, so the linear map parametrised by s has rank at least $k > c$, i.e. $U \subset \text{rk}^{-1}(c, \infty)$), so $\dim(\pi(V_t)) = \text{rk}(\pi \circ A_t)$ is lower semicontinuous, which means $t \mapsto \dim(U \cap V_t) = p - \dim(\pi(V_t))$ is upper semicontinuous.

For $t \in \mathbb{S}^1$, we set $(s_0(t), \dots, s_n(t)) = \text{type}_k(U, F(t))$. By upper semicontinuity, for a fixed $t_0 \in \mathbb{S}^1$ we get a neighborhood W of it such that $s_i(t) \leq s_i(t_0)$ for all $t \in W$. To satisfy the equation in the previous proposition, this means $s_i(t) = s_i(t_0)$ for all i , so being a certain intersection type is locally constant, so constant for all $t \in \mathbb{S}^1$ because $\mathbb{S}^1$ is connected. $\square$

Lastly, we show no *bad vertex* can appear, effectively proving 5.1:

Proposition 5.6. *For a Hitchin representation $\rho : \pi_1(\Sigma) \longrightarrow \text{PSL}(n, \mathbb{R})$ with flag curve $F : \partial_\infty \pi_1(\Sigma) \longrightarrow \text{Flag}(\mathbb{R}^n)$ and associated boundary map $\xi : \mathbb{S}^1 \longrightarrow \partial_F X \subset \partial_\infty X$, there is no vertex $\eta \in \partial_\infty X$ that satisfies $\angle(\eta, \xi(t)) = \pi/2$ for all $t \in \mathbb{S}^1$.*

Proof. Suppose such vertex exists and recover our previous notations. We can be only in one of four cases:

- If $s_1 = 1$, then $\dim(U \cap F^1(t)) = 1$, so $F^1(t) \subset U$ for all $t \in \mathbb{S}^1$. But this contradicts transversality in the osculating flag:

$$\mathbb{R}^n = \bigoplus_{j=1}^n F^j(t_j) \subset U$$

but $\dim(U) = k < n$ (otherwise it would not represent a vertex).

- If $s_i = k$ for $i < n$, then $U \subset W = \bigcap_{t \in \mathbb{S}^1} F^i(t) \neq \mathbb{R}^n$ since $\dim(F^i(t)) < n$. Then W is non-empty and ρ -invariant by equivariance of F , contradicting irreducibility of ρ .
- If there exists i, j such that $i + j \leq n$ and $s_i + s_j > k$, for $t \neq t'$ we have $\dim(U \cap F^i(t)) + \dim(U \cap F^j(t')) = s_i + s_j > k = \dim(U)$, so $(U \cap F^i(t)) \cap (U \cap F^j(t')) \neq \emptyset$, and this contradicts the fact that the sum $F^i(t) \oplus F^j(t')$ is direct, as should correspond to a osculating flag.
- The rest of the cases have to satisfy $s_1 = 0$, $s_i < k$ for $i < n$ and $s_i + s_j \leq k$ for $i + j \leq n$. We will check that this contradicts 5.4.

For n odd, use $s_1 = 0$, $s_{n-1} < k$ and $s_i + s_{n-1} \leq k$ to show:

$$\sum_{i=1}^n s_i = s_1 + s_{n-1} + s_n + \sum_{i=2}^{n-2} s_i < k + k + \sum_{i=2}^{(n-1)/2} (s_i + s_{n-i}) \leq 2k + \left(\frac{n-1}{2} - 1\right) k = \frac{(n+1)k}{2}$$

For n even, use furthermore that $2s_{n/2} \leq k$, and compute:

$$\sum_{i=1}^n s_i = s_1 + s_{n-2} + s_{n-1} + s_n + \sum_{i=2}^{n/2-1} s_i < \frac{k}{2} + k + k + \left(\frac{n}{2} - 1\right) k = \frac{(n+1)k}{2}$$

□

Because no bad vertex exists, our analysis of the slope yields a strictly positive value, so the measure is stable. This means we can apply the barycenter method in the setting of Hitchin representations:

Corollary 5.7. *Given a Hitchin representation $\rho : \pi_1(\Sigma) \rightarrow PSL(n, \mathbb{R})$, the natural map:*

$$f_{\theta, \rho} : \mathbb{H}^2 \longrightarrow X, \quad z \mapsto \text{bar}(\mu_z^{\theta, \rho})$$

is well defined, smooth on z , ρ and θ and intertwines θ and ρ .

Proof. • It is well-defined because the measure is stable.

- If $\eta \in \partial_\infty X$ and $g \in G$:

$$b_{g\eta}(x) = \lim_{t \rightarrow \infty} d(x, gc(t)) - t = \lim_{t \rightarrow \infty} d(g^{-1}x, c(t)) - t = b_\eta(g^{-1}x) = (b_\eta \circ g^{-1})(x)$$

where $c(\infty) = \eta$ and in the calculation we might lose our chosen normalisation of Busemann functions, but this is fine since they are well-defined up to a global constant for our purposes. By the definition of pushforward measure:

$$B_{g_*\mu}(x) = \int_{\partial_\infty X} b_\eta(x) d(g_*\mu)(\eta) = \int_{\partial_\infty X} b_\eta(x) d\mu(g^{-1}\eta) = \int_{\partial_\infty X} b_{g\eta}(x) d\mu(\eta) =$$

$$= \int_{\partial_\infty X} b_\eta(g^{-1}x) d\mu(\eta) = B_\mu(g^{-1}x) = (B_\mu \circ g^{-1})(x)$$

Then, $\text{bar}(g_*\mu) = g\text{bar}(\mu)$. Since $F_{\theta,\rho} : \partial_\infty \mathbb{H}^2 \longrightarrow \partial_\infty X$ is also equivariant and same for the visual measures ν_z :

$$\begin{aligned} f_{\theta,\rho}(\theta(\gamma)z) &= \text{bar}((F_{\theta,\rho})_*\nu_{\theta(\gamma)z}) = \text{bar}((F_{\theta,\rho})_*(\theta(\gamma))_*\nu_z) = \text{bar}((\rho(\gamma))_*(F_{\theta,\rho})_*\nu_z) = \\ &= \rho(\gamma)\text{bar}((F_{\theta,\rho})_*\nu_z) = \rho(\gamma)f_{\theta,\rho}(z) \end{aligned}$$

for all $\gamma \in \pi_1(\Sigma)$ and $z \in \mathbb{H}^2$, so it intertwines θ and ρ .

- We will just define what smoothness means in this context and will shun the proof (it can be found in [Sle21]), to avoid having to take care of technicalities that escape our purposes.

The fundamental group of a surface Σ of genus g is generated by $2g$ elements $\gamma_i \in \pi_1(\Sigma)$. For an open $\Omega \subset \mathbb{R}^m$, we say the family of representations $(\rho_u)_{u \in \Omega}$ of $\pi_1(\Sigma)$ into G is **smooth** if the map $u \in \Omega \mapsto \rho_u(\gamma_i) \in G$ is smooth for every $i = 1, \dots, 2g$.

The smoothness in our case then means that for a family of Fuchsian representations $(\theta_u)_{u \in \Omega_1}$ and Hitchin representations $(\rho_v)_{v \in \Omega_2}$, the map:

$$f : \Omega_1 \times \Omega_2 \times \mathbb{H}^2 \longrightarrow, \quad (u, v, z) \mapsto \text{bar}(\mu_z^{u,v})$$

is smooth. The proof of this relies on the smoothness of weighted Busemann functions $B_{\mu_z^{u,v}}$ on $(u, v, z) \in \Omega_1 \times \Omega_2 \times \mathbb{H}^2$. □

If our Hitchin representation is Fuchsian: $\rho : \pi_1(\Sigma) \longrightarrow \text{PSL}(2, \mathbb{R})$, then the map $f_{\theta,\rho}$ is the one given by [DE86]. If $\theta = \rho$, the flag curve is the identity and $\text{bar}(\nu_0) = \text{bar}(\mathcal{L}/2\pi) = 0$, and similarly $\text{bar}(\nu_z) = z$ for all $z \in \mathbb{H}^2$. This translates into $f_{\theta,\theta} = \text{id}_{\mathbb{H}^2}$.

If $\rho : \pi_1(\Sigma) \longrightarrow \text{PSL}(2, \mathbb{R}) \longrightarrow \text{PSL}(n, \mathbb{R})$ is $\text{PSL}(n, \mathbb{R})$ -Fuchsian, recall from 1.2.3 that the image of H via the irreducible embedding is, modulo rescaling, what we later called the barycenter b (after normalising, the barycenter of a Weyl chamber at infinity). Turns out, the irreducible embedding $\text{PSL}(2, \mathbb{R}) \subset \text{PSL}(n, \mathbb{R})$ corresponds to an isometric embedding of $\mathbb{H}^2 \subset P(n, \mathbb{R})$ and the boundary at infinity $\partial_\infty \mathbb{H}^2$ is embedded in $\partial_F P(n, \mathbb{R}) = \text{PSL}(n, \mathbb{R}) \cdot b$ (it is then intuitively clear that in $\text{PSL}(n, \mathbb{R})$ -Fuchsian representations we avoid the flats associated to walls of Weyl chambers and hyperconvexity is satisfied; by showing stability of measures we just ensured that via deforming $\text{PSL}(n, \mathbb{R})$ -Fuchsian representations within the Hitchin component, we do not get too close to the walls of the Weyl chambers, when looking at the image of $F_{\theta,\rho}$). Therefore, if ρ is the composition of a Fuchsian $\tilde{\theta}$ with the principal map, $f_{\theta,\rho}$ is the composition of $f_{\theta,\tilde{\theta}}$ with the induced isometric embedding $\mathbb{H}^2 \subset P(n, \mathbb{R})$. In particular, $\theta = \tilde{\theta}$ means $f_{\theta,\rho}$ is precisely that isometric embedding.

At last, we can write down the parametrisation of Hitchin component that crowns this discussion. We consider the space $\mathcal{C} = C^0(\mathbb{H}^2, X)/G$ of continuous maps from $\mathbb{H}^2$ to X up to composition with an isometry of X (recall $\text{Iso}(X) = G$), with the topology obtained by quotienting the compact-open topology on $C^0(\mathbb{H}^2, X)$. We have the following topological embedding of the Hitchin component for $\text{PSL}(n, \mathbb{R})$ (see [Sle21]):

$$\text{Hit}_n \longrightarrow \mathcal{C}, \quad [\rho] \longrightarrow [f_{\theta,\rho}]$$

The parametrisation resembles the one we ascribe to Corlette, and is mentioned in Remark 2.24 of [li19], where each representation is associated a unique equivariant harmonic map that solves a PDE and thus is not explicit in general. Our Section 3.2 suggests that ours could be a more geometrically relevant replacement.

We now put forward an improvement/generalisation of this result that follows naturally from the setup we have built up. The idea is that the result for $\mathrm{PSL}(n, \mathbb{R})$ can be directly generalised for the groups $\mathrm{SO}(n, n+1)$, $\mathrm{PSp}(2n, \mathbb{R})$ and G_2^{split} . We also highlight why it is not so easy in the rest of the split real groups.

Theorem 5.8. *Fix a Fuchsian representation $\theta : \pi_1(\Sigma) \rightarrow \mathrm{PSL}(2, \mathbb{R})$. For a Hitchin representation $\rho : \pi_1(\Sigma) \rightarrow G$, where G is either $\mathrm{PSL}(n, \mathbb{R})$, $\mathrm{SO}(n, n+1)$, $\mathrm{PSp}(2n, \mathbb{R})$ or G_2^{split} , denote its flag curve $F_{\theta, \rho} : \partial_\infty \mathbb{H}^2 \rightarrow \partial_\infty X$, with $X = G/K$.*

Then, the measure $\mu_z^{\theta, \rho} = (F_{\theta, \rho})_ \nu_z$ is stable. Thus, for the Hitchin component $\mathrm{Hit}(G)$ in the character variety $\mathrm{Rep}(\pi_1(\Sigma), G)$, we obtain a topological embedding:*

$$\mathrm{Hit}(G) \rightarrow \mathcal{C}, \quad [\rho] \mapsto [f_{\theta, \rho}]$$

where the maps $f_{\theta, \rho}$ are smooth and intertwine θ and ρ , as in 5.7.

Proof. We just need to plate what we cooked up. First, observe that these groups are all subgroups of $\mathrm{PSL}(m, \mathbb{R})$ for a certain m . Moreover, by 1.57 and 1.58, the G -Fuchsian representations are $\mathrm{PSL}(M, \mathbb{R})$ -Fuchsian representations (they share their principal maps). Because Hitchin component is the connected component of the character variety constructed by taking deformations of G -Fuchsian representations, we get the natural inclusion $\mathrm{Hit}(G) \subset \mathrm{Hit}_m = \mathrm{Hit}(\mathrm{PSL}(m, \mathbb{R}))$.

Consequently, hyperconvexity properties are inherited. Since 1.57 and 1.58 also show irreducibility of these representations, we obtain the analog of 4.11 for any Hitchin representation in G .

On another note, 2.39 tells us that the symmetric space $G/K \subset P(m, \mathbb{R})$ embeds, and same for its structures at infinity. In particular, the osculating flag of ρ looks like:

$$F_{\theta, \rho} : \partial_\infty \mathbb{H}^2 \rightarrow G/B = \partial_F X \subset \partial_\infty X \cap \partial_F P(m, \mathbb{R}) \subset \partial_\infty X \subset \partial_\infty P(n, \mathbb{R})$$

Thus, the previous propositions concerning the barycenter construction follow directly by changing the Furstenberg boundary of $P(m, \mathbb{R})$ by such subset of it, namely $\partial_F X \subset \partial_F P(m, \mathbb{R})$. This implies that $(F_{\theta, \rho})_* \nu_z$ is stable and the analog of 5.7 still holds by *mimicking* the proofs (vertices at infinity for X are vertices of Weyl chambers at infinity for $P(m, \mathbb{R})$, by the embedding of chambers that comes from the fact that flats in X are flats in $P(m, \mathbb{R})$ intersected with X , along the lines of 2.38; there cannot be such a bad vertex, since it would be a bad vertex for $\partial_\infty P(m, \mathbb{R})$, and so on). $\square$

Remark 5.9. *In view of 2.39, we might wonder why we cannot do the same for the rest of split real groups. The obstacle is that we are not able to embed their Hitchin components into the one associated to the linear case $\mathrm{PSL}(n, \mathbb{R})$.*

For example, 1.59 shows that the Hitchin representations in $\mathrm{PSO}(n, n)$ do not act irreducibly on $\mathbb{R}^{2n}$, so the second item in the proof of 5.6 breaks down. This is reflected in $\mathrm{Hit}(\mathrm{PSO}(n, n)) \not\subset \mathrm{Hit}_{2n}$

Aside from that, Table 1 of [Sam20] suggests that one could prove $\mathrm{Hit}(F_4^{\mathrm{split}}) \subset \mathrm{Hit}(E_6^{\mathrm{split}})$ is a nice embedding of Hitchin components in the above sense, so if we were to show a similar statement for E_6 as the one we did for $\mathrm{PSL}(n, \mathbb{R})$, the result would be inherited by F_4 .

5.2 Further developments

This terminology should not be blamed on me. It was obtained by a democratic process in my course of 1976-77. An orbifold is something with many folds; unfortunately, the word “manifold” already has a different definition. I tried “foldamani”, which was quickly displaced by the suggestion of “manifolded”. After two months of patiently saying “no, not a manifold, a manifolded”, we held a vote, and “orbifold” won.

THE GEOMETRY AND TOPOLOGY OF 3-MANIFOLDS, *William Thurston*

As denouement, let us mention some possible directions of future study:

- The focal aim of the genesis of this parametrisation was to replace the one given in terms of harmonic maps that does not convey any geometric information. Accordingly, we should now check how much we can learn about Hitchin components by analysing this parametrisation: action of the mapping class group, compactifications, etc.
- The results concerning positive representations [FG06] suggest that we should be able to extend the scope of 5.8 to new split real groups, but for this we need to explore other techniques. For example, in the case of $\mathrm{PSO}(n, n)$, perhaps we should revisit Lusztig notion of positivity [Lus19].

By the same token, we could proceed to investigate applying the barycenter method in other contexts associated to higher Teichmüller spaces, since we have seen Hitchin components for split real groups are just a fraction of the whole picture and the notion of discrete and faithful representations is notably more general.

- Another approach would be to *change what we represent*: we could step out of surface groups and go for fundamental groups of other spaces, such as compact hyperbolic manifolds of any dimension (as we mentioned in the previous section).

In this spirit, we can already point out some results concerning fundamental groups of compact 2-dimensional orbifolds [ALS22]. This could be inspired by the fact that the classical study of Teichmüller space is naturally generalised to surfaces with punctures and boundary components.

An **orbifold** is locally a quotient of Euclidean space by a finite group. It is a fundamental source of examples for singular spaces, such as the moduli space of Riemann surfaces of genus one (related to the theory of modular forms), described in 0.7.

We can look at those with negative (orbifold) Euler characteristic, where Thurston already laid the foundations in the classical sense of Teichmüller theory [Thu78], and obtain analogous results as the ones here (relating being Hitchin to hyperconvexity, a version of non-abelian Hodge correspondence for orbifolds, etc.).

All in all, this essay might convince the reader that higher Teichmüller theory is an active area, and we can find many possible research options, more or less challenging and with different applications or utility in varying disciplines. Perhaps it might be difficult to discern which directions are more valuable, but time will tell, and its worth could maybe be already established in the basis of the interest of academics on it and the contemporary flux of researchers participating in its blossoming. In any case, it cannot be questioned that it provides the newcomers with motivation for developing their knowledge on a broad spectrum of mathematics.

References

- [Alb99] Paul Albuquerque. “Patterson-Sullivan Theory in Higher Rank Symmetric Spaces”. In: *Geometric And Functional Analysis* 9 (Mar. 1999), pp. 1–28. DOI: 10.1007/s000390050079.
- [ALS22] Daniele Alessandrini, Gye-Seon Lee, and Florent Schaffhauser. “Hitchin components for orbifolds”. In: *Journal of the European Mathematical Society* (Feb. 2022). DOI: 10.4171/JEMS/1210.
- [BGS85] Werner Ballmann, Mikhail Gromov, and Viktor Schroeder. “Manifolds of Nonpositive Curvature”. In: 61 (Jan. 1985). DOI: 10.1007/978-1-4684-9159-3_11.
- [BCG95] G. Besson, Gilles Courtois, and Sylvestre Gallot. “Entropies et rigidités des espaces localement symétriques de courbure strictement négative”. In: *Geometric and Functional Analysis* 5 (Sept. 1995), pp. 731–799. DOI: 10.1007/BF01897050.
- [BH09] Martin Bridson and Andr’e Haefliger. *Metric Spaces of Non-Positive Curvature*. Vol. 319. Jan. 2009. ISBN: 978-3-642-08399-0. DOI: 10.1007/978-3-662-12494-9.
- [CG93] Suhyoung Choi and William Goldman. “Convex real projective structures on closed surfaces are closed”. In: *Proc. Am. Math. Soc* 118 (June 1993), pp. 657–661. DOI: 10.1090/S0002-9939-1993-1145415-8.
- [CF01] Chris Connell and Benson Farb. “The Degree Theorem in Higher Rank”. In: *Journal of Differential Geometry* 65 (Feb. 2001). DOI: 10.4310/jdg/1090503052.
- [CF02] Chris Connell and Benson Farb. “Some recent applications of the barycenter method in geometry”. In: (May 2002). DOI: 10.1090/pspum/071/2024628.
- [Cor88] Kevin Corlette. “Flat G -bundles with canonical metrics”. In: *Journal of Differential Geometry* 28 (Nov. 1988). DOI: 10.4310/jdg/1214442469.
- [DZ19] Jeffrey Danciger and Tengren Zhang. “Affine actions with Hitchin linear part”. In: *Geometric and Functional Analysis* 29 (Oct. 2019). DOI: 10.1007/s00039-019-00511-6.
- [DE86] Adrien Douady and Clifford Earle. “Conformally natural extension of homeomorphisms of the circle”. In: *Acta Mathematica* 157 (Dec. 1986), pp. 23–48. DOI: 10.1007/BF02392590.
- [Ebe97] Patrick Eberlein. “Geometry of Nonpositively Curved Manifolds”. In: *Bibliovault OAI Repository, the University of Chicago Press* (Jan. 1997).
- [FG06] Vladimir Fock and Alexander Goncharov. “Moduli spaces of local systems and higher Teichmüller theory”. In: *Publications Mathématiques. Institut de Hautes Études Scientifiques, v.103, 1-211 (2006)* 103 (June 2006). DOI: 10.1007/s10240-006-0039-4.
- [FH91] William Fulton and J. Harris. *Representation Theory: A First Course*. Vol. 79. Jan. 1991. DOI: 10.2307/3618117.
- [GLW21] Olivier Guichard, François Labourie, and Anna Wienhard. “Positivity and representations of surface groups”. In: (June 2021).
- [GW07] Olivier Guichard and Anna Wienhard. “Convex Foliated Projective Structures and the Hitchin Component for $\mathrm{PSL}(4, \mathbb{R})$ ”. In: (June 2007).
- [Hal03] Brian Hall. *Lie groups, Lie algebras, and representations. An elementary introduction*. Jan. 2003. DOI: 10.1007/978-0-387-21554-9.
- [Hit92] N. Hitchin. “Lie groups and Teichmüller space”. In: *Topology* 31 (July 1992), pp. 449–473. DOI: 10.1016/0040-9383(92)90044-I.

- [Jac22] Audibert Jacques. “Zariski-dense surface groups in non-uniform lattices of split real Lie groups”. In: (June 2022).
- [Jag06] E. M. de Jager. “On the origin of the Korteweg-de Vries equation”. In: (2006). DOI: 10.48550/ARXIV.MATH/0602661. URL: <https://arxiv.org/abs/math/0602661>.
- [KLM03] Misha Kapovich, Bernhard Leeb, and John Millson. “Convex functions on symmetric spaces, side lengths of polygons and stability inequalities for weighted configurations at infinity”. In: *Journal of Differential Geometry* 81 (Dec. 2003). DOI: 10.4310/jdg/1231856263.
- [Kos59] Bertram Kostant. “The Principal Three-Dimensional Subgroup and the Betti Numbers of a Complex Simple Lie Group”. In: *American Journal of Mathematics* 81 (Oct. 1959). DOI: 10.1007/b94535_11.
- [KM08] Vincent Koziarz and Julien Maubon. “The Toledo invariant on smooth varieties of general type”. In: *Journal für die Reine und Angewandte Mathematik* 2010 (Nov. 2008). DOI: 10.1515/CRELLE.2010.093.
- [Lab04] Francois Labourie. “Anosov Flows, Surface Groups and Curves in Projective Space”. In: *Inventiones mathematicae* 165 (Feb. 2004). DOI: 10.1007/s00222-005-0487-3.
- [li19] Qionglin li. “An Introduction to Higgs Bundles via Harmonic Maps”. In: *Symmetry, Integrability and Geometry: Methods and Applications* (May 2019). DOI: 10.3842/SIGMA.2019.035.
- [Lus19] George Lusztig. “Total Positivity in Reductive Groups, II”. In: *Bulletin of the Institute of Mathematics Academia Sinica NEW SERIES* 14 (Dec. 2019). DOI: 10.21915/BIMAS.2019402.
- [Mul22] Motohico Mulase. “Algebraic theory of the KP equations”. In: (Aug. 2022).
- [Sam20] Andrés Sambarino. “Infinitesimal Zariski closures of positive representations”. In: (Dec. 2020).
- [Sle21] Ivo Slegers. “Natural maps in higher Teichmüller theory”. In: (July 2021).
- [Tho22] Alexander Thomas. “A Gentle Introduction to the Non-Abelian Hodge Correspondence”. In: (Aug. 2022). DOI: 10.48550/arXiv.2208.05940.
- [Thu78] W. Thurston. “The Geometry and Topology of 3-Manifolds”. In: (Jan. 1978).
- [Wie19] Anna Wienhard. “An invitation to higher Teichmüller theory”. In: May 2019, pp. 1013–1039. DOI: 10.1142/9789813272880_0086.